

IHARQ BenchGuard Stretch C

Experiment, Ablation, Evaluation, Execution, and Analysis Protocol v1.0
Single Canonical Cumulative Authority Through Phase 02
Single Canonical Protocol Authority — Cumulative Through P02

Canonical Protocol	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1
Authority Model	SINGLE_CANONICAL_CUMULATIVE_DOCUMENT_WITH_PHASE_PROJECTIONS
P01 / P02 Execution Decision	ACCEPTED; P02 ACCEPTED_FOR_PROTOCOL_AND_PHASE_ANALYSIS
Protocol Decision	IHARQ_PROTOCOL_V1_THROUGH_P02_MERGE_STATUS: COMPLETE - FROZEN
Generation Timestamp	2026-08-14T18:31:00+03:30

One current authority • P00 preserved • P01 preserved • P02 accepted • A4 scope bounded • A14 absent

Document map

- Part I - Project-Wide Protocol Authority
- Part II - Phase 00 Protocol (exact inherited P00 Annex R2)
- Part III - Phase 01 Protocol (accepted execution contract)
- Part IV - Phase 02 / Layer 2 Protocol Record
- Part V - Cumulative Run Matrix
- Part VI - Cumulative Analysis Contract
- Part VII - Cumulative Ablation and Control Register
- Part VIII - Amendments, Deviations, Supersessions, and Reruns
- Part IX - Environment and Reproducibility Register
- Part X - Cumulative Dataset, Split, Model, and Method State
- Parts XI-XII - External Artifacts, Limitations, Handoffs, and Current State
- Appendices A-I - Cumulative structured authority, source-use, merge-consistency, and integrity views

The canonical cumulative Markdown through P02 is the single current Protocol authority. This Word file is a non-authoritative presentation derivative; phase annex projections and cumulative machine-readable companions are governed subordinate surfaces and must not drift from the master.

Authority singularity rule. The cumulative Markdown through P02 is the single current Protocol v1.0 authority. P00/P01 predecessor material and the frozen standalone P02 R2 annex remain provenance sources. The packaged phase projections and machine-readable companions are subordinate governed representations, not competing Protocol authorities.

Document control

Field	Current controlling value
protocol_id	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1
version	1.0-CUM-P02-R1
authority_representation	SINGLE_CANONICAL_CUMULATIVE_DOCUMENT_WITH_PHASE_PROJECTIONS
phases	P00, P01, P02
generation_timestamp	2026-08-14T18:31:00+03:30
governance	Governance V6.1 single-track
P01 scientific freeze	P01-L1-OFFICIAL-RUN-FREEZE-R2
P01 config ID	d03f0a7c869dd95a2a67e5a32edc501d52bfc3851d8e761ef56595d8bd1ff52f
P01 execution status	ACCEPTED
unresolved blockers	0
future update rule	P03 and later extend this same cumulative Protocol authority
P02 accepted run	P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d
P02 execution status	ACCEPTED_FOR_PROTOCOL_AND_PHASE_ANALYSIS
P02 evidence status	SUFFICIENT_WITH_EXPLICIT_RESOURCE_CONSTRAINED_A4_LIMITS

Owner-directed document-architecture consolidation

The project owner has directed a transition from a fragmented Protocol-v1 distribution (master + phase annexes + separate machine-readable run/analysis surfaces) to **one cumulative canonical Protocol v1.0 document**. This is an authority-representation consolidation only.

Consolidation question	Disposition
scientific contract changed	NO
historical execution changed	NO
phase identity changed	NO
ablation identity changed	NO
analysis rules silently removed	NO
traceability reduced	NO
authority representation consolidated	YES

Predecessor representation and supersession

- IHARQ-PROTOCOL-V1-MASTER-R3 — predecessor project-wide component; its substantive current rules are internalized in Part I.
- IHARQ-PROTOCOL-V1-P00-ANNEX-R2 — exact P00 phase history; preserved verbatim in Part II.
- IHARQ-PROTOCOL-V1-P01-ANNEX-R1 — exact P01 execution-and-analysis contract; preserved in Part III.
- IHARQ-PROTOCOL-V1-P02-ANNEX-R2 — frozen validated P02/L2 source annex; integrated as Part IV without semantic reconstruction.
- predecessor cumulative run_matrix.yaml, analysis_contract.yaml, ablation/readiness and amendment/repair ledgers — preserved and extended through P02 in cumulative Parts V–XII and machine-readable companions.
- IHARQ-PROTOCOL-V1-MASTER-R2 remains a historical predecessor with SHA-256 938fcaab2be30d2e59ccc5082a9c3cc6c2ccc3e8889b352099aa514b96e046d4; it is not current workflow governance.

No prior Protocol history is deleted. The P00/P01 cumulative predecessor and standalone P02 R2 annex remain provenance sources; the current cumulative through-P02 authority internalizes their valid substance.

How to read this cumulative Protocol

- Part I is the current project-wide Protocol framework and global scientific-integrity authority.
 - Part II preserves the Phase 00 Protocol record.
 - Part III preserves the Phase 01 Protocol record and its state at P01 closure.
 - Part IV integrates the validated P02/L2 Protocol record; Parts V–XII provide cumulative cross-phase registers through P02.
 - Appendices A–I contain cumulative structured metadata, merge/source-use consistency views, and integrity state.
- Historical terms (for example P00 timing_mode: B) remain valid descriptions of historical evidence classification but do not reintroduce obsolete workflow modes into current Governance V6.1.

PART I — PROJECT-WIDE PROTOCOL AUTHORITY

Provenance note: The substantive rules below are inherited from predecessor Master R3 (IHARQ-PROTOCOL-V1-MASTER-R3). Under the owner-directed consolidation, they now operate through IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1 rather than through a separate current master file.

Document control

Field	Frozen value
project_id	IHARQ-BENCHGUARD-STRETCH-C
protocol_id	IHARQ-PROTOCOL-V1-MASTER-R3
protocol_version	1.0-R3
document_scope	PROJECT_WIDE_MASTER
effective_date	2026-08-07
registration_timestamp	2026-08-07T23:36:00+03:30
registration_or_archive_uri	LOCAL_PROTOCOL_PACKAGE_SNAPSHOT
identity_mode	CONTENT_HASHED_LOCAL_PACKAGE
predecessor_id	IHARQ-PROTOCOL-V1-MASTER-R2
predecessor_file_sha256	938fcaab2be30d2e59ccc5082a9c3cc6c2ccc3e8889b352099aa514b96e046d4
predecessor_protocol_freeze_sha256	881d7f705bfdacc0c00b4fb547a35e9b8dad60114314ca13f4fec115c3cb9d82
current_governance	Governance V6.1 single-track
current_workflow	ZIP-first; Kaggle-centered; evidence-insufficiency loopback only; Protocol-after-sufficient-execution
p00_annex	IHARQ-PROTOCOL-V1-P00-ANNEX-R2 — preserved unchanged as historical/current inherited authority
p01_annex	IHARQ-PROTOCOL-V1-P01-ANNEX-R1
p01_execution_status	ACCEPTED
current_status	FROZEN_WITH_NONBLOCKING_LIMITATIONS
registry_version	R44
review_mode	MULTI_PASS_FAIL_CLOSED_DOCUMENT_AND_EXECUTION_AUDIT
human_review_used	false
unresolved_freeze_critical_objection	NONE

R3 successor and governance-migration control

R3 is required because Master R2 contains **Governance V4-era timing-mode and publication/workflow machinery** that no longer describes the controlling project procedure. Governance V6.1 now owns a single direct phase workflow: inspect cumulative project state, implement/execute, assess evidence sufficiency, repair only evidence-insufficient scope, then complete Protocol v1.0, the Phase Evidence/Results/Interpretation Report, Layer 0 review, Evidence Map, Layer 10, and the updated cumulative ZIP. R2 remains immutable historical evidence.

R3 makes the minimum global semantic migration necessary:

- replaces timing-mode workflow machinery as a **current governing process** with V6.1 single-track chronology;
- preserves P00 Mode B only as historical evidence classification attached to the P00 execution record;
- removes any implication that GitHub synchronization/CI is required for normal intermediate-phase continuation;
- updates the phase register to record the existence/freeze of the P01 annex;

- retains all stable project-wide scientific rules, phase/layer identities, A0-A13 ladder, no-A14 lock, fairness, leakage, matching, denominator, negative-result, Layer 0 and Layer 10 boundaries;
 - adds no P01-specific dataset/configuration details to the master; those remain annex-owned.
- No P00 scientific or engineering history is rewritten.

1. Purpose and authority

This master freezes project-wide Protocol identities, evidence-status rules, amendment/deviation governance, global run-cell/comparison requirements, A0-A13 semantics, evidence-product interfaces, and the inheritance contract for phase annexes. It does not redesign Architecture, select methods post hoc, invent Registry objects, implement code, report scientific findings, approve claims, or permit Layer 10 recomputation.

The master plus phase annexes are binding only when human-readable and machine-readable identities agree, deterministic validation passes, source snapshots/hashes resolve, and no freeze-critical objection remains.

2. Scope-partitioned authority and conflict resolution

Authority	Owned surface	Protocol boundary
Governance V6.1	current document/workflow governance, continuation, Protocol-after-execution, cumulative ZIP strategy	supersedes older workflow/timing machinery; cannot alter scientific authorities
Architecture	P00-P15, L0-L10, A0-A13 and system boundaries	identities may not be renamed by configs/reports
Registry R44	canonical records, fields, aliases, lifecycle/status, lineage/interfaces	Protocol references; does not invent
Execution and Evidence Plan R41	phase products, gates, evidence roles, exit criteria	Protocol makes required identities/execution exact
Protocol v0.1/R42 + Protocol v1.0	fairness, matching, leakage, denominators, exclusions, negative evidence, ablations, analysis governance	v1 records actual execution and freezes future analysis without post-hoc upgrading
Phase Playbook R41	operational order, lawful repair and handoff	minimum-scope repair; failed evidence preserved
Method Selection R2	accepted datasets/methods/strategies	Protocol may not select new methods based on results
Nuts-and-Bolts R2	technical behavior, validators, failure semantics	Protocol freezes accepted profile identities and parameters
Implementation Build Book	executable realization, paths, environment intent, configs, run cells/gates	pre-run intent; actual execution evidence may document lawful runtime amendments
Final phase execution bundle	actual environment/run IDs/failures/amendments/accepted outputs	controls what actually happened on execution-owned surfaces

Conflicts are classified by affected surface and routed to the owning authority. Historical values are preserved; the current controlling value is explicit; incompatible values are never averaged or silently merged.

3. Master-plus-annex inheritance

There is one current cumulative Protocol authority with phase-indexed records and preserved phase-annex projections. Project-wide sections own global identities/rules; phase records own exact questions, inputs, configs, execution identities, analysis contracts, evidence ceilings, amendments and handoffs. Phase projection files support readability/traceability and do not compete with the cumulative master.

4. Current review, publication and project-state governance

Review is fail-closed and deterministic evidence outranks prose. During intermediate phases, the cumulative ZIP/package and its external immutable pointers are the controlling project state. GitHub and Hugging Face are optional complements for oversized/public artifacts; routine GitHub pull/synchronization/CI is not a phase-continuation requirement.

5. Current chronology and historical evidence classification

Current governing chronology has no Protocol timing modes. For every phase: pre-run frozen intent is preserved; actual execution is recorded; failed attempts and amendments remain visible; evidence sufficiency is evaluated; only the minimum evidence-insufficient scope is repaired/rerun; Protocol v1.0 is then completed from the accepted execution without post-hoc manipulation.

Historical labels such as P00 Mode B remain valid only as immutable descriptions of historical evidence chronology. They are not current workflow branches and cannot be used to create a second process track. Evidence class/claim ceiling still depends on chronology and whether choices were made before or after observing results.

6. Freeze surfaces

- FS-ID: phase, ablation, source, dataset/split/artifact and authority identities.
- FS-SCI: scientific questions, comparisons, eligibility, matching, estimands, exclusions and analysis rules.
- FS-EXEC: source snapshot/commit, environment, configs, seeds, budgets, simulator/stress/reward profiles.
- FS-ANALYSIS: analysis implementation, metric dictionaries, estimators, tables/figures and failure views.
- FS-RELEASE: accepted/invalid inventories, attrition release, source bundles, claim/evidence and Layer 10 handoffs.

A change to any frozen surface creates a successor or amendment and triggers topological invalidation of affected descendants.

7. Status axes

Document maturity, evidence class, record lifecycle, run validity and claim disposition remain separate. Document maturity never upgrades evidence class. Claim approval is owned by Layer 0 after the Phase Evidence Report. Template-local aliases map one-to-one to accepted Registry values and may not replace them silently.

8. Official phase register

Phase	Official name	Master role	Status
P00	Repository, Configuration, and Record Schema	Global identity only; exact values annex-owned	Created
P01	Public Data and Split Protocol	Global identity only; exact values annex-owned	Annex frozen in P01 R1
P02	Baseline Decoders	Global identity only; exact values annex-owned	Annex R2 frozen; accepted
P03	Calibration and Uncertainty	Global identity only; exact values annex-owned	Master ready
P04	IHARQ-lite Evidence Verification	Global identity only; exact values annex-owned	Master ready
P05	RegimeRisk Temporal Trust	Global identity only; exact values annex-owned	Master ready
P06	Evidence-Quality Estimator and Supervised Adaptive-IHARQ	Global identity only; exact values annex-owned	Master ready
P07	Learning-to-Defer	Global identity only; exact values annex-owned	Master ready
P08	Simulated Closed-Loop Readiness	Global identity only; exact values annex-owned	Master ready
P09	StressForge-Lite	Global identity only; exact values annex-owned	Master ready
P10	Contextual Bandit	Global identity only; exact values annex-owned	Master ready
P11	Reinforcement-Learning Policy	Global identity only; exact values annex-owned	Master ready
P12	MyoSuite Embodiment Demo	Global identity only; exact values annex-owned	Master ready
P13	OpenSim Replay or Optional Comparison	Global identity only; exact values annex-owned	Master ready
P14	Dashboard and Cards	Global identity only; exact values annex-owned	Master ready
P15	Final Thesis Integration	Global identity only; exact values annex-owned	Master ready

9. Official layer register

Layer	Owned surface	Protocol rule
L0	Claim safety, evidence sufficiency, wording, downgrade, limitations, lifecycle	Owner preserved; Protocol freezes applicable exact references
L1	Data, labels, splits, windows, preprocessing, manifests	Owner preserved; Protocol freezes applicable exact references
L2	Decoder families, controls, predictions, model identity	Owner preserved; Protocol freezes applicable exact references
L3	Calibration, uncertainty, selective prediction, registered operating points	Owner preserved; Protocol freezes applicable exact references
L4	IHARQ evidence verification, reasons, combination, fallback, unsafe states	Owner preserved; Protocol freezes applicable exact references
L5	Temporal features, trust, regime logic, stop-loss	Owner preserved; Protocol freezes applicable exact references
L6	Evidence quality, policy, deferral, costs, supervised/adaptive readiness	Owner preserved; Protocol freezes applicable exact references
L7	State, action, transition, reward, cost, session, rollout, simulator diagnostics	Owner preserved; Protocol freezes applicable exact references
L8	Stress taxonomy, profiles, schedules, injection, clean/stressed matching	Owner preserved; Protocol freezes applicable exact references
L9	Simulation platform, command mapping, safety gates, outcomes, embodiment evidence	Owner preserved; Protocol freezes applicable exact references
L10	Read-only provenance, dashboards, cards, exports, reproduction and release presentation	Owner preserved; Protocol freezes applicable exact references

Layer 0 governs claims after evidence exists. Layers 1–9 produce scientific or technical evidence. Layer 10 is read-only and cannot repair, rematch, retune, recompute, strengthen limitations, approve claims or create primary evidence.

10. Evaluation-mode registry

EM-OFFLINE, EM-EVIDENCE, EM-TEMPORAL, EM-POLICY, EM-CLOSEDLOOP, EM-STRESS, EM-EMBODIMENT, and EM-DASHBOARD are distinct modes. Every run cell declares one primary mode and any secondary modes; incompatible modes are not pooled without a registered estimand and claim boundary.

11. Official A0–A13 ladder and no-A14 lock

ID	Official identity	P00 disposition	Global constraint
A0	Raw Decoder / Accept-All Raw Decoder Reference	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A1	Calibrated Decoder / Calibration Visibility	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A2	Simple Registered Threshold / Confidence-Threshold Baseline	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A3	Uncertainty and Selective Prediction	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A4	Longer-Window, Multi-Window Voting/Averaging, and Ordinary Ensemble Controls	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A5	IHARQ-lite / Rule-Based Evidence Verification	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A6	IHARQ + Evidence-Quality Estimator	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A7	IHARQ + RegimeRisk Temporal Trust	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A8	Learning-to-Defer / Deferral Comparison	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A9	Supervised Adaptive-IHARQ / Adaptive Readiness Policy	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A10	Contextual Bandit / Simulation-Bounded Immediate-Reward Adaptation	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited

ID	Official identity	P00 disposition	Global constraint
A11	Reinforcement-Learning Policy / Simulation-Bounded Sequential Policy Learning	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A12	StressForge Stress Tests / Controlled Stress Robustness	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited
A13	Layer 9 MyoSuite/OpenSim/Static-Replay Embodiment Demo	READINESS_ONLY_NOT_ACTIVATED in P00	A14 prohibited

A12.5 is a local synchronized Layer 8/Layer 9 comparison under A12/A13, not a global A14. A14 is invalid in configs, schemas, CLI, validators, reports and handoffs.

12. Global data, split, chronology and visibility controls

Dataset/source eligibility, split roles, calibration/threshold fitting roles, chronology, dual clocks, field visibility and reuse decisions must be frozen in the applicable annex. Test/evaluation fields cannot be visible to training, calibration, threshold selection or decision-time logic. Temporal joins require strict predecessor legality and explicit clock uncertainty. Reuse is by exact identity; semantic change requires derivation or rerun.

13. Global run-cell contract

Every run cell records cell ID, phase, evaluation mode, source snapshot/commit, environment, config hash, input artifacts, seed or NOT_APPLICABLE, budgets, command, expected outputs, inclusion/invalidity rules, evidence class, limitations, rerun rule and downstream consumers. Missing required identity blocks the cell.

14. Matching, comparison and denominator governance

A matched comparison freezes eligible set, key dimensions, conditional keys, independent unit, common support, arm compatibility, attrition, unmatched disposition and estimand. Denominators are conserved through expected → discovered → eligible → attempted → valid → matched → analyzed → released. Missing/unmatched/invalid records remain explicit.

15. Metric, estimand and analysis governance

Every metric binds to a MetricDictionary identity, formula, direction, unit, denominator, aggregation, tie/endpoint/smoothing rules, inference method and claim role. Statistical units and dependence profiles are explicit. Multiplicity, selection, equivalence/noninferiority/no-harm margins, missingness, subgroup and sensitivity plans are annex-owned and frozen before claim-bearing execution.

16. Negative, null, harmful and failed evidence

Negative, null, equivalent, harmful, invalid, blocked, unmatched and diagnostic-only results remain visible. They cannot be dropped because they weaken a narrative. Failed runs are preserved with exact reason, invalidation and rerun disposition. Layer 10 must include negative/failure panels where applicable.

17. Specialized A10–A13 clauses

A10 requires actual propensities, support/ESS, estimator sensitivity, cross-fitting where applicable, frozen outer evaluation and no future leakage. A11 requires demonstrated sequential necessity, legal masks, reward/cost versions, valid trajectories, checkpoint identity and held-out frozen evaluation. A12 requires versioned stress profiles/schedules/seeds and clean/stressed matching. A13 requires platform/task/mapping/safety/outcome identities and explicit simulation/embodiment-proxy limitations.

18. Layer 0 and Layer 10 interfaces

The Protocol supplies claim ceilings, candidate statement interfaces, mandatory limitations and evidence identities; Layer 0 alone issues disposition after the Phase Evidence Report. Layer 10 consumes accepted saved evidence and mapped claims read-only. Any recomputation, repair, rematching, retuning or limitation weakening invalidates the Layer 10 package.

19. Evidence products and lifecycle

The immutable run bundle includes source/config/environment identities, inputs/outputs, logs, tests, gates, limitations and hashes. Later products follow Governance V6.1 single-track order: Protocol v1.0 phase annex → Phase Evidence, Results, and Interpretation Report → Layer 0 disposition → Evidence Map → Layer 10 package → cumulative project ZIP/handoff. This master does not create those final products.

20. Amendment, deviation and supersession

Pre-run amendments may change scientific/execution rules after owning-authority review. Post-execution Protocol completion records what actually happened and may not upgrade retrospective evidence. Deviations do not change the contract silently. Post-hoc analyses are exploratory and separately identified. Scientific corrections after release create successor artifacts and invalidate descendants.

21. Structured authority and machine-readable singularity

The authoritative cumulative machine companions are packaged under protocol_v1_0/machine_readable/ and may be placed at the governed repository path docs/authorities/protocol_v1_0/machine_readable/ when the cumulative project state is updated.

22. Validation and definition of done

Acceptance requires structural, parsing, semantic, cross-authority, hash, human/machine no-drift, execution-conformance, no-post-hoc, A0-A13/no-A14, lineage/integrity, security, downstream-usability and adversarial validation; no unresolved freeze-critical objection; no A14; and exact phase-level evidence identities.

Appendix A — Protocol profile ledger

Profile	Purpose	P00 status	Future owner
PV1-001	Threshold applicability profile	REFERENCE_ONLY_FUTURE_PHASE	P03
PV1-002	Dependence profile	APPLICABLE_FOUNDATION_PROFILE	P01/P03 and later
PV1-003	Temporal history profile	REFERENCE_ONLY_FUTURE_PHASE	P05
PV1-004	Priority/guard profile	REFERENCE_ONLY_FUTURE_PHASE	P04/P06/P07
PV1-005	Budget profile	REFERENCE_ONLY_FUTURE_PHASE	P04/P06-P11
PV1-006	Cost/burden profile	REFERENCE_ONLY_FUTURE_PHASE	P04/P06-P11
PV1-007	Combination relation profile	REFERENCE_ONLY_FUTURE_PHASE	P02-P04
PV1-008	A5 identity profile	REFERENCE_ONLY_FUTURE_PHASE	P04
PV1-009	Local A5 profile	REFERENCE_ONLY_FUTURE_PHASE	P04
PV1-010	Matched comparison profile	APPLICABLE_FOUNDATION_PROFILE	all later comparison phases
PV1-011	Metric profile	APPLICABLE_FOUNDATION_PROFILE	all later evidence phases
PV1-012	A7 causal profile	REFERENCE_ONLY_FUTURE_PHASE	P05 and later
MET-R11	Protocol R42 metric-interface closure ledger	APPLICABLE_FOUNDATION_PROFILE	P00 readiness; values phase-annex-owned

Appendix B — Source coverage

The source snapshot manifest includes Governance V4, all seven core authorities, Build Book R3 and Phase 0 annex successors, all Phase 0 prompts/audits/finalization records, the R3 Protocol template, and this creation prompt. Requirement-level mappings are in protocol_requirement_ledger.csv.

Appendix C — Nonblocking limitations

1. The portable registry-resolved cross-version uv.lock remains incomplete and fail-closed; the exact verified Python 3.13.5 runtime lock is available.
2. Python 3.11 and 3.12 were unavailable locally; Python 3.13.5 passed the full predecessor suite and Protocol validations.
3. Historical P00 conformance remains Mode B engineering/retrospective evidence; freezing this master does not upgrade it.

Appendix D — R3 migration ledger summary

- predecessor master: IHARQ-PROTOCOL-V1-MASTER-R2
- predecessor file SHA-256: 938fcaab2be30d2e59ccc5082a9c3cc6c2ccc3e8889b352099aa514b96e046d4
- predecessor P00 annex: IHARQ-PROTOCOL-V1-P00-ANNEX-R2 (unchanged)
- migration owner: Governance V6.1 on workflow/timing/project-state surfaces
- P01 addition: IHARQ-PROTOCOL-V1-P01-ANNEX-R1
- scientific effect of master migration: **NONE**
- P00 execution/history invalidated: **NO**
- additional P00 computation required: **NO**

PART II — PHASE 00 PROTOCOL

Exact preservation note: The complete P00 Annex R2 below is reproduced verbatim from the Phase 0 Protocol v1.0 package. Source SHA-256: a0dbabfc1c5be739955696f8b5d32d9cdbdedf563a1c9225d9767599bbfbd7b5. Its master_protocol_id and Mode-B fields are historical frozen metadata. They do not create a second current Protocol authority.

```
title: "IHARQ Protocol v1.0 Phase Annex — P00 Repository, Configuration, and Record Schema"
document_id: "IHARQ-PROTOCOL-V1-P00-ANNEX-R2"
```

master_protocol_id: "IHARQ-PROTOCOL-V1-MASTER-R2" version: "1.0-R2" status: "FROZEN_WITH_NONBLOCKING_LIMITATIONS" timing_mode: "B" timing_subtype: "ADMINISTRATIVE_FOUNDATION" evidence_ceiling: "ENGINEERING_FOUNDATION_CONFORMANCE" protocol_freeze_sha256: "881d7f705bfdacc0c00b4fb547a35e9b8dad60114314ca13f4fec115c3cb9d82"

Protocol v1.0 Phase Annex — P00 Repository, Configuration, and Record Schema

Document control

Field	Frozen value
annex_id	IHARQ-PROTOCOL-V1-P00-ANNEX-R2
master_protocol_id	IHARQ-PROTOCOL-V1-MASTER-R2
phase_id	P00
official_name	Repository, Configuration, and Record Schema
version	1.0-R2
registration_timestamp	2026-08-03T15:07:37+03:30
timing_mode	B
timing_subtype	ADMINISTRATIVE_FOUNDATION
evidence_ceiling	ENGINEERING_FOUNDATION_CONFORMANCE
identity_mode	LOCAL_PACKAGE_SNAPSHOT
source_snapshot_sha256	bcbf5c70e4b0ca09caa73c3f0963d8cc1bc0adbc0741fad60ef6cd936ca66c95
protocol_freeze_sha256	881d7f705bfdacc0c00b4fb547a35e9b8dad60114314ca13f4fec115c3cb9d82
publication_strategy	LOCAL_FIRST_SINGLE_PUBLICATION
github_ci_required	false
active_empirical_ablations	[]
claim_bearing_empirical_cells	[]
scientific_effectiveness_claims_allowed	false
status	FROZEN_WITH_NONBLOCKING_LIMITATIONS

Independent final-audit successor control

This R2 annex supersedes the frozen R1 annex only on Protocol-owned audit and package-governance surfaces. The Phase 0 engineering scope, Mode B timing determination, evidence ceiling, no-empirical-cell boundary, and local-first identity remain unchanged. The successor adds source-exhaustive requirement traceability, normalized review provenance, field-level human/machine drift evidence, explicit placeholder classification, and a passing synthetic P01 inheritance test.

1. Phase declaration and purpose

P00 freezes the administrative and engineering foundation contract for authority intake, source/package identity, schemas, record families, configurations, typed IDs, JCS/SHA-256 hashing, manifests, lineage, lifecycle, fixtures, validators, tests, integration, local reproduction, package integrity and downstream document readiness. It contains no empirical claim-bearing cell and does not activate A0–A13.

2. Timing-mode audit and evidence consequence

The selected mode is **B**. Broad requirements and gates existed before execution, but the exact Protocol annex, complete fixture taxonomy, validation rules, manifest exclusions, runtime-lock corrections, bounded runner behavior and package freeze rules were finalized through observed failure-and-repair cycles. That history does not satisfy Mode C no-result-contingent-change requirements.

Historical runs therefore remain ENGINEERING / RETROSPECTIVE foundation-conformance evidence. They may support bounded infrastructure and reproducibility findings, but not confirmatory scientific, clinical, deployment, safety or regulatory claims. A future prospective Mode C rerun is required only if the project wants prospective administrative-foundation registration; it is not required to preserve the historical engineering record.

3. Source and implementation freeze

- source snapshot: IHARQ_Phase_0_Local_First_Finalization_and_Readiness_COMPLETE_R1.zip
- source snapshot SHA-256: bcbf5c70e4b0ca09caa73c3f0963d8cc1bc0adbc0741fad60ef6cd936ca66c95
- authority manifest: IHARQ-PV1-AUTHORITY-SNAPSHOT-R1
- local-finalization handoff: P00-LOCAL-FIRST-FINALIZATION-HANDOFF-R1
- environment: Python 3.13.5 locally verified
- exact local dependency closure: REQUIREMENTS-LOCK-LOCAL-EXACT-R3
- portable uv lock: COMPLETE_UV_LOCK_DEFERRED_ENVIRONMENTALLY
- schemas/configs/records/fixtures/validators/tests/gates: exact catalogs in the source snapshot
- freeze identity: 720a38906046738d29d0bf63087eeda0c03211a7ee9d397b0dd2870037050605

4. Exact scope

Applicable: authority resolution; repository-package structure; schema/config/record coverage; typed IDs; canonical serialization and SHA-256; seeds or explicit NOT_APPLICABLE; manifests; lineage/lifecycle/supersession/invalidation; valid and malformed fixtures; validator/test coverage; L0–L10 foundation integration; A0–A13 readiness and A14 rejection; local CI-equivalent execution; clean reproduction; package checksum reconciliation; future phase/document contracts.

Not applicable: real datasets and scientific splits; model training/inference; calibration fitting or threshold selection; IHARQ effectiveness evaluation; temporal inference; policy learning; closed-loop scientific simulation; stress or embodiment execution; scientific metric estimation; statistical superiority; clinical or real-world claims.

5. Scientific applicability dispositions

Surface	P00 status	Reason	Future owner
Dataset eligibility	NOT_APPLICABLE	Only authority packages and explicit mock/non-empirical fixtures are used	P01 and later dataset-bearing phases
Calibration/threshold selection	NOT_APPLICABLE	No predictions or operating points are fitted in P00	P03

Surface	P00 status	Reason	Future owner
Scientific estimands	NOT_APPLICABLE	P00 uses deterministic engineering inventory/conformance only	Applicable later annex
Statistical inference	NOT_APPLICABLE	No scientific population or treatment effect exists	Applicable later annex
A10 policy/OPE	NOT_APPLICABLE	No adaptation or OPE occurs	P10
Stress and embodiment outcomes	NOT_APPLICABLE	Only schemas, fixtures and interfaces are validated	P09/P12/P13

6. Engineering run matrix

Cell ID	Purpose	Registered command	Pass criterion
P00-CELL-AUTHORITY-INTAKE	Verify authority/source identities and requirement disposition	python scripts/run_local_first_finalization_audit.py	All authority hashes resolve; requirement ledger has no missing applicable disposition
P00-CELL-SCHEMA-COVERAGE	Validate JSON Schema and record-family coverage	python scripts/run_static_checks.py	All schemas parse and catalog references resolve
P00-CELL-CONFIG-RESOLUTION	Resolve strict P00 configuration	python -m iharq.cli phase validate-inputs --phase P00 --profile configs/phases/p00.yaml	Strict configuration validates; phase identity matches P00
P00-CELL-IDENTITY-JCS-HASH	Validate typed IDs, canonical serialization, SHA-256 and golden vectors	python -m pytest -q tests/test_canonical.py tests/test_lineage_lifecycle.py	All deterministic identity/hash tests pass
P00-CELL-VALID-FIXTURES	Accept all valid and integrated non-empirical fixtures	python -m pytest -q tests/test_valid_fixtures.py	Every registered valid/integrated bundle passes
P00-CELL-MALFORMED-FIXTURES	Reject the complete malformed taxonomy	python -m pytest -q tests/test_negative_fixtures.py tests/test_audit1_negative_fixtures.py tests/test_audit2_negative_fixtures.py tests/test_audit3_negative_fixtures.py	Every malformed category produces a deterministic failure
P00-CELL-VALIDATOR-TEST-COVERAGE	Run complete deterministic suite	python -m pytest -q -p no:cacheprovider	Complete suite passes with no hidden deselection
P00-CELL-A0-A13-READINESS	Verify A0-A13 readiness and reject A14	python scripts/run_phase0_final_implementation_audit.py	A0-A13 foundation hooks complete; A14 rejected
P00-CELL-L0-L3-INTEGRATION	Verify L0-L3 integration foundation	python scripts/run_official_layer_audit_1.py	Audit 1 regression passes
P00-CELL-POLICY-UPDATE	Verify update-enabled policy traceability	python scripts/run_official_layer_audit_2.py	Update trace, before/after policy IDs, reward/config/seed and limitations preserved
P00-CELL-FROZEN-EVALUATION	Verify frozen-evaluation immutability	python scripts/run_official_layer_audit_2.py	No mutation; disabled-update evidence and mode warning present
P00-CELL-L8-STRESS-LINEAGE	Verify clean-to-stressed lineage and limitations	python scripts/run_official_layer_audit_3.py	Stress lineage and matching pass
P00-CELL-L9-EMBODIMENT-PROXY	Verify simulation-only embodiment proxy contracts	python scripts/run_official_layer_audit_3.py	Proxy limitations and safety/reward lineage pass
P00-CELL-L10-READ-ONLY	Verify Layer 10 source-only behavior	python scripts/run_official_layer_audit_3.py	No upstream mutation, rematching, retuning, claim approval or primary evidence creation
P00-CELL-MANIFEST-RECONCILIATION	Regenerate and compare repository manifest	python scripts/reconcile_repository_manifest.py && python scripts/reconcile_repository_manifest.py --check	Manifest matches governed tree after transient exclusions
P00-CELL-LOCAL-REPRODUCTION	Reproduce from clean isolated local copy	python scripts/run_local_reproduction.py	Isolated reproduction passes using exact verified local dependency closure
P00-CELL-PACKAGE-INTEGRITY	Build and verify repository-ready archive	python -m iharq.cli package build --output protocol_package_test.zip && python -m iharq.cli package verify --archive protocol_package_test.zip	Archive CRC, file count and hashes reconcile
P00-CELL-FUTURE-PHASE-CONTRACTS	Verify P00-P15 reusable contracts and L0-L10 foundations	python scripts/run_phase0_final_implementation_audit.py	All phase contracts and layer foundations have complete dispositions
P00-CELL-NEXT-DOCUMENT-READINESS	Verify six downstream readiness packages	python scripts/run_local_first_finalization_audit.py	Six readiness packages exist and are clearly non-final

All cells use `ablation_id = NOT_APPLICABLE`, `scientific_matching_key = NOT_APPLICABLE`, `scientific_estimand = NOT_APPLICABLE`, and `scientific_seed_set = NOT_APPLICABLE`, except deterministic fixture seeds where a test explicitly requires a stable identity.

7. Engineering analysis contract

The registered analysis is descriptive and deterministic. It reports exact expected/discovered/validated/failed/excluded/invalid/accepted inventories for artifacts, schemas, configs, records, fixtures, validators, tests, gates, integration chains, hashes, manifests, environment/lock evidence, clean reproduction and package integrity. It does not compute scientific performance, p-values, treatment effects, superiority, calibration effectiveness, stress robustness or simulator performance.

8. PV1/MET applicability

Profile	Purpose	P00 disposition	Future owner
PV1-001	Threshold applicability profile	REFERENCE_ONLY_FUTURE_PHASE	P03
PV1-002	Dependence profile	APPLICABLE_FOUNDATION_PROFILE	P01/P03 and later
PV1-003	Temporal history profile	REFERENCE_ONLY_FUTURE_PHASE	P05
PV1-004	Priority/guard profile	REFERENCE_ONLY_FUTURE_PHASE	P04/P06/P07
PV1-005	Budget profile	REFERENCE_ONLY_FUTURE_PHASE	P04/P06-P11
PV1-006	Cost/burden profile	REFERENCE_ONLY_FUTURE_PHASE	P04/P06-P11
PV1-007	Combination relation profile	REFERENCE_ONLY_FUTURE_PHASE	P02-P04
PV1-008	A5 identity profile	REFERENCE_ONLY_FUTURE_PHASE	P04
PV1-009	Local A5 profile	REFERENCE_ONLY_FUTURE_PHASE	P04
PV1-010	Matched comparison profile	APPLICABLE_FOUNDATION_PROFILE	all later comparison phases
PV1-011	Metric profile	APPLICABLE_FOUNDATION_PROFILE	all later evidence phases
PV1-012	A7 causal profile	REFERENCE_ONLY_FUTURE_PHASE	P05 and later
MET-R11	Protocol R42 metric-interface closure ledger	APPLICABLE_FOUNDATION_PROFILE	P00 readiness; values phase-annex-owned

No future scientific numerical value is populated. P00 validates only profile existence, IDs, versions, hooks, validators and the absence of hidden defaults.

9. A0–A13 readiness and A14 rejection

ID	Official identity	P00 status
A0	Raw Decoder / Accept-All Raw Decoder Reference	READINESS_ONLY_NOT_ACTIVATED
A1	Calibrated Decoder / Calibration Visibility	READINESS_ONLY_NOT_ACTIVATED
A2	Simple Registered Threshold / Confidence-Threshold Baseline	READINESS_ONLY_NOT_ACTIVATED
A3	Uncertainty and Selective Prediction	READINESS_ONLY_NOT_ACTIVATED
A4	Longer-Window, Multi-Window Voting/Averaging, and Ordinary Ensemble Controls	READINESS_ONLY_NOT_ACTIVATED
A5	IHARQ-lite / Rule-Based Evidence Verification	READINESS_ONLY_NOT_ACTIVATED
A6	IHARQ + Evidence-Quality Estimator	READINESS_ONLY_NOT_ACTIVATED
A7	IHARQ + RegimeRisk Temporal Trust	READINESS_ONLY_NOT_ACTIVATED
A8	Learning-to-Defer / Deferral Comparison	READINESS_ONLY_NOT_ACTIVATED
A9	Supervised Adaptive-IHARQ / Adaptive Readiness Policy	READINESS_ONLY_NOT_ACTIVATED

ID	Official identity	P00 status
A10	Contextual Bandit / Simulation-Bounded Immediate-Reward Adaptation	READINESS_ONLY_NOT_ACTIVATED
A11	Reinforcement-Learning Policy / Simulation-Bounded Sequential Policy Learning	READINESS_ONLY_NOT_ACTIVATED
A12	StressForge Stress Tests / Controlled Stress Robustness	READINESS_ONLY_NOT_ACTIVATED
A13	Layer 9 MyoSuite/OpenSim/Static-Replay Embodiment Demo	READINESS_ONLY_NOT_ACTIVATED

A14 is prohibited. A12.5 remains a local synchronized comparison under A12/A13 and is not a global ablation.

10. Layer participation and noninterference

Layer	P00 foundation role	Noninterference rule
L0	Claim safety, evidence sufficiency, wording, downgrade, limitations, lifecycle	Foundation contracts only; no scientific execution or ownership reassignment
L1	Data, labels, splits, windows, preprocessing, manifests	Foundation contracts only; no scientific execution or ownership reassignment
L2	Decoder families, controls, predictions, model identity	Foundation contracts only; no scientific execution or ownership reassignment
L3	Calibration, uncertainty, selective prediction, registered operating points	Foundation contracts only; no scientific execution or ownership reassignment
L4	IHARQ evidence verification, reasons, combination, fallback, unsafe states	Foundation contracts only; no scientific execution or ownership reassignment
L5	Temporal features, trust, regime logic, stop-loss	Foundation contracts only; no scientific execution or ownership reassignment
L6	Evidence quality, policy, deferral, costs, supervised/adaptive readiness	Foundation contracts only; no scientific execution or ownership reassignment
L7	State, action, transition, reward, cost, session, rollout, simulator diagnostics	Foundation contracts only; no scientific execution or ownership reassignment
L8	Stress taxonomy, profiles, schedules, injection, clean/stressed matching	Foundation contracts only; no scientific execution or ownership reassignment
L9	Simulation platform, command mapping, safety gates, outcomes, embodiment evidence	Foundation contracts only; no scientific execution or ownership reassignment
L10	Read-only provenance, dashboards, cards, exports, reproduction and release presentation	Foundation contracts only; no scientific execution or ownership reassignment

Layer 0 cannot modify measurements, metrics, denominators, matching, predictions or empirical records. Layer 10 cannot recompute, repair, rematch, retune, hide negatives, weaken limitations, approve claims or create primary evidence.

11. Local-first gate crosswalk

Gate	Status	Protocol evidence
P0-GATE-01	PASS	P00-CELL-AUTHORITY-INTAKE
P0-GATE-02	PASS	P00-CELL-SCHEMA-COVERAGE
P0-GATE-03	PASS	P00-CELL-CONFIG-RESOLUTION
P0-GATE-04	PASS	P00-CELL-IDENTITY-JCS-HASH
P0-GATE-05	PASS	P00-CELL-VALID-FIXTURES
P0-GATE-06	PASS	P00-CELL-MALFORMED-FIXTURES
P0-GATE-07	PASS	P00-CELL-VALIDATOR-TEST-COVERAGE
P0-GATE-08	PASS	P00-CELL-A0-A13-READINESS
P0-GATE-09	PASS	P00-CELL-L0-L3-INTEGRATION

Gate	Status	Protocol evidence
P0-GATE-10	PASS	P00-CELL-POLICY-UPDATE
P0-GATE-11	PASS	P00-CELL-FROZEN-EVALUATION
P0-GATE-12	PASS	P00-CELL-L8-STRESS-LINEAGE
P0-GATE-13_FOUNDATION	PASS	P00-CELL-L9-EMBODIMENT-PROXY
P0-GATE-14_FOUNDATION	PASS	P00-CELL-L10-READ-ONLY
P0-GATE-15	PASS	P00-CELL-MANIFEST-RECONCILIATION
P0-GATE-16_IMPLEMENTATION	PASS_WITH_NONBLOCKING_LIMITATIONS	P00-CELL-LOCAL-REPRODUCTION
P0-GATE-17_LOCAL	PASS	P00-CELL-PACKAGE-INTEGRITY
P0-GATE-18	DEFERRED_TO_LATER_GOVERNED_PHASE0_CLOSURE	P00-CELL-FUTURE-PHASE-CONTRACTS

GitHub CI is NOT_APPLICABLE_BY_ACCEPTED_WORKFLOW_STRATEGY; it is neither attempted nor simulated.

12. No-drift, amendment and deviation

Mode C no-drift is not claimed for historical runs. From this freeze forward, any affected source, schema, config, fixture, test, gate, environment, run-cell or analysis change requires an amendment or successor, descendant invalidation, rerun disposition and regenerated hashes. Clerical corrections require before/after digests and proof of unchanged semantics.

13. Evidence and downstream handoffs

The annex prepares, but does not create, the final Phase Analysis, Phase Evidence Report, Layer 0 disposition, accepted Evidence Map, Layer 10 package, final release or Phase 1 handoff. Downstream input specifications are provided under docs/authorities/protocol_v1_0/downstream_readiness/.

14. Limitations

- 4. Portable cross-version uv.lock is incomplete and explicitly fail-closed.
- 5. Python 3.11 and 3.12 were unavailable locally; Python 3.13.5 passed.
- 6. Historical conformance is Mode B engineering/retrospective, not prospective Mode C.
- 7. The annex governs deterministic foundation evidence only; it does not authorize scientific claims.

15. Freeze decision

P00_PROTOCOL_V1_MASTER_AND_ANNEX_FROZEN_WITH_NONBLOCKING_LIMITATIONS.

All freeze-critical P00 fields are resolved. Human and machine IDs match. The remaining limitations do not prevent exact local use of the annex but must be inherited by future execution/analysis records.

P00 cumulative consolidation disposition

- P00 scientific/engineering meaning: **PRESERVED.**
- P00 historical Mode-B evidence ceiling: **PRESERVED.**
- P00 empirical ablations: **NONE executed.**
- P00 run matrix: **19 cells preserved in Part IV / Appendix B.**
- P00 analysis contract: **6 deterministic engineering analyses preserved in Part V / Appendix C.**
- Current workflow machinery: **Governance V6.1**, not historical Governance V4 timing modes.
- Additional P00 computation: **NOT REQUIRED.**

PART III — PHASE 01 PROTOCOL

Predecessor integration note: The Phase 01 Annex R1 content below is the accepted P01 contract and is internalized here. Its historical master_protocol_id: IHARQ-PROTOCOL-V1-MASTER-R3 records its creation lineage; the current controlling Protocol identity is this cumulative document.

```
title: "IHARQ Protocol v1.0 — Phase 01 Annex: Public Data and Split Protocol"
document_id: "IHARQ-PROTOCOL-V1-P01-ANNEX-R1"
```

version: "1.0-P01-R1" status: "FROZEN_WITH_EXPLICIT_EXECUTION_AMENDMENTS_AND_DOWNSTREAM_REQUIREMENTS" master_protocol_id: "IHARQ-PROTOCOL-V1-MASTER-R3" phase_id: "P01" primary_layer: "L1" scientific_freeze: "P01-L1-OFFICIAL-RUN-FREEZE-R2" config_id: "d03f0a7c869dd95a2a67e5a32edc501d52bfc3851d8e761ef56595d8bd1ff52f" registration_timestamp: "2026-08-07T23:36:00+03:30" execution_status: "ACCEPTED"

IHARQ BenchGuard Stretch C

Experiment, Ablation, and Evaluation Protocol v1.0 — Phase 01 Annex

6.1 Document control

Field	Frozen/actual value
annex_id	IHARQ-PROTOCOL-V1-P01-ANNEX-R1
master_protocol_id	IHARQ-PROTOCOL-V1-MASTER-R3
phase_id	P01
official_phase_name	Public Data and Split Protocol
primary_implementation_layer	L1
version	1.0-P01-R1
status	FROZEN_WITH_EXPLICIT_EXECUTION_AMENDMENTS_AND_DOWNSTREAM_REQUIREMENTS
registration/freeze timestamp	2026-08-07T23:36:00+03:30
source snapshot	final P01 execution bundle SHA-256 09c3bb0442d79ddf110cf66fc4fe61fe63e94c7d8ed9f198f606639b4191f16e
implementation Build Book	IHARQ-IBB-R10-P01-L1-INDEPENDENT-AUDIT-REPAIRED / IHARQ-IBB-P01-L1-ANNEX-R4
execution bundle	IHARQ_P01_L1_Phase_Execution_Bundle_d03f0a7c869d.zip
config_id	d03f0a7c869dd95a2a67e5a32edc501d52bfc3851d8e761ef56595d8bd1ff52f
executed notebook source SHA-256	54b84d9cd29eb57bb22f45b7b7251e76e4947863c832528dba9c51b96189b023
final runtime revision	R49-MATCHED-A4-R2-COMPLETE
base notebook manifest ID	P01-L1-KAGGLE-NOTEBOOK-R26
scientific_freeze	P01-L1-OFFICIAL-RUN-FREEZE-R2
gate decision	ACCEPTED; P01-G01..P01-G16 PASS; 0 blockers
evidence status	SUFFICIENT_FOR_PROTOCOL_AND_PHASE_REPORT; DATA/ENGINEERING FOUNDATION ONLY
external core pointer	csthv999z/iharq-p01-l1-derived-windows-d03f0a7c869d-20260806222242-68a91473 / provider v2 / logical rev1 / dc21f418e9a1add62adb346f627d5d729735a5f1ecaa1d771f6fa5cfede627e1

Field	Frozen/actual value
external A4 pointer	csthv999z/iharq-p01-l1-a4-d03f0a7c-9a7c6108 / provider v1 / logical family R2 / 29124d59907610b1545e0a2b5d1811a4d4a2bf1df64cad49aa880f4721c47305
supersession status	First P01 annex; inherits P00 and Master R3; A4 R2 supersedes only the failed proposed A4 R1 alternative profile

6.2 Phase declaration and purpose

P01 is the governed **public-data, source-provenance, labeling, subject-group split, preprocessing, event/window materialization, quality-validation, canonical-record, external-persistence and downstream-readiness phase** for Layer 1. Its claim-bearing ceiling is data/protocol integrity and reproducible downstream readiness. P01 does **not** own decoder training, decoder superiority, calibration effectiveness, clinical efficacy, deployment safety, policy benefit, stress robustness or embodiment effectiveness.

6.3 Prior-state inheritance from P00

P01 reuses, without rewriting P00 history: the project authority stack; Registry-backed record/schema infrastructure; canonical hashing/canonicalization; lifecycle/lineage rules; validators/fixtures; phase/layer interfaces; global A0-A13 identities; the no-A14 lock; Protocol master inheritance; and Build Book/reproduction infrastructure. P01 extends these surfaces with actual L1 DatasetRecords, LabelMapRecords, SplitRecord, PreprocessingRecord, WindowRecords, quality records, validation reports, readiness artifacts, external Dataset pointers and P01 → P02 handoffs.

P00 Annex R2 is copied byte-for-byte into this package and remains associated with Master R2 as its historical frozen context. Master R3 does not rewrite that artifact; it supplies current workflow governance for new/current annexes.

6.4 Pre-run Phase 1 freeze

The controlling pre-run intent was P01-L1-OFFICIAL-RUN-FREEZE-R2, under Build Book IHARQ-IBB-R10-P01-L1-INDEPENDENT-AUDIT-REPAIRED and Annex IHARQ-IBB-P01-L1-ANNEX-R4. It froze:

- active datasets: PhysioNetMI, BNCI2014_001, Lee2019_MI; inactive screened sources: Cho2017, GuttmanFlury2025_MI;
- official binary labels: left_hand/right_hand; exclusions: rest/no_action, feet, tongue, motor execution, technical markers and unlabeled online/test events as source-appropriate;
- P01-L1-PREPROCESS-OFFICIAL-R2;
- P01-L1-SPLIT-OFFICIAL-R2, subject grouping, 60/20/10/10, seed 20260804;
- P01-L1-LOW-CAL-OFFICIAL-R2, budgets 1/2/4/8/16/32 per class from calibration only, seed 20260804;
- P01-L1-WINDOW-OFFICIAL-R2, one +0.5..+3.5 s window/event at 160 Hz;
- P01-L1-QUALITY-OFFICIAL-R2, annotate-not-repair;
- original Kaggle environment intent, exact package pins, deterministic thread variables, original 60/90 GiB disk policy;
- P01-G01..P01-G16, dual persistence and P02 handoff expectations.

The Build Book itself explicitly stated that the official run was **not executed in that package**; therefore actual execution values below are taken from the final execution bundle.

6.5 Actual source datasets

Dataset	Release/ source revision	Scientific role	License	Access	Observed aggregate SHA-256	Subjects	Sessions	Runs	Sampling Hz	Source event inventory count	Observed channel-set width*	Accepted core-window role counts	DatasetRecord	Semantic hash	Lifecycle	Limitations
PhysioNetMI	1.0.0	source/rest/ provenance anchor	Open Data Commons Attribution License 1.0 (ODC-By-1.0)	MOABB_1_5_ 0_OFFICIAL_ PHYSIONET_ DOWNLOAD	28cd2062983b6 236 f9a0e7fdee91fc 9d 8d5aad8eee3ef 561 cff5828ae89bf2 ba	109	0	12, 4, 8	128.0, 160.0	9509	65	T 2949 / C 979 / V 495 / Test 495	IHARQ- DATASETREC ORD- 20260806- 66309cda68771 bef	66309cda68771 bef 9bd7a3aebdac8 19d 91201cedca79a b11 2ea993fb61f55 8cf	VALIDATED	Run context is mandatory for T1/T2 semantics; Executed- movement and baseline runs are excluded from the official binary MI branch; Subject 88 rate exception must be recorded and resampled deterministically ; MOABB bulk loading across subject 88 and 160 Hz subjects is prohibited; per-subject acquisition is mandatory.
BNCI2014_001	001-2014 provider file set A01T/A01E through A09T/ A09E	standard four- class MI benchmark companion; official P01 binary left/right branch	Creative Commons Attribution- NoDerivatives 4.0 International (CC BY-ND 4.0)	MOABB_1_5_ 0_OFFICIAL_ BNCI_ DOWNLOAD	04a5390f8f36e aad bc0c480ec9377 ce1 b99caf0b7ab53 bad 9fda12347995b c49	9	0train, 1test	0, 1, 2, 3, 4, 5	250.0	5184	26	T 1440 / C 576 / V 288 / Test 288	IHARQ- DATASETREC ORD- 20260806- 42c424800627b 6ee	42c424800627b 6ee 0be7f47ad602b a49 cf4c24a632c1fa 69 c8c750a4ba77e 163	VALIDATED	Feet and tongue are preserved as excluded source labels in the official binary branch; EOG channels are metadata/ quality channels and not model input; CC BY- ND restrictions prohibit redistributed derived raw- signal variants
Lee2019_MI	GigaDB dataset DOI 10.5524/ 100542; MOABB 1.5.0 Lee2019_MI wrapper; labeled offline/ train MI runs only	maximum- scope two- session left/right MI target companion	GNU General Public License v3.0 as documented by the maintained MOABB source card; source terms retained in DatasetCard	MOABB_1_5_ 0_OFFICIAL_ GIGADB_ DOWNLOAD	3a07b2f302da9 49e fd418a0712d5a 942 7df34dc8b027 ca5 53fae8e67a849 f78	54	1	1train	1000.0	5400	67	T 3200 / C 1100 / V 500 / Test 600	IHARQ- DATASETREC ORD- 20260806- adb91f25a65e5 88e	adb91f25a65e5 88e ce06884a9598c c92 bd932e91c4bab c3c acb98c8290159 6f1	VALIDATED	Online/test runs are excluded from supervised P01 records; High-rate source requires deterministic anti-aliased resampling; Source phase and session must remain in lineage

*Channel-set width is the observed source record channel-set string width, including metadata/stim/EOG channels where present; the official signal tensor remains EEG-only under preprocessing. Source event_count is the admitted source event inventory and is not identical to the accepted left/right core denominator. Accepted core-window counts are the supervised P01 denominator after labels/exclusions and all validity rules.

6.6 Task, labels and exclusions

The P01 task is binary left-hand versus right-hand motor imagery. Non-target events are **excluded, not relabeled as negative**. Unknown source events fail closed. Dataset-specific mappings are:

Dataset	Mapping	Explicit exclusions	Unknown-event behavior	LabelMapRecord	Semantic hash	Lifecycle
PhysioNetMI	{"run_4_8_12:T1": "left_hand", "run_4_8_12:T2": "right_hand"}	run_4_8_12:T0	BLOCK_SOURCE_EVENT_WITH_UNKNOWN_LABEL	IHARQ-LABELMAPRECORD-20260806-4379781a3b5f5ea9	4379781a3b5f5ea91f70c560e007de42d8624b4ee0e6a792921079bd069a663	VALIDATED
BNCI2014_001	{"769": "left_hand", "770": "right_hand"}	1023, 1072, 276, 277, 32766, 768, 771, 772, 783	BLOCK_SOURCE_EVENT_WITH_UNKNOWN_LABEL	IHARQ-LABELMAPRECORD-20260806-587dcff81307768	587dcff813077685eaa34b9b204eae5791be7c5d25f2a500daff39b0348f84	VALIDATED
Lee2019_MI	{"left_hand": "left_hand", "right_hand": "right_hand"}	NONE	BLOCK_SOURCE_EVENT_WITH_UNKNOWN_LABEL	IHARQ-LABELMAPRECORD-20260806-b551cfd20335896c	b551cfd20335896c762d508e5950c8598772b7d4c75bec5d9dde8e177bfe105	VALIDATED

Additional source-level exclusions include PhysioNet executed-movement/baseline runs outside MI runs 4/8/12, Lee2019 online/test/unlabeled runs, and BNCI technical/rejected/feet/tongue/ambiguous events. All excluded/unknown behavior remains traceable.

6.7 Split protocol and leakage contract

The final split is P01-L1-SPLIT-OFFICIAL-R2, canonical record IHARQ-SPLITRECORD-20260806-e4e371d332c61e36, semantic hash e4e371d332c61e36699f07cb6bed6d0820e14b22d5685dc353d89c1de144c148. The grouping unit is (dataset_id, subject_id); subjects are atomic; source events/windows cannot cross roles. Ratios are train 0.60, calibration 0.20, validation 0.10, test 0.10 using deterministic SHA-256 ranking, largest remainder/minimum-one allocation and seed **20260804**.

Dataset	Train subjects	Calibration subjects	Validation subjects	Test subjects	Train windows	Calibration windows	Validation windows	Test windows	Total windows
PhysioNetMI	65	22	11	11	2949	979	495	495	4918
BNCI2014_001	5	2	1	1	1440	576	288	288	2592
Lee2019_MI	32	11	5	6	3200	1100	500	600	5400

Global accepted core windows by role: train **7589**, calibration **2655**, validation **1283**, test **1383**. Leakage audit status: **PASS** for GROUP_DISJOINTNESS, DUPLICATE_SAMPLE, OVERLAP_GROUP, FIT_SCOPE, BUDGET_TEST_CONTAMINATION. Subject-role disjointness: **PASS**, with no intersections and no missing roles.

Fit/visibility rules: preprocessing has no learned fit requirement; future model fitting may use train; low-calibration subsets may use calibration only; validation is reserved for governed selection/checking as defined by downstream annexes; test is never visible to training, calibration, threshold selection or decision-time tuning.

6.8 Low-calibration budget contract

Registered budgets are **1, 2, 4, 8, 16, 32 source events per class**, calibration role only, seed **20260804**, nested deterministic SHA-256-ranked prefixes, exact class balance where feasible, and no test visibility. These are **infrastructure identities for future downstream evaluation**, not evidence that calibration experiments were executed in P01.

6.9 Preprocessing contract

Canonical PreprocessingRecord: IHARQ-PREPROCESSINGRECORD-20260806-a11b59eeb3861a08; semantic hash: a11b59eeb3861a0801cd0540702dca2f0e96b9a55fe8d50d078ea2b3a48acb8c. The accepted sequence is:

- 8. validate/normalize signal units to volts;
- 9. capture original source event indices/onsets before channel dropping/resampling;
- 10. select EEG signal channels and preserve deterministic ordering;
- 11. demean the continuous run;
- 12. apply average reference;
- 13. jointly resample signal and event samples to **160 Hz** with `mne.io.Raw.resample`, polyphase method, Kaiser beta **5.0**, reflect padding, `n_jobs=1`, and use the returned jointly resampled event array (independent float-time rounding prohibited);
- 14. apply 8–32 Hz fourth-order Butterworth SOS zero-phase filtering with `scipy.signal.sosfiltfilt`, odd padding, exact padlen **27**, minimum input samples **29**, effective forward/backward order **8**;
- 15. cast signal output to **float32**.

Fit scope is NOT_REQUIRED_FOR_OFFICIAL_LAYER1_PROFILE; the generated fit-state artifact is a deterministic infrastructure/provenance surface, not held-out learned normalization. Any run violating units, event lineage, required filter length, shape or nonfinite rules fails/invalidates according to the quality/validation contracts.

6.10 Official core-window contract

The official core profile remains P01-L1-WINDOW-OFFICIAL-R2: MI cue onset +0.5 s to +3.5 s, 3.0 s duration, 160 Hz, start offset 80 samples, duration/stride 480 samples, exactly one official window per included source event, out-of-bounds **REJECT**, clipping **PROHIBITED**, parent-event overlap-group identity retained. Actual closure: **12,910 core windows / 12,910 accepted parent events, 0 invalid windows**. Signal dtype is float32.

The core profile was **not changed** by the A4 repairs.

6.11 Core numerical Dataset persistence

P01 uses dual persistence: compact governed records/manifests/pointers remain in the project bundle, while large lossless numerical tensors remain in a private Kaggle Dataset.

- artifact ID: P01-L1-DERIVED-WINDOWS-d03f0a7c869dd95a-20260806222242-68a91473
- provider/handle: Kaggle / csthv999z/iharq-p01-l1-derived-windows-d03f0a7c869d-20260806222242-68a91473
- logical immutable revision: **1**
- actual scientific provider Dataset version: **2**
- provider version 1: historical short-title shell only; provider version 2: verified scientific artifact
- manifest SHA-256: dc21f418e9a1add62adb346f627d5d729735a5f1ecaa1d771f6fa5cfede627e1
- format: lossless HDF5 subject shards; compression gzip-1; dtype float32
- shards: **172**; windows: **12910**
- logical float32 bytes: **1356625920** (1.263 GiB)
- actual uploaded HDF5 bytes: **1166652764** (1.087 GiB)
- local shard state: SHARDS_DELETED_AFTER_VERIFIED_KAGGLE_BLOB_UPLOAD
- creation/adoption: ADOPTED_VERIFIED_EXISTING_DATASET; scientific artifact recomputed = False; core HDF5 reuploaded = False.

This Protocol explicitly preserves the distinction between **logical immutable revision 1** and **Kaggle provider Dataset version 2** to avoid falsely representing the shell version as the scientific artifact.

6.12 Quality and validation contract

Quality profile: P01-L1-QUALITY-OFFICIAL-R2; policy: **ANNOTATE_NOT_REPAIR**. Hard-invalid rules include nonfinite data, incorrect tensor rank/shape, insufficient duration and missing source-event lineage. Soft diagnostics include flat/repeated signals, large voltage excursions and provider quality flags. Silent interpolation/repair is prohibited.

Observed quality closure: **489 quality summaries**, quality available for **489, 20 soft/provider flags** represented by **20 ArtifactFlagRecords**, and **0 hard-invalid summaries**. Core invalid-window count is **0**. These are data-quality/protocol results, not decoder-performance claims.

6.13 A0-A13 readiness matrix

Ablation	Official identity	Owner	Activated P01	Scientifically executed P01	P01 output	Downstream	Readiness
A0	Raw Decoder / Accept-All Raw Decoder Reference	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A1	Calibrated Decoder / Calibration Visibility	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A2	Simple Registered Threshold / Confidence-Threshold Baseline	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A3	Uncertainty and Selective Prediction	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY

Ablation	Official identity	Owner	Activated P01	Scientifically executed P01	P01 output	Downstream	Readiness
A4	Longer-Window, Multi-Window Voting/Averaging and Ordinary Ensemble Controls	DOWNSTREAM_AUTHORITY	False	False	A4 R2 external data substrate + registered views; Protocol synchronized here; downstream experiment still not executed in P01	P02-P15	READY_WITH_PROTOCOL_SYNC_REQUIRED
A5	IHARQ-lite / Rule-Based Evidence Verification	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A6	IHARQ + Evidence-Quality Estimator	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A7	IHARQ + RegimeRisk Temporal Trust	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A8	Learning-to-defer / Deferral Comparison	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A9	Supervised Adaptive-IHARQ / Adaptive Readiness Policy	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A10	Contextual Bandit / Simulation-Bounded Immediate-Reward Adaptation	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A11	Reinforcement-Learning Policy / Simulation-Bounded Sequential Policy Learning	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A12	StressForge Stress Tests / Controlled Stress Robustness	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY
A13	Layer 9 Simulation-Only Embodiment Demo	DOWNSTREAM_AUTHORITY	False	False	Layer-1 matched-key/data foundation; no P01 effectiveness result	P02-P15	FOUNDATION_READY

All A0-A13 rows are **foundation/readiness** dispositions. No downstream ablation is reclassified as experimentally executed merely because L1 prepared matched keys/data.

6.14 A14 prohibition

A14 = PROHIBITED / ABSENT. Final machine evidence: selector present = false; run present = false; result present = false; claim present = false; audit status = PASS. Local identities such as A12.x remain subordinate/local and are not renamed A14.

6.15 A4 R2 Protocol synchronization — COMPLETE FOR FUTURE USE

The final P01 execution prepared an additive Layer-1 data substrate for A4 and explicitly marked it DATA_READY_PROTOCOL_SYNC_REQUIRED. This annex completes that **identity synchronization**, but does **not** convert A4 into a P01 effectiveness experiment.

Final future A4 profiles:

- family: P01-L1-A4-WINDOW-FAMILY-FREEZE-R2;
- longer matched profile: A4_LONG_MATCHED_3P5S_R2, cue +0.0 to +3.5 s, **560 samples at 160 Hz**;
- multi profile: A4_MULTI_3X2S_UNIFORM_0P75S_R2;
- M1 A4_MULTI_3X2S_M1_R2: slice 0:320, +0.00..+2.00 s;
- M2 A4_MULTI_3X2S_M2_R2: slice 120:440, +0.75..+2.75 s;
- M3 A4_MULTI_3X2S_M3_R2: slice 240:560, +1.50..+3.50 s.
- one physical 560-sample tensor/event; the three 2 s members are registered immutable views; overlapping bytes are not duplicated;

- exact matched parent denominator: **12,910/12,910**; no clipping, padding or silent event drop;
- longer records: **12,910**; multi-member records: **38,730**; total A4 records: **51,640**; groups **12,910**; shards **172**;
- external handle: csthv999z/iharq-p01-l1-a4-d03f0a7c-9a7c6108 provider version **1**, logical family revision **2**, manifest SHA-256 29124d59907610b1545e0a2b5d1811a4d4a2bf1df64cad49aa880f4721c47305; remote manifest verification **PASS**.

A4 historical deviation and no-post-hoc boundary

The earlier proposed A4 R1 profile used cue +0.0..+4.0 s (640 samples). A universal denominator audit established that valid parent event `PhysioNetMI:104:0:8:event:24` has only 560 available samples after cue; 4.0 s would require **80 nonexistent samples**. Padding, clipping, fabricating or dropping the event would violate the matched-denominator contract. R2 therefore changed the **alternative A4 profile only** to +0.0..+3.5 s and uniformly registered 2 s subviews, retaining all 12,910 parents and the same +3.5 s endpoint as core.

This R1 → R2 change occurred after feasibility evidence and is classified as `SCIENTIFIC_CONTRACT_CHANGE` for the **future A4 alternative profile**, not as a change to the official P01 core. It is **not described as preregistered** and P01 may not use it as retrospective confirmatory A4 effectiveness evidence. Future claim-bearing A4 execution must inherit the exact R2 identities frozen here and register the relevant downstream analysis/estimand before execution. `executed_in_p01 = false`.

6.16 Actual Kaggle environment

Actual accepted environment, not the original Build Book image intent:

- Python: **3.12.13**; platform `Linux-6.12.90+-x86_64-with-glibc2.35`; CPU count **4**; RAM **33659379712 bytes**; observed total disk **20957446144 bytes**; Stage-01 free disk **20336979968 bytes**;
- package versions exactly matched required execution pins: `moabb=1.5.0`, `mne=1.12.1`, `numpy=2.2.6`, `scipy=1.15.3`, `pandas=2.3.1`, `scikit-learn=1.7.1`, `h5py=3.14.0`, `pooch=1.8.2`, `pyyaml=6.0.2`, `pydantic=2.11.7`, `jsonschema=4.25.0`, `nbformat=5.10.4`; pin mismatches = **0**; required import failures = **0**;
- deterministic environment: `PYTHONHASHSEED=20260804`, `OMP/MKL/OPENBLAS threads = 1`;
- runtime environment amendment: `P01-L1-KAGGLE-ENV-FREEZE-R5`, compatibility/connection only; scientific values changed = none.

The earlier Build Book targeted Python `>=3.11,<3.12`/Kaggle image intent. The accepted run used 3.12.13 through an explicit compatibility successor. This is recorded rather than pretending the pre-run environment ran verbatim.

6.17 Adaptive-disk runtime amendment

Resource amendment: `P01-L1-KAGGLE-ADAPTIVE-DISK-R1`. Historical policy: minimum 60 GiB / recommended 90 GiB. Accepted runtime used an adaptive startup floor of **6.0 GiB**, calculated requirement approximately **3908544705 bytes**, observed free **18.94 GiB**, soft-warning 4.0 GiB, hard-emergency 1.5 GiB, export reserve 1.5 GiB. Read-only Kaggle inputs could not be deleted; automatic source removal was prohibited; only lawful reverified writable caches could be evicted. Scientific datasets/configs/split/labels/preprocessing/core windows were unchanged. Classification: `RESOURCE_POLICY_CHANGE`, non-scientific.

6.18 Executed notebook and stage matrix

The accepted stage identity is exactly **00-26 once each**. The final canonical Stage-26 state is **PASS** only after R54 external packaging repair; historical blocked/failure outputs are preserved separately.

Stage	Purpose	Final status	Key outputs	Blockers	Disposition
00	Corrected bootstrap and persistent isolated worker	PASS	authority_manifest.json	0	FINAL_ACCEPTED
01	Environment	PASS	environment_manifest.json	0	FINAL_ACCEPTED
02	Project and input intake	PASS	No direct file / state transition	0	FINAL_ACCEPTED
03	Authority and configuration	PASS	config_snapshot/official_run_freeze_manifest.json	0	FINAL_ACCEPTED
04	Phase 0 regression	PASS	reports/phase_01/tests/phase0_and_runtime_regression.json	0	FINAL_ACCEPTED
05	Source resolution	PASS	No direct file / state transition	0	FINAL_ACCEPTED

Stage	Purpose	Final status	Key outputs	Blockers	Disposition
06	Dataset registry	PASS	No direct file / state transition	0	FINAL_ACCEPTED
07	Pass 1: verified source acquisition and bounded loading	PASS	No direct file / state transition	0	FINAL_ACCEPTED
08	Metadata normalization	PASS	No direct file / state transition	0	FINAL_ACCEPTED
09	Label mapping	PASS	No direct file / state transition	0	FINAL_ACCEPTED
10	Preprocessing compilation	PASS	No direct file / state transition	0	FINAL_ACCEPTED
11	Split construction and frozen fit population	PASS	No direct file / state transition	0	FINAL_ACCEPTED
12	Low-calibration budgets	PASS	No direct file / state transition	0	FINAL_ACCEPTED
13	Pass 2A: bounded preprocessing fit	PASS	No direct file / state transition	0	FINAL_ACCEPTED
14	Adopt verified core and materialize matched A4 R2	PASS	external_artifact_pointers/derived_windows_dataset.json, reports/phase_01/storage/a4_derived_output_storage_actual_precommit.json	0	FINAL_ACCEPTED
15	Validate and commit the separate A4 R2 Dataset	PASS	external_artifact_pointers/derived_windows_dataset.json, external_artifact_pointers/a4_window_family_dataset.json	0	FINAL_ACCEPTED
16	Record validation	PASS	No direct file / state transition	0	FINAL_ACCEPTED
17	Leakage audit	PASS	No direct file / state transition	0	FINAL_ACCEPTED
18	A0–A13 readiness	PASS	No direct file / state transition	0	FINAL_ACCEPTED
19	Cards	PASS	No direct file / state transition	0	FINAL_ACCEPTED
20	Manifests	PASS	No direct file / state transition	0	FINAL_ACCEPTED
21	Negative register	PASS	negative_and_failed_results/run_failures_and_blockers.json	0	FINAL_ACCEPTED
22	P02 and later compatibility	PASS	No direct file / state transition	0	FINAL_ACCEPTED
23	Evidence sufficiency	PASS	reports/phase_01/preliminary_gate_evaluation.json	0	FINAL_ACCEPTED
24	Repair metadata	PASS	reports/phase_01/repair_reentry.json	0	FINAL_ACCEPTED
25	Final export preparation	PASS	No direct file / state transition	0	FINAL_ACCEPTED
26	Terminal decision and bundle export	PASS	/kaggle/working/iharq_p01_l1/IHARQ_P01_L1_Phase_Execution_Bundle_d03f0a7c869d.zip, /kaggle/working/iharq_p01_l1/IHARQ_P01_L1_Phase_Execution_Bundle_d03f0a7c869d.zip.sha256, /kaggle/working/iharq_p01_l1/IHARQ_P01_L1_GitHub_Ready_Repository_d03f0a7c869d_20260807143013-663eab13.zip	0	FINAL_ACCEPTED

The execution notebook source also contains material failed/recovery cells outside the 00-26 stage identity. They are historical evidence, not extra scientific stages.

6.19 Failure, repair, rerun and supersession ledger

Repair	Affected stage	Class	Defect	Owner	Science changed	Data changed	Core changed	Rerun scope	Final resolution
R45 / SAME-SESSION CORE-ADOPTION NORMALIZATION	14	CANONICALIZATION_FIX	Existing verified core adoption blocked because date-bearing upstream record IDs changed dependent semantic hashes despite otherwise equivalent scientific records.	Implementation/Registry integration	False	False	False	fresh worker/bundle plus affected adoption path; preserved persistent checkpoints	Dependency-order full-record equivalence with exact upstream-ID mapping plus exact remote manifest hash; core Dataset not mutated/reuploaded.
R46	runtime overlay	IMPLEMENTATION_BUG_FIX	Missing datetime/timzone runtime imports.	Implementation Build Book/code	False	False	False	affected runtime path only	Required imports added.
R47	A4 profile canonicalization	CANONICALIZATION_FIX	Float literals in hash-bearing A4 profile could cause governed representation drift.	Registry/implementation canonicalization	False	False	False	profile serialization/validation only	Governed decimal strings used for hash-bearing seconds fields.
R48	14-15 A4 materialization/persistence	IMPLEMENTATION_BUG_FIX + PERSISTENCE_FIX	A4 child interface/set handling and storage identity/read verification gaps; resumability and exact remote manifest closure hardened.	Implementation/Nuts-and-Bolts	False	False	False	A4 child/interface and persistence path	Child interface corrected; reader/storage identity closure, synthetic E2E, resumable subject checkpoints and remote manifest verification added.
R49 / P01-L1-A4-WINDOW-FAMILY-FREEZE-R2	14-15 A4 profile	SCIENTIFIC_CONTRACT_CHANGE (A4 ALTERNATIVE ONLY)	Original proposed A4 +0.0..+4.0 s window is impossible for one valid released parent event (PhysioNetMI:104:0:8: event:24) without 80 nonexistent samples.	Protocol/Method/Nuts-and-Bolts/Implementation for A4 alternative profile	True	True	False	A4 materialization path with exact 12,910 matched parent events	R2 freezes +0.0..+3.5 s 560-sample tensor and three registered 2 s slices; no drop/pad/clip; confirmatory use prohibited until Protocol sync and downstream execution.
R50	07	INTEGRATION_FIX	Stale local DISPLAY_REVISION guard expected R42 while live runtime was R49; Stage 07 had not yet been submitted.	Notebook integration	False	False	False	submit Stage 07 once using live worker; no 00-06 replay	Revision guard corrected; Stage 07 PASS.
R51-R53 / P01-L1-R53-STAGE18-WORKER-ENV-IMPORT-PROBE-R1	18	INTEGRATION_FIX	A4 readiness wrapper imported write_json from nonexistent module; two recovery-cell validation mistakes then occurred (malformed shim newline and notebook-kernel import probe outside worker PYTHONPATH).	Notebook/runtime integration	False	False	False	Stage 18 only after worker-environment import probe	Compatibility shim re-exported authoritative manifests.write_json and was probed under exact worker_env; Stage 18 PASS and readiness artifact verified.
P01-L1-R54-FINAL-EXPORT-SECRET-REDACTION-R1	26 external release closure	SECURITY/PACKAGING_FIX	Live Kaggle credential was serialized into environment_amendment and the secret scanner correctly blocked the contaminated export.	Packaging/security	False	False	False	redaction/repack/final integrity only; original worker Stage 26 not duplicated	Secret-like environment values redacted; final bundle/repository rebuilt; exact-token scan, manifests and checksums PASS; contaminated failed release is non-authoritative.

Minor tracebacks subsumed by a governed episode are not promoted to independent scientific revisions; the material lineage remains reconstructible from notebook outputs and preserved runtime repair files.

6.20 Scientific versus non-scientific amendment classification

Repair	Classification	Estimand/profile science changed	Data changed	Core changed	Denominator changed	Rerun/repair needed	Resolution
R45 / SAME-SESSION CORE-ADOPTION NORMALIZATION	CANONICALIZATION_FIX	False	False	False	False	YES	Dependency-order full-record equivalence with exact upstream-ID mapping plus exact remote manifest hash; core Dataset not mutated/reuploaded.
R46	IMPLEMENTATION_BUG_FIX	False	False	False	False	YES	Required imports added.
R47	CANONICALIZATION_FIX	False	False	False	False	YES	Governed decimal strings used for hash-bearing seconds fields.
R48	IMPLEMENTATION_BUG_FIX + PERSISTENCE_FIX	False	False	False	False	YES	Child interface corrected; reader/storage identity closure, synthetic E2E, resumable subject checkpoints and remote manifest verification added.
R49 / P01-L1-A4-WINDOW-FAMILY-FREEZE-R2	SCIENTIFIC_CONTRACT_CHANGE (A4 ALTERNATIVE ONLY)	True	True	False	False	YES	R2 freezes +0.0..+3.5 s 560-sample tensor and three registered 2 s slices; no drop/pad/clip; confirmatory use prohibited until Protocol sync and downstream execution.
R50	INTEGRATION_FIX	False	False	False	False	YES	Revision guard corrected; Stage 07 PASS.
R51-R53 / P01-L1-R53-STAGE18-WORKER-ENV-IMPORT-PROBE-R1	INTEGRATION_FIX	False	False	False	False	YES	Compatibility shim re-exported authoritative manifests.write_json and was probed under exact worker_env; Stage 18 PASS and readiness artifact verified.
P01-L1-R54-FINAL-EXPORT-SECRET-REDACTION-R1	SECURITY/PACKAGING_FIX	False	False	False	False	YES	Secret-like environment values redacted; final bundle/repository rebuilt; exact-token scan, manifests and checksums PASS; contaminated failed release is non-authoritative.

Only the A4 R1 → R2 alternative profile is a scientific-contract/profile change; it neither changes the official core nor supplies an effectiveness result. All other listed repairs are execution/integration/canonization/persistence/security fixes with no scientific estimand, split, label, preprocessing, core-window or metric change.

6.21 Security/release repair history

During original Stage 26 packaging, a live Kaggle credential was inadvertently serialized into an environment artifact. The secret scanner correctly blocked the contaminated export. R54 (P01-L1-R54-FINAL-EXPORT-SECRET-REDACTION-R1) redacted secret-like environment values, rebuilt final packaging/integrity surfaces, and produced PASS for manifest verification, checksum verification and exact-token scanning. The secret value is **not reproduced anywhere in this Protocol package**. Science, data, split, labels, preprocessing and results were unchanged. The contaminated failed release is non-authoritative; only the R54-corrected package is accepted.

6.22 Gate and validation closure

Gate	Purpose	Repair owner	Status	Failure codes	Evidence	Disposition
P01-G01	authority_phase0_intake	GOVERNANCE_AND_PHASE0	PASS	0	manifests/phase_01/test_manifest.json; authority_manifest.json	NONBLOCKING/PASS
P01-G02	source_provenance_license	METHOD_SELECTION_AND_OWNER	PASS	0	reports/phase_01/sources/source_version_license_report.json; inputs/source_inventory.json	NONBLOCKING/PASS

Gate	Purpose	Repair owner	Status	Failure codes	Evidence	Disposition
P01-G03	schema_canonical_object	REGISTRY	PASS	0	reports/phase_01/validation/; records/	NONBLOCKING/PASS
P01-G04	metadata_completeness	L1_METADATA	PASS	0	reports/phase_01/metadata/metadata_completeness.json	NONBLOCKING/PASS
P01-G05	label_mapping	L1_LABELS	PASS	0	reports/phase_01/labels/label_map_validation.json; records/labels/	NONBLOCKING/PASS
P01-G06	preprocessing_fit_scope	PROTOCOL_AND_L1_PREPROCESSING	PASS	0	reports/phase_01/preprocessing/fit_scope.json; records/preprocessing/	NONBLOCKING/PASS
P01-G07	split_disjointness	PROTOCOL_AND_L1_SPLITS	PASS	0	reports/phase_01/splits/disjointness.json; records/splits/	NONBLOCKING/PASS
P01-G08	leakage_chronology	PROTOCOL_AND_L1_LEAKAGE	PASS	0	reports/phase_01/leakage/leakage_contamination.json	NONBLOCKING/PASS
P01-G09	low_calibration_budgets	PROTOCOL_AND_L1_BUDGETS	PASS	0	reports/phase_01/splits/low_calibration_budgets.csv	NONBLOCKING/PASS
P01-G10	window_identity	L1_WINDOWS	PASS	0	reports/phase_01/windows/window_timing_overlap.json; records/windows/	NONBLOCKING/PASS
P01-G11	quality_coverage	L1_QUALITY	PASS	0	reports/phase_01/quality/quality_coverage.json; records/quality/	NONBLOCKING/PASS
P01-G12	matched_keys_ablation_readiness	PROTOCOL_AND_L1_READINESS	PASS	0	manifests/phase_01/layer1_ablation_readiness_l1_v1.json; reports/phase_01/readiness/matched_key_completeness.csv	NONBLOCKING/PASS
P01-G13	cards_limitations	L0_BOUNDARY_AND_L1_CARDS	PASS	0	docs/cards/datasets/; docs/cards/protocols/	NONBLOCKING/PASS
P01-G14	manifest_path_hash_closure	BUILD_BOOK_AND_L1_MANIFESTS	PASS	0	manifests/phase_01/execution_bundle_manifest.json; checksums.sha256	NONBLOCKING/PASS
P01-G15	phase2_compatibility	BUILD_BOOK_AND_P02_CONSUMER	PASS	0	phase2_handoff/phase_01_to_phase_02.yaml	NONBLOCKING/PASS
P01-G16	complete_artifact_closure	EXECUTION_PLAN_AND_L1_BUNDLE	PASS	0	manifests/phase_01/layer1_manifest.json; phase_execution_handoff.yaml	NONBLOCKING/PASS

Additional closure: test suite **50 passed / 0 failed**; schema/canonical-record validation PASS; subject split disjointness PASS; leakage checks PASS; window timing/denominator closure PASS; external core and A4 manifest verification PASS; execution-bundle checksums . sha256 independently reverified for **13,164/13,164 entries with 0 missing and 0 mismatches**; unresolved blocker count **0**.

6.23 Negative, failed, invalid and diagnostic evidence

P01 preserves distinct classes: historical failed attempts, superseded implementation defects, valid negative/diagnostic observations, invalid data items, and current blockers. The final accepted core contains **0 invalid windows**; quality contains 20 recorded soft/provider flags and 0 hard-invalid summaries. Material Stage 07/18/26 failures and repair evidence remain preserved. Current unresolved blockers: **0**. Failure history is not deleted or converted into success evidence.

6.24 Evidence status and claim ceiling

P01 can support factual/protocol claims about: source provenance and checksum closure; reproducible intake; exact labels/exclusions; subject-grouped split; preprocessing/window implementation; validated canonical data products; 12,910-window denominator conservation; quality/lineage/integrity; external persistence; A0-A13 foundation readiness; A14 absence; and downstream technical readiness.

P01 **cannot by itself support** decoder superiority, calibration effectiveness, clinical effectiveness, deployment safety, real-world control, downstream A4 effectiveness, policy-learning benefit, temporal-trust benefit, stress robustness or embodiment claims. Mandatory inherited limitations include PUBLIC_EEG_ONLY, NON_CLINICAL, NO_DEPLOYMENT_CLAIM. Layer 0 later controls claim wording/disposition.

6.25 External artifact registry

Artifact	Provider/handle	Provider rev	Logical rev	Format/access	Size/count	SHA-256 identity	Producer/consumers	Local-copy state/retrieval
P01-L1-DERIVED-WINDOWS-d03f0a7c869dd95a-20260806222242-68a91473	csthv999z/iharq-p01-l1-derived-windows-d03f0a7c869d-20260806222242-68a91473	2	1	LOSSLESS_HDF5_SUBJECT_SHARDS / PRIVATE	172 shards; 12910 windows; 1166652764 bytes	dc21f418e9a1add62adb346f627d5d729735a5f1ecaa1d771f6fa5cfede627e1	P01 / P02-P15	SHARDS_DELETED_AFTER_VERIFIED_KAGGLE_BLOB_UPLOAD; Attach the private Kaggle Dataset at immutable version 1; load IHARQ_P01_L1_WINDOW_TO_SHARD_INDEX.jsonl (or an ordinal-prefixed suffix match); resolve the shard filename; read the declared HDF5 group and row.
P01-L1-A4-DERIVED-WINDOWS-4cd080393345e8aa-20260807143013-663eab13	csthv999z/iharq-p01-l1-a4-d03f0a7c-9a7c6108	1	2	LOSSLESS_HDF5_MATCHED_3P5S_SUBJECT_SHARDS_WITH_REGISTERED_VIRTUAL_MULTIWINDOW_VIEWS / PRIVATE	172 shards; 12910 materialized events; 51640 records; 1357362334 bytes	29124d59907610b1545e0a2b5d1811a4d4a2bf1df64cad49aa880f4721c47305	P01 / P02 A4 and later governed consumers	A4_SHARDS_DELETED_AFTER_VERIFIED_KAGGLE_BLOB_UPLOAD; attach exact provider version and verify manifest SHA before use

A handle without exact revision and checksum is insufficient for governed consumption.

6.26 Phase 2 handoff constraints

P02 may consume the verified core pointer, exact split/label/preprocessing/window identities, config ID and limitation tags. A4 may be consumed only under the exact R2 family/profile identities synchronized here and with a downstream Protocol/analysis cell that distinguishes it from core. Required matching keys include dataset, subject, session/run, parent event/window/group, split role, preprocessing, label map, config and profile identity as applicable. P02 may not silently rewindow, relabel, change split membership, expose test data during fitting/selection, or reinterpret Layer-1 contracts. Any such change requires a governed amendment and descendant invalidation/rerun analysis.

6.27 Limitations

Supported limitations only:

- project-wide: public benchmark/research evidence is not clinical or deployment validation; downstream claims remain bounded by owner layers and Layer 0;
- P01: selected public EEG portfolio and binary MI branch only; excluded source events remain out of the supervised denominator; source-specific licensing/redistribution restrictions apply;
- environment: accepted execution is tied to the recorded Kaggle/Python/package/runtime-amendment identities;
- external access: core and A4 numerical HDF5 Datasets are private Kaggle artifacts and require access plus exact revision/hash verification;
- evidence ceiling: no decoder/calibration/policy/robustness/embodiment effectiveness result was generated by P01;
- A4: R2 profile emerged from feasibility repair, is synchronized prospectively for future use, and is not retrospective confirmatory evidence;
- security: the superseded contaminated Stage-26 release is non-authoritative; credential rotation was recommended.

6.28 Freeze decision

P01_PROTOCOL_V1_ANNEX_R1_FROZEN_WITH_EXPLICIT_EXECUTION_AMENDMENTS_AND_A4_R2_FUTURE_CONFIRMATORY_SYNC

- current execution status: **ACCEPTED**;
- Protocol annex status: **FROZEN**;
- unresolved blockers: **0**;
- nonblocking limitations: PUBLIC_EEG_ONLY, NON_CLINICAL, NO_DEPLOYMENT_CLAIM, private external Dataset access, recorded runtime amendments, A4 retrospective-history limitation;
- A4 R2 identity synchronization: **COMPLETE** for future downstream contract use; downstream A4 effectiveness execution: **NOT PERFORMED IN P01**;
- additional P01 Kaggle execution required: **NO**;
- next lawful output: Phase 1 Evidence, Results, and Interpretation Report using this annex plus analysis_contract.yaml and the accepted execution bundle;
- P02 technical handoff: **READY**, subject to completion of the documentary closure chain required by Governance V6.1 before formal phase transition.

Integration provenance. This Part preserves the scientifically validated standalone source annex IHARQ-PROTOCOL-V1-P02-ANNEX-R2 inside the current cumulative authority. Documentary placement changes only; Stage 11/12 amendment history, A0/A4 evidence, Stage 18S status, failures, exact identities, limitations and external-artifact contracts are not semantically rewritten.

PART IV — PHASE 02 / LAYER 2 PROTOCOL RECORD

7.1 Document control

Field	Frozen / actual value
annex_id	IHARQ-PROTOCOL-V1-P02-ANNEX-R2
supersedes	IHARQ-PROTOCOL-V1-P02-ANNEX-R1 for documentary/machine-readable corrections only; no scientific result changed
parent cumulative Protocol at standalone-annex freeze	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P01-R1 (historical parent at standalone freeze)
current integration behavior	Integrated into IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1; no competing master
phase_id	P02
official phase role	Baseline decoders / decoder measurement spine
primary layer	L2 — Decoder and Baseline Measurement Spine
version	1.0-P02-R2
status	FROZEN_WITH_EXPLICIT_SCIENTIFIC_RUNTIME_AMENDMENTS_RESOURCE_CONSTRAINED_A4_POST_HOC_SENSITIVITY_DUAL_HF_HANDOFF_AND_AUDIT_CLOSURE
accepted run_id	P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d
run configuration SHA-256	9e181d2e935d2e9674ca6e05572f49520ad0306a3761362b770f8bee8c78ce13
scientific freeze	P02-PLANNED-SCIENTIFIC-EXECUTION-FREEZE-R5
scientific freeze SHA-256	7449216c988eec14191ba85300d720547f06fbc6ac7020e9e142bf12a4b0a598
notebook logical identity	IHARQ-P02-COMPLETE-EXECUTION-AND-ANALYSIS-R4
runtime-revision source_sha256 fingerprint	a54bfa816acc6763c1d1042b8c65cf54ae04ec774c57ec50de86014a6730c5fa
runtime-revision config SHA-256	d82334535e397c681d6172694da96d3fff0143ffb1b63a18876ee1f7778f60a1
stage-plan SHA-256	043eb30e008af910d37c563e8be8c719ef881cf009aec3e34687dc97bdc34aad
owner-supplied final post-restart continuation notebook SHA-256	498b837ffaabc945160cd2b8482528c17ed85e978501119a19c88073d72b543a
packaged implementation notebook SHA-256	4c36e340721156de2f58e03b1568f7b1072db3d31145aaba566b283305e7fecf
G23 evidence sufficiency	PASS; missing=[]
G24 bundle/finalization gate	PASS; blockers []
unresolved blockers	0
execution status	ACCEPTED_FOR_PROTOCOL_AND_PHASE_ANALYSIS
evidence status	SUFFICIENT_FOR_PROTOCOL_AND_PHASE_REPORT_WITH_EXPLICIT_RESOURCE_CONSTRAINED_A4_LIMITS
canonical phase source	Frozen Phase-02 source annex preserved under

Field	Frozen / actual value
	protocol_v1_0/phases/phase_02_annex.md
current cumulative presentation derivatives	Current cumulative DOCX/PDF derive from the cumulative Protocol authority
phase-source machine-readable companions	P02 R2 source companions integrated into cumulative machine-readable contracts

7.1.1 Hash and notebook-identity nomenclature guard

The accepted P02 state exposes several distinct identities. They are **not interchangeable**:

run_config_hash = 9e181d2e935d2e9674ca6e05572f49520ad0306a3761362b770f8bee8c78ce13 identifies the accepted scientific/runtime run configuration exposed by Stage 06 and the Protocol handoff;

scientific_freeze_sha256 = 7449216c988eec14191ba85300d720547f06fbc6ac7020e9e142bf12a4b0a598 identifies the frozen pre-execution scientific plan state;

runtime_revision_config_sha256 = d82334535e397c681d6172694da96d3fff0143ffb1b63a18876ee1f7778f60a1 is the serialized runtime/config revision fingerprint carried by the stage ledger;

runtime_revision_source_sha256 = a54bfa816acc6763c1d1042b8c65cf54ae04ec774c57ec50de86014a6730c5fa is the stage-ledger **runtime-revision source fingerprint field**. It is retained exactly as recorded, but R2 no longer labels it as the physical SHA-256 of the owner-supplied notebook file;

stage_plan_sha256 = 043eb30e008af910d37c563e8be8c719ef881cf009aec3e34687dc97bdc34aad identifies the executed stage-plan revision;

the owner-supplied notebook110a2128e2 (18).ipynb is the final **post-restart continuation notebook**, SHA-256 498b837ffaabc945160cd2b8482528c17ed85e978501119a19c88073d72b543a; its executed cells intentionally begin from the survived canonical Stage-18 state and continue through Stage18S/18U/19–24 rather than replaying Stages 00–18;

the packaged implementation notebook IHARQ_Phase_02_Complete_Execution_and_Analysis_R4.ipynb has physical SHA-256 4c36e340721156de2f58e03b1568f7b1072db3d31145aaba566b283305e7fecf and represents the integrated implementation notebook source distributed in the cumulative project state.

The runtime stage ledger, accepted stage artifacts, and post-restart continuation together reconstruct the actual execution history. No single physical notebook SHA is substituted for a different fingerprint class.

7.2 Phase declaration, Layer 2 role, record surface, and claim boundary

Phase 02 converts the frozen P01 public-data, split, preprocessing, label, and window substrate into the project's first governed **decoder and baseline measurement spine**. Layer 2 is a measurement layer: it trains/qualifies plural decoder families, records native prediction/score evidence, measures raw accept-all performance, constructs low-label and participant/session evidence, executes ordinary A4 longer/multi-window/ensemble controls, preserves model/run provenance and failures, and emits a fail-closed downstream readiness handoff.

The Architecture fixes eight official Layer-2 modules: **Baseline trainer; Prediction logger; Low-calibration curve builder; Subject difficulty profiler; Model-family registry; Ensemble comparison builder; Compact SSL adapter; Downstream readiness validator**. P02 executed or dispositioned each of these responsibilities through its stage/record/handoff surfaces.

Layer 2 does **not** own calibration, uncertainty estimation, thresholding, abstention, selective prediction, readiness policy, clinical interpretation, deployment safety, temporal trust, stress robustness, closed-loop simulation, or embodiment claims. Those remain downstream responsibilities. P02 supplies the prediction substrate and baseline evidence against which later layers operate; it does not convert decoder performance into a safety or deployment decision.

The Registry-backed P02 record surface includes PredictionRecord, ModelRegistryRecord, BaselineMetricRecord, LowCalibrationCurveRecord, SubjectProfileRecord, EnsembleControlRecord, FailureCaseIndex, Layer2ReadinessReport, NegativeResultNote, and DiagnosticOnlyFlag, with owner-routed LeakageWarningRecord and MatchedComparisonReport where applicable. The accepted runtime contains record partitions/instances for the relevant P02 families and preserves prediction partitions separately through the prediction manifest surface.

Status axes remain separate: method role, run terminal state, record validation state, lifecycle state, consumer-readiness state, diagnostic/negative-evidence classification, and claim limitation are not collapsed into one generic “success” label.

7.3 Authority intake and source-utilization reconciliation

R2 was produced only after a source-complete audit of the supplied project state. Governance, all seven authorities, the P00/P01 project state and cumulative documentary products, the P02 Build Book/implementation package, the owner-supplied final continuation notebook, the accepted P02 cumulative execution state, and the R1 Protocol package were independently inventoried and reconciled. Archive duplicates/cumulative copies were dispositioned as duplicate evidence rather than counted as independent scientific results.

Source authority / evidence surface	Protocol use
Governance V6.1	workflow, Protocol-after-execution role, no-post-hoc rewriting, evidence sufficiency/repair loop, external-pointer requirements
Architecture	Layer 2 purpose, eight-module scope, interfaces and downstream boundaries
Canonical Registry R44	record/status/identity/interface vocabulary and lifecycle rules
Execution & Evidence Plan	P02 required outputs/evidence/gates/handoffs and closure criteria
Protocol v0.1 R42	exact A0–A13 identities, A14 prohibition, fairness/leakage/matching/statistical discipline
Phase Execution Playbook R41	operational phase ordering, validation and closure behavior
Method Selection R2	selected datasets/model/method/platform families and rationale
Nuts-and-Bolts R2	accepted Layer-2 internals, matching, metrics, fail-closed behavior and interfaces
P00/P01 state + cumulative Protocol	inherited scientific identities and cross-phase documentary precedent
P02 Build Book R4	pre-execution P02 implementation/run intent and the 89-row responsibility traceability matrix
P02 runtime/notebook/evidence	actual execution, amendments, successors, terminal states, gates, analysis inputs and release history
R1 Protocol package	documentary target audited line-by-line and superseded by R2 where corrections were required

7.3.1 Supplied top-level source snapshot identities

Source	File / package	SHA-256
Governance V6.1	00_IHARQ_Document_Stack_Governance_and_Creation_Guide_V6_1_Single_Track_Full_Depth_Consolidated_Notebook(8).md	c811373c19a7c2c3f6d72cf2aed984e02ffcb07bb448cfc3bdbdf26a35a4f1d9
Seven-authority package	IHARQ_Clean_Final_Seven_Documents_Layers0_to_10_R1_20260801_COMPLETE(20260814-122103).zip	beb00f47e4a790242d62405dcca799647d849c8dc2ff043c5196cee372607128
Phase 00 final whole-stack package	IHARQ_Phase_0_Final_Whole_Stack_Advanced_Internal_Repository_COMPLETE_R1(5).zip	2c554eb57602bb8286338a682ff4172a6768480e126edbee9b35077603277bf7
P01 final download package	IHARQ_FINAL_DOWNLOAD_PACKAGE.zip	9226cb780710b7e7b84b35ea960a98a0c08c4a67288281fb36a7a9f63b9abc72
P01 executed notebook	Phase_01_Notebook(2).ipynb	54b84d9cd29eb57bb22f45b7b7251e76e4947863c832528dba9c51b96189b023
Prior cumulative Protocol through P01	IHARQ_Protocol_v1_0_Final_Through_Phase_01(3).zip	fef708febf6b0cad18b6c3e82767c31b27da4794587432ace799c3995b9075eb
Cumulative Phase Analysis through P01	IHARQ_Cumulative_Phase_Analysis_Through_P01_Final(3).zip	bb0cdaffe9ea813611e731e7ae907f419a2b7a90b198bb44151aa5adc9162227

Source	File / package	SHA-256
Layer-0-embedded Phase Analysis through P01	IHARQ_Cumulative_Phase_Analysis_Embedded_Layer0_Through_P01_Final(3).zip	2c95e96d961d820a3cb0db1661f315498c825eb59f849b0756fba90892e29eda
Cumulative Evidence Map through P01	IHARQ_Cumulative_Evidence_Map_Through_P01_Final(2).zip	f5d435a4429fb0a376cca6958c66924208cea0a4fbbd09c281e6aa3c173221aa
Cumulative Layer 10 through P01	IHARQ_Cumulative_Layer10_Through_P01_Final(2).zip	02377e2e67ea553efe6bb3e43b95c9351d32b3efe5283674807b47549f65f632
Cumulative GitHub-ready through P01	IHARQ_Cumulative_GitHub_Ready_Through_P01_R1(20260814-123111).zip	dc2708ab70e7746499ae09760456825b54efcf587cae3db3410e6f72b0024542
P02 Build Book package R4	IHARQ_P02_L2_Implementation_Build_Book_Package_R4(20260814-123109).zip	2d19167370b70772fe9533e627f85ef83f1e1b68ae33f878955eef1348c9f31e
Owner-supplied final P02 continuation notebook	notebook110a2128e2 (18).ipynb	498b837ffaabc945160cd2b8482528c17ed85e978501119a19c88073d72b543a
P02 cumulative GitHub-ready execution state	IHARQ_Cumulative_GitHub_Ready_Through_P02_Stage24Replay_R1.zip	4e779859fc5ab2c07cb1cdb079f1e9d08cfe41641142d51e093f6c3936bfec81
Protocol package under audit	IHARQ_Protocol_v1_0_Phase_02_Layer_02_Final_R1.zip	7bab07e7f662182ea5681366e0d8a5bf8c245aae0edec31c7036832007350575

The final execution evidence controls over stale planning text when the Build Book and actual runtime differ, within authority ownership. Earlier failed/superseded attempts remain provenance evidence. The complete source-coverage ledger and archive/parse validation are carried in audit/ and validation/.

7.4 Prior-state inheritance from P00/P01

P02 inherits without mutation the project authority stack, canonical hashing/lineage infrastructure, A0–A13 ladder, no-A14 lock, and the P01 scientific data substrate. Specifically, the accepted P01 handoff fixes:

datasets: PhysioNetMI, BNCI2014_001, Lee2019_MI;

binary task: left_hand versus right_hand, with source-specific non-target events excluded rather than relabeled;

split: P01-L1-SPLIT-OFFICIAL-R2, grouped by dataset/subject and preserving train/calibration/validation/test roles;

core window: P01-L1-WINDOW-OFFICIAL-R2, +0.5..+3.5 s, 3.0 s, 480 samples at 160 Hz;

12,910 governed core windows;

low-label profile: P01-L1-LOW-CAL-OFFICIAL-R2, exact inherited per-class subsets at 1/2/4/8/16/32, seed 20260804, with no new membership generation;

A4 family: P01-L1-A4-WINDOW-FAMILY-FREEZE-R2, long profile A4_LONG_MATCHED_3P5S_R2 and three virtual views [0:320, 120:440, 240:560], with 12,910/12,910 matched parents;

inherited limitations: public benchmark/research scope, non-clinical evidence, and no deployment claim.

P02 may train new models and generate new Layer 2 records, but it may not silently relabel, resplit, rewind, alter P01 membership, or use test data for model/hyperparameter/policy selection.

7.5 Pre-run scientific freeze and actual change boundary

The controlling scientific freeze was P02-PLANNED-SCIENTIFIC-EXECUTION-FREEZE-R5 with SHA-256 7449216c988eec14191ba85300d720547f06fbc6ac7020e9e142bf12a4b0a598. It froze A0 and A4 as the two P02 full-execution ablations; prohibited result-dependent additional-ablation unlock; declared A1–A3 and A5–A13 as downstream confirmatory execution; and retained A14 as absent/prohibited.

The initial freeze also fixed the principal metric/statistical framework, seed derivation, low-label membership, candidate caps, and model-family implementation plan. A separate owner-authorized pre-test amendment, P02-PREEEXEC-TRAINING-POLICY-AMENDMENT-R2, resolved the EEGNet segmentation/reconstruction diagnostic challenger and class-weight policy before test evaluation.

The accepted runtime later contains additional **post-observation scientific amendments** at Stages 11 and 12 and one **resource-constrained A4 protocol deviation** at Stage 18. The Protocol does not backdate those changes. Each is classified below with its trigger, changed surface, unchanged fairness boundaries, superseded evidence, and claim consequence.

7.6 Actual inherited source data, task, split, preprocessing, windows, and evaluation roles

P02 reused the validated P01 scientific substrate **without mutation**. The inherited source identities below are therefore part of the P02 Protocol because P02 model evidence is conditional on them.

7.6.1 Dataset identities

Dataset	Release/source	License/source terms recorded by P01	Access route	Observed aggregate SHA-256	Subjects	P01 accepted core windows T/C/V/Test	DatasetRecord / semantic hash
PhysioNetMI	release 1.0.0	ODC-By-1.0	MOABB_1_5_0_OF FICIAL_PHYSIONE T_DOWNLOAD	28cd2062983b6236f9a0e7fdee91fc9d8d5aad8eee3ef561cff5828ae89bf2ba	109	2949 / 979 / 495 / 495	IHARQ-DATASETRECORD-20260806-66309cda68771bef / 66309cda68771bef9bd7a3aebdac819d91201cedca79ab112ea993fb61f558cf
BNCI2014_001	provider file set A01T/A01E... A09T/A09E	CC BY-ND 4.0	MOABB_1_5_0_OF FICIAL_BNCI_DOW NLOAD	04a5390f8f36eaadb c0c480ec9377ce1b99caf0b7ab53bad9fda12347995bc49	9	1440 / 576 / 288 / 288	IHARQ-DATASETRECORD-20260806-42c424800627b6ee / 42c424800627b6ee0be7f47ad602ba49cf4c24a632c1fa69c8c750a4ba77e163
Lee2019_MI	GigaDB DOI 10.5524/100542; MOABB 1.5.0 wrapper; labeled offline/train MI	GPLv3 as recorded by maintained P01 source card; source terms retained	MOABB_1_5_0_OF FICIAL_GIGADB_D OWNLOAD	3a07b2f302da949efd418a0712d5a9427df34dcb8b027ca553fae8e67a849f78	54	3200 / 1100 / 500 / 600	IHARQ-DATASETRECORD-20260806-adb91f25a65e588e / adb91f25a65e588ec e06884a9598cc92bd932e91c4babc3cacb98c82901596f1

Task/labels remain binary left_hand versus right_hand. PhysioNet MI uses run-context T1/T2 for runs 4/8/12 with T0 excluded; BNCI maps 769/770 and excludes feet/tongue/technical/non-target events; Lee uses left/right and excludes non-supervised source runs. Unknown source events fail closed rather than being relabeled as negative.

7.6.2 Split identity and denominators

Split profile: P01-L1-SPLIT-OFFICIAL-R2; canonical SplitRecord IHARQ-SPLITRECORD-20260806-e4e371d332c61e36; semantic hash e4e371d332c61e36699f07cb6bed6d0820e14b22d5685dc353d89c1de144c148. Grouping unit is (dataset_id, subject_id) with atomic subjects, deterministic SHA-256 ranking, 60/20/10/10 train/calibration/validation/test allocation and seed 20260804.

Dataset	Train subj	Cal subj	Val subj	Test subj	Train windows	Cal windows	Val windows	Test windows
PhysioNetMI	65	22	11	11	2949	979	495	495
BNCI2014_001	5	2	1	1	1440	576	288	288
Lee2019_MI	32	11	5	6	3200	1100	500	600
Global	102	35	17	18	7589	2655	1283	1383

P02 fit visibility is exact: FULL_TRAIN fits use train only; LOW_LABEL uses inherited calibration-role budget membership; validation owns governed selection/early stopping; test is final evaluation only. P01 leakage audits for group disjointness, duplicates, overlap groups, fit scope and budget-test contamination passed and P02 did not reopen the split.

7.6.3 Preprocessing and window identity

Canonical PreprocessingRecord: IHARQ-PREPROCESSINGRECORD-20260806-a11b59eeb3861a08; semantic hash a11b59eeb3861a0801cd0540702dca2f0e96b9a55fe8d50d078ea2b3a48acb8c. Inherited sequence: units normalized to volts; original event indices captured; deterministic EEG channel selection; continuous-run demeaning; average reference; joint signal/event resampling to 160 Hz using MNE polyphase/Kaiser beta 5/reflect/n_jobs=1; 8–32 Hz fourth-order Butterworth SOS zero-phase filtering with odd padding/padlen 27/minimum 29 samples; cast to float32.

Core profile: P01-L1-WINDOW-OFFICIAL-R2, cue +0.5...+3.5 s, 3.0 s, 480 samples at 160 Hz, one official window per included source event, out-of-bounds REJECT, clipping PROHIBITED; closure 12,910/12,910 accepted parents, 0 invalid. A4 family: P01-L1-A4-WINDOW-FAMILY-FREEZE-R2; A4_LONG_MATCHED_3P5S_R2 is cue +0.0...+3.5 s (560 samples), with registered 2 s subviews V0 [0:320], V1 [120:440], V2 [240:560], retaining the same 12,910 matched parents.

Low-label profile: P01-L1-LOW-CAL-OFFICIAL-R2; exact nested inherited calibration subsets at 1/2/4/8/16/32 per class, one frozen membership set per budget. No P02 resampling/replacement membership is allowed.

7.7 Model portfolio, branch-level configuration, terminal accounting, and checkpoint semantics

The complete accepted A0 portfolio contains 16 branch IDs. Every planned branch has an explicit terminal state; unsuccessful families are not omitted.

Branch	Family/role	A0 terminal accounting	Executed scope	Actual/frozen configuration summary	A4 status
SAN-MAJ	sanity	3 S	FULL only; deterministic	training-frequency majority hard label; prior vector separate	No
SAN-STRAT	sanity	60 S	FULL; 20 repeats/dataset	seeded stratified random using train-role frequencies	No
SAN-PERM	sanity diagnostic	300 S	100 grouped permutations/dataset	group/source-event atomic full-pipeline label permutation; base simple pipeline	No

Branch	Family/role	A0 terminal accounting	Executed scope	Actual/frozen configuration summary	A4 status
SAN-PRIOR	sanity diagnostic	3 S	FULL only	training-prior probability vector	No
DIAG-LOGVAR	diagnostic	3 S	FULL only	log-bandpower/log-variance + L2 logistic	No
CLS-CSP-LDA	classical	18 S / 3 F	FULL + six low-label budgets; deterministic	CSP OAS; shrinkage LDA lsqr/auto; validation-selected CSP components 4/6/8	Yes; FULL pool + low-label
CLS-FBCSP-LR	classical	3 S	FULL primary	bands 8–12/12–16/16–20/20–24/24–28/28–32; order 4; CSP 4 OAS; L2 LR max_iter 2000; validation C in {1,10}	Yes; FULL pool
RIE-TS-LR	Riemannian	21 S	FULL + six low-label budgets; deterministic	OAS covariance -> Riemann tangent -> L2 LR max_iter 2000; validation C in {0.1,1,10}	Yes; FULL + low-label
RIE-EA-TS	Riemannian	3 S	FULL admitted	train-safe Euclidean Alignment + tangent LR; validation C observed in {0.1,1}	Yes when admitted; FULL pool
RIE-MDM	Riemannian diagnostic	3 S	FULL diagnostic	MDM/FgMDM distance-derived semantics, not calibrated probability	No
DNN-EEGNET	compact neural	105 S	FULL + six low-label budgets; 5 seeds	EEGNet/Lawhern-centered P01-compatible variant; Adam lr .001, dropout .5, kernels 40/20, pools 5/10; Stage11 mixed accepted stopping recipes	Yes; FULL + low-label
DNN-FBCNET	conditional neural	15 S	FULL; 5 seeds	FBCNet author-centered: batch16 Adam .001 wd0 constant; max1500/pat200 restore-best validation BACC	Yes if admitted; FULL pool
DNN-SEQ	conditional sequence	15 S	FULL; 5 seeds	DBConformer preferred admitted slot; batch32 Adam .001 wd0 constant max150/pat40 restore-best	Yes if admitted; FULL pool
DNN-EGTC	fallback	15 CS	fallback slots explicit	EEG-TCNet fallback not activated because sequence slot admitted	No in accepted run
SSL-CBRAMOD	compact SSL	105 S	FULL + six low-label	AdamW body1e-4/head5e-4	Qualified A4; LONG fail-

Branch	Family/role	A0 terminal accounting	Executed scope	Actual/frozen configuration summary	A4 status
			budgets; 5 seeds	wd.05, label smoothing .1, input scale 10000, 160 → 200 adapter, cosine eta_min1e-6, clip1, max150/pat40; audited checkpoint	closed
SSL-REVE	SSL diagnostic	3 DB	FULL diagnostic attempts	known P02 overlap/dependency block preserved	No

Legend: S=SUCCESS, F=FAILED, CS=CONDITIONAL_SKIP, DB=DEPENDENCY_BLOCKED. These totals close the 678 official A0 terminal cells: 657 SUCCESS, 3 FAILED, 15 CONDITIONAL_SKIP and 3 DEPENDENCY_BLOCKED. Stage 13 closed **672 successful model checkpoints with zero checkpoint round-trip failures**; Stage 14 closed 672 prediction model partitions with zero missing partitions. Individual checkpoint IDs/SHA-256 values are preserved in ModelRegistryRecord, checkpoint manifests and the external artifact index rather than duplicated as a 672-row prose appendix.

7.7.1 A4 representative-selection pools

Validation-only A4 representative pools were frozen by role: classical FULL_TRAIN CLS-CSP-LDA / CLS-FBCSP-LR and low-label CLS-CSP-LDA; Riemannian FULL_TRAIN RIE-TS-LR / RIE-EA-TS and low-label RIE-TS-LR; neural FULL_TRAIN DNN-EEGNET / admitted DNN-FBCNET / admitted DNN-SEQ and low-label DNN-EEGNET; SSL admitted SSL-CBRAMOD. SSL-REVE, sanity/diagnostic branches, MDM and fallback-only EGTC do not enter claim-bearing A4 role selection. The selected representative is frozen on validation evidence only, with test outcomes prohibited from selection.

7.8 Seeds, repetitions, candidate selection, and training-policy contract

7.8.1 Exact seed derivation

Master seed: 20260804. For each stochastic purpose, the implementation forms the frozen canonical UTF-8 identity containing master seed, phase, dataset, run family, budget identity, budget repeat, model repeat, purpose and—where applicable—auxiliary repeat. It computes SHA-256, interprets the first 8 hexadecimal digits as an unsigned integer, reduces modulo 2^31-1, and maps zero to one. Registered purposes include MODEL_INIT, SAMPLER, AUGMENT, DATALOADER, PERMUTATION, STAT_BOOTSTRAP, STAT_TEST, and CHECKPOINT_TIEBREAK. Actual per-cell seed values remain in run-cell/registry records; the Protocol freezes the derivation rule rather than copying thousands of seed values.

7.8.2 Repetition contract

Surface	Frozen / actual rule
deterministic classical/Riemannian	1 model repeat
neural / sequence / SSL A0	5 model repeats where active
budget membership	1 inherited frozen subset (P01_FROZEN_SINGLE_SUBSET)
SAN-PERM	100 grouped permutation repeats per dataset
SAN-STRAT	20 stochastic repeats per dataset
canonical A4 refit repeat	original 0–4 planned; only repeat 0 executed after Stage18 deviation
Stage18S descriptive anchor extension	additional repeats 1 and 2 for selected post-hoc anchor cells only; not five-repeat confirmatory evidence

7.8.3 Pre-test class-weight policy

P02-PREEXEC-TRAINING-POLICY-AMENDMENT-R2 resolves class weighting only on legal fit/validation evidence. Equal class counts short-circuit to uniform weighting. Where counts are unequal and a branch has native/verified class-weight support, uniform versus sklearn-balanced weighting may be compared under the same selected hyperparameters/seed; validation BACC is primary, macro-F1 secondary, and uniform wins ties. Test access is prohibited. Unsupported classifiers are not silently converted into a different algorithm family.

7.8.4 EEGNet segmentation/reconstruction diagnostic challenger

The S&R challenger is diagnostic-only, FULL_TRAIN-only, not an A-number, does not replace primary A0, is not A4-eligible, and is not a P03 primary branch. It uses the same five EEGNet model seeds and validation-only probability candidates {0.25,0.5,0.75} with median BACC, then macro-F1, then distance-to-0.5 and lower-probability tie breaks. Final pre-test selections: BNCI2014_001 = 0.75; Lee2019_MI = 0.25; PhysioNetMI = 0.25. All 45 calibration fits and 15 final challenger cells succeeded. Challenger cells remain outside the official 678 A0-cell and 1,218 A4-cell contracts.

7.9 Stage 11 scientific amendment and canonical EEGNet history

Stage 11 is a **post-observation scientific runtime amendment**, not a packaging-only repair and not an untouched Build Book execution. Earlier generic/prior stopping-policy behavior raised source-fidelity/collapse-risk concerns. The owner-authorized continuation aligned the EEGNet architecture/input treatment more closely with Lawhern/EEGNet source context while preserving P01 data/split/test isolation. The source-centered record explicitly states that Lawhern's 500-epoch fact is **context**, and that P02's true patience-120-from-epoch-1 continuation is **not claimed as an original Lawhern stopping rule**.

The accepted artifact is P02-STAGE11-R7H7-EEGNET-AUTHOR-CENTERED-10000E-TRUE-P120-CONTINUATION-R1; runtime successor P02-RUNTIME-SUCCESSOR-R7H7-STAGE11-EEGNET-AUTHOR-CENTERED-10000E-TRUE-P120-CONTINUATION; Stage-11 attempt 77533b224dad. P01 bytes were not mutated; test was not used for selection; no performance-threshold gate was introduced.

Crucially, **canonical Stage 11 is a mixed grandfathered recipe lineage**, not a claim that all 105 primary A0 EEGNet cells were retrained under the final R7 stopping rule:

Recipe lineage	Recipe SHA-256	Accepted count	Canonical role
P500_FLOOR500 / grandfathered source-centered P500	76a601fee7b69395453f240507abc843c63722c548211f43cf2b88647ac7af09	101	sealed primary A0 cells retained rather than needlessly retrained
P120_FLOOR500_R6	74fa2011fa0b94a537338792ce9669350e4ca3153b5f1a3ee50ad556d312690e	4	sealed primary A0 cells retained
TRUE_P120_FLOOR1_R7	18cbe259c05bfcea6b26934118216d5a60ff14d659906e6e8e2f590ffc502866	15	newly fitted diagnostic S&R challenger cells; max 10,000, patience 120 from epoch 1

Thus the 105 canonical primary A0 EEGNet cells are **101 P500 + 4 R6 P120/FLOOR500**, while the 15 R7 TRUE_P120 cells are diagnostic challenger cells. Stage 11 also completed 45 internal S&R probability-calibration fits. All three recipe SHA-256 values remain accepted lineage evidence. The Protocol must not say or imply that the final R7 recipe retroactively replaced every primary A0 model.

History chain: original/generic policy -> observed source-fidelity/collapse-risk concern -> owner-authorized author/source-centered repair -> grandfather valid sealed primary artifacts -> execute true-P120 new/unsealed challenger fits -> validate G11 PASS -> revalidate downstream Stage 12 dependency. Earlier predecessor archives remain preserved as superseded history.

7.10 Stage 12 scientific amendment and author/source-centered external-family history

Stage 12 underwent a scientifically material, post-Stage12 forensic policy repair. Primary amendment artifact: P02-STAGE12-R6-SCIENTIFIC-POLICY-AMENDMENT-R2; explicit change_authority = USER_AUTHORIZED_POST_STAGE12_VALIDATION_ONLY_FORENSIC_REPAIR; candidate policy

ONE_PREDECLARED_AUTHOR_CENTERED_RECIPE_PER_ACTIVE_EXTERNAL_BRANCH. The amendment froze source/author-centered recipes while retaining P01 compatibility, validation-only early stopping/selection, test isolation, uniform class weighting, and explicit dependency/conditional terminal states.

Accepted Stage-12 attempt: b9d3bbec230b; final runtime successor P02-RUNTIME-SUCCESSOR-R6R5-STAGE12-FBCNET-AUTHOR-HORIZON-1500-200. After canonical Stage 11 was replaced/revalidated, Stage 12 was revalidated; the final G12 record states stage12_retraining_implied=false and stage12_scientific_artifacts_byte_identical=true for that later revalidation step.

Branch	Source/recipe lineage	Accepted training policy	Terminal surface
DNN-FBCNET	ravikiran-mane/FBCNet, commit de1bbdd8a54cb1e466830e3d47070e0e56761a37, codes/classify/cv.py	batch16, Adam, lr .001, wd0, constant, max1500/pat200 , restore-best; P01 split/validation BACC/test firewall override incompatible author protocol details	15 SUCCESS
DNN-SEQ	DBConformer preferred admitted final sequence slot; source-centered compatibility repair	batch32, Adam, lr .001, wd0, constant, max150/pat40, restore-best	15 SUCCESS
SSL-CBRAMOD	source-centered CBraMod with audited checkpoint/adaptation	batch64, AdamW, body1e-4/head5e-4, wd .05, label smoothing .1, scale10000, cosine eta_min1e-6, gradclip1, max150/pat40	105 SUCCESS
DNN-EGTC	governed low-resource fallback	not activated while DBConformer final slot admitted	15 CONDITIONAL_SKIP
SSL-REVE	known P02 overlap/dependency risk	fail-closed scientific overlap block preserved	3 DEPENDENCY_BLOCKED

Total Stage 12 surface: **153 attempted; 135 SUCCESS; 15 CONDITIONAL_SKIP; 3 DEPENDENCY_BLOCKED**. P00/P01 artifacts were not mutated, and test data were not used for hyperparameter selection. The amendment is reported as post-observation; it is not backdated as the original Build Book plan.

7.11 Actual Kaggle environment and security boundary

Resource / package	Actual accepted environment
Python	3.12.13
CPU	4 logical CPUs
RAM	31.348 GiB total; 29.439 GiB available at Stage 01 probe
GPU	Tesla T4; CUDA available; 14.562 GiB VRAM
disk at Stage 01	19.518 GiB total; 19.494 GiB free
moabb	1.5.0
mne	1.12.1
numpy	2.2.6
scipy	1.15.3
pandas	2.3.1
scikit-learn	1.7.1
h5py	3.14.0
pooch	1.8.2
PyYAML	6.0.2

Resource / package	Actual accepted environment
pydantic	2.11.7
jsonschema	4.25.0
nbformat	5.10.4
pytest	8.4.1
torch	2.13.0
braindecode	1.6.1
pyriemann	0.12
huggingface-hub	1.24.0
safetensors	0.8.0

Credential presence was required and resolved from an `ENVIRONMENT_SECRET_PROVIDER`. **No credential value is Protocol evidence.** Any literal Hugging Face access token, including values that may have appeared in an interactive notebook source during execution, is explicitly excluded from this document, all machine-readable companions, validation outputs, and release surfaces. The security contract is to inject credentials at runtime, never print them, never persist them to manifests/logs, and secret-scan release material.

7.12 Executed notebook and stage matrix

Stage	Function	Gate/output status	Actual Protocol disposition
00	Authority/preflight	PASS	Authority and phase identity intake; deterministic preflight.
01	Environment	PASS	Python 3.12.13; Tesla T4; exact package pins and secret-provider credential presence recorded.
02	Cumulative project intake	PASS	P00/P01 state and P01 → P02 inputs resolved.
03	External pointer and conditional asset resolution	PASS	External assets resolved/fail-closed without changing P01 bytes.
04	P01 validation	PASS	Immutable P01 data/split/preprocessing/window substrate revalidated.
05	Scientific freeze verification	PASS	P02-PLANNED-SCIENTIFIC-EXECUTION-FREEZE-R5; owner-authorized pre-test training-policy amendment retained.
06	Schema/config/import closure	PASS	10 schemas; run_config_hash=9e181d2e935d2e9674ca6e05572f49520ad0306a3761362b770f8bee8c78ce13; clean import PASS.
07	Loader/leakage/role discipline	PASS	P01 roles retained; test isolated from fitting/selection.
08	Sanity/diagnostic family	PASS	369 attempted; 369 SUCCESS.

Stage	Function	Gate/output status	Actual Protocol disposition
09	Classical family	PASS	24 attempted; 21 SUCCESS; 3 FAILED.
10	Riemannian family	PASS	27 attempted; 27 SUCCESS.
11	EEGNet + training-policy challenger	PASS	Canonical author-centered successor; 105 primary SUCCESS; 45 internal S&R calibration fits SUCCESS; 15 diagnostic challenger SUCCESS.
12	External neural/SSL families	PASS	Canonical author-centered/source-centered successor; 153 attempted; 135 SUCCESS; 15 CONDITIONAL_SKIP; 3 DEPENDENCY_BLOCKED.
13	Checkpoint closure	PASS	672 successful model checkpoints; 0 roundtrip failures.
14	Prediction closure	PASS	672 model partitions emitted; 0 missing; score semantics complete.
15	A0 + challenger closure	PASS	678 planned/terminal; 657 SUCCESS; 3 FAILED; 15 CONDITIONAL_SKIP; 3 DEPENDENCY_BLOCKED; 3942 participant rows; 2625 metric records.
16	Low-label curves	PASS	36 records; 216 attempted points; 213 SUCCESS; 3 FAILED; 207 VALID + 9 EXPLICIT_NON_SUCCESS; single frozen subset limitation retained.
17	Subject/session profiles	PASS	18 descriptive profile records.
18	A4 ordinary controls	PASS	1218 terminal slots; 591 SUCCESS; 576 CONDITIONAL_SKIP; 9 INPUT_INCOMPATIBLE; 42 NOT_APPLICABLE_REPEAT_SLOT; resource-constrained deviation explicitly frozen.
18U	Additional-ablation dispatcher	PASS	NO_ADDITIONAL_FULL_EXECUTION_ABLATION_UNLOCKED; result-dependent unlock prohibited.
19	Failure/negative evidence	PASS	648 failures, 340 negative-result notes, 309 diagnostic flags aggregated and preserved.
20	Downstream readiness	PASS	1014 prediction partitions checked; P03 contract complete; readiness record emitted.
21	Figure/table source data	PASS	Governed source tables/figures prepared without Layer10 recomputation.
22	Handoffs	PASS	Protocol, Phase Analysis, Layer0, Evidence Map, Layer10 and P03 handoffs emitted.
23	Evidence sufficiency	PASS	missing=[]; all required state flags true.

Stage	Function	Gate/output status	Actual Protocol disposition
24	Bundle/finalization	PASS	Runtime bundle ready; governed G24 PASS; subsequent release packaging/replay changed no science.

Every governing stage ledger reports SUCCESS with its output/gate status PASS for the accepted lineage. The stage matrix includes 18U as the same-notebook additional-ablation dispatcher. Stage 18S is a later supplemental sensitivity surface and is intentionally not substituted for canonical Stage 18/G18.

7.13 A0 raw accept-all decoder contract and actual closure

A0 asks what raw accept-all decoder evidence is produced under frozen P01 inputs without calibration, rejection, abstention, or downstream policy. Its exact datasets, split, windows, budgets, model/run lineage, seed identities, and test-after-selection discipline are frozen above.

Required A0 metrics/source evidence:

- primary: balanced accuracy (BACC);
- complementary: macro-F1;
- secondary: accuracy;
- conditional: ROC-AUC only when score semantics permit;
- required source data: confusion, class support, prediction completeness, branch success;
- participant-first aggregation and exact denominator accounting.

Actual A0 closure:

Field	Actual value
planned / terminal cells	678 / 678
SUCCESS	657
FAILED	3
CONDITIONAL_SKIP	15
DEPENDENCY_BLOCKED	3
participant metric rows	3,942
metric records	2,625
prediction record complete	YES
raw score semantics complete	YES
denominator accounting complete	YES
checkpoint provenance complete	YES
minimum floor	at least one successful classical + one successful Riemannian branch for every dataset

The three failed and all skipped/blocked cells remain part of the A0 accounting and are not imputed as successes.

7.14 Low-label curve contract and actual closure

Low-label evidence inherits the exact P01 calibration-role nested subsets at 1/2/4/8/16/32 examples per class. No new random membership is generated in P02. The curve therefore measures behavior on **one frozen subset realization per budget**, not repeated budget-membership sampling.

Actual Stage 16 closure: 36 curve records; 216 attempted points; 213 successful run points and 3 failed run cells; 207 VALID points and 9 EXPLICIT_NON_SUCCESS points after terminal-state mapping. All budgets are accounted for. A derived-evidence repair was allowed only to restore/close the analysis surface; it did not retrain models, regenerate predictions, alter scientific configuration, or select on test evidence.

Claim limitation: the resulting low-label curves may support behavior under the inherited single-subset protocol, but not variance attributable to repeated low-label subset resampling.

7.15 Subject/session heterogeneity and descriptive profile contract

Stage 17 emitted 18 SubjectProfileRecord-type descriptive records. These profiles are common-support descriptive evidence for participant/session heterogeneity and difficulty patterns. They are not diagnostic labels, clinical phenotypes, routing policies, or eligibility decisions. Any downstream use must preserve the exact contributing prediction/model/split/budget identities and must not elevate descriptive difficulty into a medical or deployment classification.

7.16 A4 ordinary-control contract and resource-constrained deviation

7.16.1 Frozen A4 scientific question and six conditions

A4 tests authorized **ordinary** longer-window, multi-window, and fixed ensemble controls against matched CORE evidence, with burden recorded. It is not a learned gating, calibration, abstention, or adaptive-policy ablation.

Condition	Frozen meaning
A4-C0-CORE	matched A0 CORE reference
A4-C1-LONG-3P5S	exact 3.5-s A4 long representation, refit with frozen recipe/seed; no pad/crop substitution
A4-C2-MULTI-HARD-VOTE	deterministic parent-level majority vote over the three fixed views
A4-C3-MULTI-PROB-AVG	probability average only when genuine aligned probabilities exist
A4-C4-MODEL-HARD-VOTE	fixed ordinary model ensemble hard vote using pre-result role membership
A4-C5-MODEL-PROB-AVG	fixed ordinary model probability average only with compatible aligned probabilities

Hard-vote ties may not use truth. Probability averaging is fail-closed when score/class semantics are incompatible. Representative selection is validation BACC → macro-F1 → lower validation burden → lexical branch ID and is frozen before test.

7.16.2 Owner-authorized Stage 18 deviation

The accepted Stage 18 artifact explicitly records P02-STAGE18-RESOURCE-CONSTRAINED-ANCHOR-BUDGET-SINGLE-REPEAT-R1, reason KAGGLE_WALL_CLOCK_CONSTRAINT_OWNER_DECISION.

This deviation **did not remove** any A4 condition, dataset, representation, or model-family role. It preserved all three datasets, all six A4 conditions, and refit roles CLASSICAL, NEURAL, RIEMANNIAN, SSL. It changed the expensive refit execution surface:

original refit repeat indices: 0–4;

executed refit repeat index: 0 only;
declared skipped repeat indices: 1–4;
deep-model anchor budgets: 1/class, 8/class, and FULL_TRAIN;
frozen plan A4 terminal slots: 1,218;
frozen refit cells: 756;
declared reduced run cells: 576 = 504 repeat-reduction slots + 72 deep-budget-reduction slots;
executable refit run cells: 180, of which 54 are deep refits;
estimated unique deep member fits: reduced from 840 to 72 (91.43% reduction);
latency measurement repeats: 1.

Per-fit training recipes, representative-selection rules, test isolation, checkpoint roundtrip validation, and statistical comparison definitions were not changed.

7.16.3 Actual A4 closure

Terminal state / closure surface	Actual value
planned terminal slots	1,218
SUCCESS	591
CONDITIONAL_SKIP	576
INPUT_INCOMPATIBLE	9
NOT_APPLICABLE_REPEAT_SLOT	42
representative groups	21
role-control expected / closed	756 / 756
C4/C5 expected / closed	210 / 210
burden rows	1,218
phase-analysis source complete	YES

A4 is therefore **executed and evidence-sufficient for the explicitly authorized resource-constrained anchor-budget/single-repeat scope**, but full_buildbook_replication_equivalence = false.

7.16.4 A4 claim lock

The deviation expressly authorizes anchor-budget deep-model comparison within the executed scope. It does **not** authorize:

five-repeat/multi-seed A4 stability claims;
dense deep-model budget-response curves across every inherited budget;
representation of P02 A4 as full replication of the original Build Book refit grid.

These limitations are mandatory in Phase Analysis, Layer 0 review, Evidence Map entries, Layer 10 cards/figures, and any later manuscript synthesis.

7.17 Stage 18S post-hoc balanced sensitivity supplement

P02-STAGE18S-BALANCED-SENSITIVITY-R1 is retained as **additional post-hoc descriptive sensitivity evidence**. It is not a replacement for canonical Stage 18 and does not modify G18.

The final handoff records:

status PASS;
 canonical Stage 18 remains the G18 source;
 216 member receipts;
 162 supplemental terminal comparison cells, with 135 SUCCESS and 27 INPUT_INCOMPATIBLE;
 post_hoc = true;
 not five-repeat confirmatory;
 not full-Build-Book-equivalent;
 original Stage 18 protocol deviation must not be overwritten.

Accordingly, Stage 18S may be used in Phase Analysis as sensitivity/robustness context with explicit post-hoc labeling, but may not upgrade the confirmatory status or erase Stage 18's scope limitation.

7.18 A0–A13 disposition matrix and A14 lock

The controlling Protocol v0.1 fixes the official ablation identities. R2 corrects the R1 mislabeling of A2 and restores the exact A5–A13 names.

Ablation	Official identity	Controlling owner	P02/L2 disposition	P02 evidence/limitation
A0	Raw Decoder / Accept-All Raw Decoder Reference	L2	APPLICABLE_EXECUTED	678 terminal A0 cells; complete with explicit non-success states
A1	Calibrated Decoder / Calibration Visibility	L3 C1	NOT_APPLICABLE_WITH_REAS ON	downstream; P02 emits raw score substrate only
A2	Simple Registered Threshold / Confidence-Threshold Baseline	L3 C2	NOT_APPLICABLE_WITH_REAS ON	downstream threshold baseline; not an L2 uncertainty cell
A3	Uncertainty and Selective Prediction	L3 C2	NOT_APPLICABLE_WITH_REAS ON	downstream uncertainty/selective-prediction science
A4	Longer-Window, Multi-Window Voting/Averaging and Ordinary Ensemble Controls	L2	APPLICABLE_EXECUTED_WITH_RECORDED_DEVIATION	all C0–C5 identities preserved; resource-constrained single refit repeat and deep anchors only; no full-grid equivalence
A5	IHARQ-lite / Rule-Based Evidence Verification	L4	NOT_APPLICABLE_WITH_REAS ON	downstream evidence verification
A6	IHARQ + Evidence-Quality Estimator	L4/L6	NOT_APPLICABLE_WITH_REAS ON	downstream evidence-quality estimation
A7	IHARQ + RegimeRisk Temporal Trust	L5	NOT_APPLICABLE_WITH_REAS ON	downstream temporal trust
A8	Learning-to-defer / Deferral Comparison	L6/L7	NOT_APPLICABLE_WITH_REAS ON	downstream deferral comparison
A9	Supervised Adaptive-IHARQ / Adaptive Readiness Policy	downstream adaptive-policy layer	NOT_APPLICABLE_WITH_REAS ON	downstream supervised adaptive policy
A10	Contextual Bandit / Simulation-Bounded Immediate-Reward	simulation-bounded downstream layer	NOT_APPLICABLE_WITH_REAS ON	downstream simulation-bounded bandit

Ablation	Official identity	Controlling owner	P02/L2 disposition	P02 evidence/limitation
	Adaptation			
A11	Reinforcement-Learning Policy / Simulation-Bounded Sequential Policy Learning	simulation-bounded downstream layer	NOT_APPLICABLE_WITH_REAS ON	downstream sequential policy learning
A12	StressForge Stress Tests / Controlled Stress Robustness	stress layer	NOT_APPLICABLE_WITH_REAS ON	downstream stress robustness
A13	Layer 9 MyoSuite/OpenSim/static-replay Embodiment Demo	L9	NOT_APPLICABLE_WITH_REAS ON	downstream embodiment/static replay
A14	absent/prohibited	governance/protocol lock	PROHIBITED	no A14 may be invented/unlocked

Stage 18U confirmed NO_ADDITIONAL_FULL_EXECUTION_ABLATION_UNLOCKED; result-dependent ablation unlocking is prohibited. Not executing a downstream A-cell in P02 does not delete its official identity from the cumulative ladder.

7.19 Metrics, statistics, matching, denominators, and analysis contract

7.19.1 Metrics

primary: BACC;

complementary: F1_MACRO;

secondary: ACC;

conditional: ROC_AUC when score semantics make it lawful;

required source fields: confusion, class support, prediction completeness, branch success; A4 additionally records burden/delay.

Not selected for P02 confirmatory use: Kappa, MCC, AUPRC, Brier, NLL, ECE, calibration slope, and risk-coverage. Their absence must not be interpreted as missing evidence for a P02-owned estimand because calibration/selective-risk ownership belongs downstream.

7.19.2 Statistical contract

participant-first aggregation;

95% participant-cluster uncertainty;

10,000 BCa bootstrap resamples, percentile fallback;

two-method paired comparison: two-sided paired Wilcoxon;

multi-method comparison: Friedman followed by predeclared paired Wilcoxon comparisons;

Holm multiplicity control within the confirmatory family;

effect sizes: paired median difference, matched-pairs rank-biserial, and Kendall W for Friedman;

minimum complete participants for an inferential comparison: 5;

cross-dataset analysis is dataset-stratified; no confirmatory meta-analysis is created by P02.

Stage 18's compute deviation does not change these comparison definitions; it changes which repeat/budget cells are actually eligible to populate them.

7.19.3 Matching and denominator discipline

Every comparison must retain enough identity to establish matched support: dataset, subject, session/run, source event/parent window, split, budget, budget repeat, branch/role, model repeat, seed/config and condition/profile identity as applicable. Attrition and incompatibility are explicit. No comparison may silently use a favorable subset without reporting the support difference and reason.

7.20 Failure, negative, invalid, blocked, skipped, and diagnostic evidence

Stage 19 aggregated **648 failure entries, 340 negative-result notes, and 309 diagnostic flags**. These include both direct run non-success states and structured diagnostic/negative evidence. They are preservation surfaces, not a single interchangeable error count.

Required distinctions:

FAILED: attempted run/operation reached a failure terminal state;

DEPENDENCY_BLOCKED: required dependency/scientific admission condition was not satisfied;

CONDITIONAL_SKIP: a frozen conditional branch/condition was lawfully not activated;

INPUT_INCOMPATIBLE: exact score/input semantics do not permit the condition;

NOT_APPLICABLE_REPEAT_SLOT: a frozen slot is declared non-applicable under the authorized Stage 18 repeat reduction;

negative/null scientific observations: valid results that do not support an improvement;

diagnostic-only evidence: useful for engineering/scientific diagnosis but not claim-bearing primary evidence.

No failed or superseded execution is converted into success evidence. The final accepted lineage may supersede an earlier attempt for current analysis while the earlier attempt remains part of the audit history.

7.21 Repair, rerun, amendment, and supersession ledger

Event	Classification	Scientific configuration changed?	Test-driven selection?	Final disposition
P02-PREEXEC-TRAINING-POLICY-AMENDMENT-R2	owner-authorized pre-test training-policy resolution	YES, before relevant final test evaluation	NO	frozen supporting policy; S&R challenger/class weighting governed
Stage 11 author-centered EEGNet successors	post-observation scientific training-policy amendment + rerun	YES	NO	final R7H7 successor canonical; predecessors retained as superseded evidence
Stage 12 author/source-centered external-family successors	post-observation scientific policy/config amendment + rerun	YES	NO	final R6R5 successor canonical; blocked/skipped branches preserved
Stage 16 derived-evidence closure repair	derived analysis/evidence repair	NO training/prediction/config change	NO	accepted; does not change primary run evidence
Stage 18 resource-constrained anchor/single-repeat execution	explicit Protocol deviation from Build Book	execution surface reduced; per-fit scientific recipe unchanged	NO	accepted for restricted A4 claim scope; not full Build Book replication
Stage 18S sensitivity	post-hoc supplemental analysis/execution	supplemental only; canonical Stage18 unchanged	NO	descriptive sensitivity only
original Stage 24 monolithic ZIP	packaging/transport failure	NO	N/A	superseded by low-disk chunked

Event	Classification	Scientific configuration changed?	Test-driven selection?	Final disposition
attempt	(ENOSPC)			HF release; failure retained
Stage 24 low-disk chunked HF release R2.2	packaging/release repair	NO	N/A	accepted archival/fallback release
Stage 24 finalization replay from full workspace snapshot	release assembly replay	NO scientific-stage or training rerun	N/A	accepted release replay; Stage11/12/18/18S/23/24 source state revalidated

This ledger is the authoritative post-execution classification for Phase 02. Later documents may summarize it but must not relabel a post-observation scientific amendment as pre-registered.

7.22 Gate closure and evidence sufficiency

Every supplied canonical gate record from G00 through G24, including G18U, was parsed. All are PASS with empty blockers; observability is PASS where recorded. Runtime-successor identity remains explicit rather than being collapsed into the final gate status.

Gate	Stage	Status	Observability	Blockers	Evidence	Runtime successor
G00	00	PASS	PASS	[]	stage_ledger/ stage_00.json	
G01	01	PASS	PASS	[]	stage_ledger/ stage_01.json	
G02	02	PASS	PASS	[]	stage_ledger/ stage_02.json	
G03	03	PASS	PASS	[]	stage_ledger/ stage_03.json	
G04	04	PASS	PASS	[]	stage_ledger/ stage_04.json	
G05	05	PASS	PASS	[]	stage_ledger/ stage_05.json	
G06	06	PASS	PASS	[]	stage_ledger/ stage_06.json	
G07	07	PASS	PASS	[]	stage_ledger/ stage_07.json	
G08	08	PASS	PASS	[]	stage_artifacts/ 08_semantic_gate_suc cessor_R1.json	P02-RUNTIME- SUCCESSOR-R1- CLASS-WEIGHT- APPLICABILITY-LIVE- CONTINUATION
G09	09	PASS		[]	diagnostics/ runtime_successor/ G09_SEMANTIC_AUDI T_R1.json	P02-RUNTIME- SUCCESSOR-R2- CANONICAL-LOW- CAL-BUDGET-ID- PARSER-LIVE- CONTINUATION

Gate	Stage	Status	Observability	Blockers	Evidence	Runtime successor
G10	10	PASS		[]	diagnostics/ runtime_successor/ G10_SEMANTIC_AUDI T_R1.json	P02-RUNTIME- SUCCESSOR-R2- CANONICAL-LOW- CAL-BUDGET-ID- PARSER-LIVE- CONTINUATION
G11	11	PASS	PASS	[]	stage_ledger/ stage_11.json	P02-RUNTIME- SUCCESSOR-R7H7- STAGE11-EEGNET- AUTHOR- CENTERED-10000E- TRUE-P120- CONTINUATION
G12	12	PASS	PASS	[]	stage_ledger/ stage_12.json	P02-RUNTIME- SUCCESSOR-R6R5- STAGE12-FBCNET- AUTHOR- HORIZON-1500-200
G13	13	PASS	PASS	[]	stage_ledger/ stage_13.json	P02-RUNTIME- SUCCESSOR-R7H7- STAGE11-EEGNET- AUTHOR- CENTERED-10000E- TRUE-P120- CONTINUATION
G14	14	PASS	PASS	[]	stage_ledger/ stage_14.json	P02-RUNTIME- SUCCESSOR-R7H7- STAGE11-EEGNET- AUTHOR- CENTERED-10000E- TRUE-P120- CONTINUATION
G15	15	PASS	PASS	[]	stage_ledger/ stage_15.json	P02-RUNTIME- SUCCESSOR-R7H7- STAGE11-EEGNET- AUTHOR- CENTERED-10000E- TRUE-P120- CONTINUATION
G16	16	PASS	PASS	[]	diagnostics/ runtime_successor/ G16_SEMANTIC_AUDI T_R1.json	P02-RUNTIME- SUCCESSOR-R7H7- STAGE11-EEGNET- AUTHOR- CENTERED-10000E-

Gate	Stage	Status	Observability	Blockers	Evidence	Runtime successor
G17	17	PASS	PASS	□	stage_ledger/ stage_17.json	TRUE-P120- CONTINUATION P02-RUNTIME- SUCCESSOR-R7H7- STAGE11-EEGNET- AUTHOR- CENTERED-10000E- TRUE-P120- CONTINUATION
G18	18	PASS	PASS	□	stage_ledger/ stage_18.json	P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET- SINGLE-REPEAT-R1
G18U	18U	PASS	PASS	□	stage_ledger/ stage_18U.json	P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET- SINGLE-REPEAT-R1
G19	19	PASS	PASS	□	stage_ledger/ stage_19.json	P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET- SINGLE-REPEAT-R1
G20	20	PASS	PASS	□	stage_ledger/ stage_20.json	P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET- SINGLE-REPEAT-R1
G21	21	PASS	PASS	□	stage_ledger/ stage_21.json	P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET- SINGLE-REPEAT-R1
G22	22	PASS	PASS	□	stage_ledger/ stage_22.json	P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET- SINGLE-REPEAT-R1
G23	23	PASS	PASS	□	stage_ledger/ stage_23.json	P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET-

Gate	Stage	Status	Observability	Blockers	Evidence	Runtime successor
G24	24	PASS	PASS	[]	stage_ledger/ stage_24.json	SINGLE-REPEAT-R1 P02-RUNTIME- SUCCESSOR-R10- STAGE18-A4- ANCHOR-BUDGET- SINGLE-REPEAT-R1

G23 closes evidence sufficiency with missing=[]; G24 closes runtime bundle/finalization with blockers []. A PASS means the governed **executed** scope is sufficient. It does not imply five-repeat A4 stability, dense deep-budget coverage, or full original Build Book A4 replication equivalence.

7.23 Evidence status, interpretation hierarchy, and claim ceiling

Phase Analysis must use:

measured result → supported interpretation → candidate claim → mechanism hypothesis (only when justified).

Candidate claims are **not approved claims** until Layer 0 disposition.

7.23.1 P02-supported candidate-claim domains

P02 evidence may support appropriately bounded candidate claims about:

- governed raw accept-all baseline decoder measurement under the frozen P01 protocol;
- model-family comparisons using exact recorded data/split/budget/seed/config/support identities;
- low-label behavior under the inherited single-frozen-subset limitation;
- participant/session heterogeneity as descriptive evidence;
- ordinary A4 longer/multi-window/fixed-ensemble controls within the resource-constrained anchor-budget/single-repeat scope;
- record completeness, checkpoint/prediction provenance, reproducibility surfaces and P03 technical readiness.

7.23.2 P02-prohibited standalone claim domains

P02 alone cannot establish:

- clinical efficacy, clinical safety, or deployment safety;
- calibrated probability reliability or calibrated uncertainty;
- abstention/selective-risk/readiness-policy effectiveness;
- policy-learning benefit;
- temporal/stress robustness;
- closed-loop or embodiment effectiveness;
- five-repeat A4 stability;
- dense deep-budget A4 response curves;
- full Build Book A4 replication equivalence.

All public wording must carry the inherited PUBLIC_EEG_ONLY, NON_CLINICAL, NO_DEPLOYMENT_CLAIM limitation family plus the P02-specific A4 and low-label limitations where relevant.

7.24 External artifact registry, integrity semantics, and storage precedence

7.24.1 Preferred complete Kaggle-workspace snapshot

Field	Frozen value
provider / type	Hugging Face / private dataset
repository	Csthv/p02-phase02-final-whole-working-20260814t052120z-b890ff92
immutable Git revision	bc14961e14f2e48690e55df3577014275f9cbf30
working-tree manifest	__IHARQ_SNAPSHOT__/WORKING_TREE_MANIFEST.jsonl
working-tree manifest SHA-256	564bbb44e67d14da8457ad83d33651ded93dfc6b89ab90f10f78a36a20b8e8f5
snapshot metadata	__IHARQ_SNAPSHOT__/SNAPSHOT_METADATA.json
source manifest file count	24,661
per-object continuation index	release_metadata/all_artifact_location_index.csv in the local final P02 project state; exact source path, bytes, SHA-256, repo, revision and HF path
per-object continuation index SHA-256 / bytes	540a0c3d4cbc686645ab1fccf088205c02c32a5c34e4b238c91f9c4ec119c12f / 10,191,241
producer / consumers	P02 / P03–P15, later layers, reproduction
access	private; authorized IHARQ_HF_TOKEN_P02 at runtime
license	mixed artifact tree; artifact-level/source licenses control; no fabricated single repository license
role	PREFERRED PRACTICAL P02 CONTINUATION SUBSTRATE
live remote status in final pre-closure audit	REMOTE_NOT_ACCESSIBLE through available HF connector; local Stage24 receipt/revision/manifest evidence validated

A multi-object repository tree does not have a truthful single file-size/SHA field. Its immutable tree identity is the pinned Git revision plus the verified working-tree manifest. Required objects are then verified individually by path, byte count and SHA-256 from the manifest/artifact-location index. Downstream consumers must never select latest.

The light cumulative P02 package intentionally externalizes runtime-heavy objects. Audit reconciliation checked the 17,089 entries in the runtime checksum manifest: 68 objects are locally present and hash-correct; all 17,021 externally omitted objects matched all_artifact_location_index.csv by **exact relative path and SHA-256**, with zero unmatched externalized objects. This establishes pointer-backed completeness of the light derivative without pretending the heavy files are local.

The full workspace snapshot is preferred because it exposes direct workspace paths and avoids reconstructing routine continuation state from multipart ZIP transports. This is a retrieval precedence, not a replacement of the archival release or a change in scientific authority.

7.24.2 Archival/fallback chunked P02 release

Field	Frozen value
artifact ID	P02-STAGE24-ARCHIVAL-CHUNKED-RELEASE-R2
repository	Csthv/iharq-p02-phase02-r2-P02-L2-OFFICIAL-RUN-R4DY-20260814t051352z
content revision	bc88dccd6518aa7b172697be1ef98344997f27ff

Field	Frozen value
manifest revision	c60c0505faa8076090439277ed1d8ef8c86734b7
final release revision	257a407b5c7ea37b6c620863ee261c010c8f197c
private / remote completeness	true / PASS in Stage24 local receipts
heavy packages	4; 9,448,804,258 source bytes / 3,968 source files
execution-bundle packages	1
local transport ZIPs	deleted after confirmed HF upload
reason for successor	original monolithic Stage24 ZIP failed ENOSPC; low-disk chunked successor repaired transport only
scientific configuration changed	NO
live remote status in final pre-closure audit	REMOTE_NOT_ACCESSIBLE; do not fabricate remote verification

Exact uploaded transport objects preserved by the accepted remote-resume state:

Remote path	Bytes	SHA-256
phase_02/execution_bundle_packages/ execution_bundle_package_0001.zip	519,305,980	c380edff3afe44137afea8a3bb70699bb3a3885c3363198eebf93aad66f23805
phase_02/heavy_packages/ heavy_artifacts_package_0001.zip	2,894,348,079	5032ab8785b36044d8f8d1635c02fbb10bfca714bb9ecb16223c3d2cf76f1dcf
phase_02/heavy_packages/ heavy_artifacts_package_0002.zip	2,893,314,005	c596b502ff154c494cdcfa047826617354962f8209c2fbcffe71f79671ed4296
phase_02/heavy_packages/ heavy_artifacts_package_0003.zip	2,898,992,543	b0778b61170d6998102bf9c471b2133697e61cafbddc16ab67e52dad40f48bf2
phase_02/heavy_packages/ heavy_artifacts_package_0004.zip	763,895,568	0b0263c867fea1edd4362deee1d7607f9713ec4135ebda4a441729b6ca4273ef
phase_02/release_manifests/ execution_bundle_multipart_manifest_R2.json	3,797	8f68c1a298414eabd413b0ae6acb49ab35c2524fb63963420376a23d1e698a48
phase_02/release_manifests/ external_artifact_pointer_manifest_R2.json	7,791,281	765e4d1e4dd12bdc85c4e7435405f04a0bf00ee65d79fa6444e2bdf47e64698b
phase_02/release_manifests/ external_artifact_pointer_manifest_R2.yaml	6,775,406	b6b2581cc00ee89669b4924c709f7c7e313afcae1425dc81349f7ffad80e979c
phase_02/release_packages/cumulative_project_state/ IHARQ_Project_State_After_Phase_02_R2.zip	118,117,660	b4f1d12b0fb37faf908a3eadd2d6a43d52efbe7a2601220a030ffd6858b9b8cc
phase_02/release_packages/github_ready/ IHARQ_Cumulative_GitHub_Ready_Through_P02_R2.zip	118,118,899	2052eff12f5ec4054305577de7557049db45a52118e65d9300b318761c406839

Retrieval of the archival release requires IHARQ_HF_TOKEN_P02, pinning the exact final revision above (with the recorded content/manifest revisions retained for reconstruction), resolving the exact object path, and verifying its byte count and SHA-256 against the table/Stage24 receipt. Mutable latest is prohibited.

The archival release remains a valid fallback for exact packaged transport/release reconstruction and is not invalidated by the preferred workspace route.

7.25 Dual-Hugging-Face credential contract for downstream phases

The supplied project state establishes separate private-artifact families: earlier-phase assets use the pre-P02 authorization boundary (including prior csthv999z private assets), while P02 external storage uses the Csthv repositories frozen in §7.24. The owner additionally states that the credentials are distinct. Future mixed-phase workflows must therefore expose **two independent symbolic secret interfaces**:

Secret interface	Artifact family	Required behavior
IHARQ_HF_TOKEN_PRE_P02	P00/P01/pre-P02 private HF assets	runtime-only injection; never serialize/log; no fallback to P02 token
IHARQ_HF_TOKEN_P02	P02 full workspace snapshot and P02 archival release	runtime-only injection; never serialize/log; no fallback to pre-P02 token

A workflow consuming both families must accept both independently and fail closed for whichever required source is inaccessible. One token must never be assumed to cover both. The historical runtime's environment-variable naming is not rewritten; these are the governed **downstream access interfaces**. Secret values are prohibited from Protocol, machine-readable contracts, logs, notebook outputs, Git/release archives and Layer-10 exports.

7.26 Frozen downstream-consumer and analysis-input handoff

Stage 20 checked 1,014 prediction partitions and emitted a PASS Layer-2 readiness surface. Downstream consumers are Phase Analysis, Layer 0, Evidence Map, Layer 10, P03 and later phases/layers, the future cumulative Protocol, thesis/paper synthesis, and reproduction/release.

They must be able to determine canonical versus superseded evidence, retrieve/verify externalized objects, preserve score/class-order semantics, understand allowed analysis scope and prohibited claims, and avoid reconstructing hidden assumptions. The minimum canonical analysis-input paths are frozen in the machine-readable P02_analysis_contract.yaml and Appendix B, including A0/A4 completion/statistics/source manifests, low-label/profile sources, challenger statistics/selection, failure summaries, Stage18 deviation, Stage18S supplement artifacts, figure/table source manifest, pre-execution amendment, and readiness record.

P03 may consume P02 raw scores/probabilities and model-family evidence for downstream calibration/uncertainty/selective-prediction work. It may not retroactively alter P02 labels/splits/windows, replace terminal states, or treat P02 readiness as safety evidence. Any new representation or score transformation must be a new governed downstream artifact with explicit lineage.

7.27 Residual limitations and claim restrictions

The documentary audit closes with the following **true residual limitations**; none is erased by G23/G24 PASS:

- scientific/public scope:** public benchmark binary-MI research evidence only; non-clinical and no deployment claim;
- inherited-data scope:** all P02 conclusions are conditional on the exact P01 labels, split, preprocessing, core/A4 windows and source availability;
- low-label evidence:** one inherited frozen membership subset per budget; no repeated subset-resampling uncertainty;
- branch non-success:** 3 classical A0 failures, 15 conditional fallback skips and 3 dependency-blocked SSL cells remain in denominators; A4 includes incompatible and explicitly skipped/non-applicable terminals;
- Stage 11/12 history:** accepted neural/external evidence includes transparent post-observation, author/source-informed amendments; this is not untouched original pre-registration;
- Stage 11 mixed lineage:** primary A0 EEGNet contains grandfathered P500/R6 recipes; R7 true-P120 applies to newly fitted diagnostic challenger cells and must not be generalized backward;
- A4 compute limitation:** only refit repeat 0 executed; deep roles use 1/class, 8/class and FULL_TRAIN anchors; no five-repeat stability, dense deep-budget curve or full Build Book replication-equivalence claim;
- Stage18S:** post-hoc descriptive sensitivity only; it does not change canonical Stage18/G18 or create five-repeat confirmatory evidence;

conditional score semantics: probability averaging/ROC-AUC are lawful only where compatible genuine score/probability semantics exist;

notebook reconstruction: the owner-supplied final notebook is a post-restart continuation, so Stages 00–18 are reconstructed from canonical ledgers/artifacts rather than a single uninterrupted submitted notebook output stream;

external access: both P02 HF repositories are private and were not live-resolvable through the available connector during this audit; local Stage24 receipts, immutable revisions, manifests, per-object hashes/sizes and pointer reconciliation were sufficient for documentary closure, but downstream remote access still requires authorization;

credential separation: mixed pre-P02/P02 private-HF consumption requires two independently authorized symbolic secrets;

environment: accepted results are tied to the recorded Python/package/GPU/runtime identities and runtime-successor chain;

downstream ownership: calibration, uncertainty, abstention, readiness policy, temporal/stress robustness, simulation and embodiment conclusions remain untested by P02 itself.

These limitations must propagate to Phase Analysis, Layer 0, Evidence Map, Layer 10 and any cumulative/public synthesis using the affected evidence.

7.28 Freeze and closure decision

P02_PROTOCOL_V1_ANNEX_R2_FROZEN_AFTER_SOURCE_COMPLETE_PRE_CLOSURE_AUDIT

Phase 02 execution: **ACCEPTED_FOR_PROTOCOL_AND_PHASE_ANALYSIS**;

evidence sufficiency: **PASS** (G23=PASS, G24=PASS, blockers []);

documentary blockers after R2 repair: **0**;

A0: **COMPLETE**, including explicit non-success terminals;

A4: **COMPLETE FOR THE GOVERNED RESOURCE-CONSTRAINED ANCHOR-BUDGET/SINGLE-REPEAT SCOPE**;

full Build Book A4 replication equivalence: **NO**;

Stage18S: **PASS, POST-HOC DESCRIPTIVE SENSITIVITY ONLY**;

Stage11 canonical evidence: **mixed grandfathered author/source-centered accepted lineage; exact recipe counts frozen in §7.9**;

Stage12 canonical evidence: **author/source-centered accepted successor with explicit post-observation change authority**;

scientific rerun required for Protocol documentary closure: **NO**;

preferred downstream P02 heavy-artifact source: Csthv/p02-phase02-final-whole-working-20260814t052120z-b890ff92@bc14961e14f2e48690e55df3577014275f9cbf30;

archival fallback: Csthv/iharq-p02-phase02-r2-P02-L2-OFFICIAL-RUN-R4DY-20260814t051352z@257a407b5c7ea37b6c620863ee261c010c8f197c;

mixed pre-P02/P02 private-HF access: **DUAL SYMBOLIC CREDENTIAL INTERFACE REQUIRED**;

live remote HF verification in this audit: **UNAVAILABLE/PRIVATE**; **locally preserved immutable receipt/manifest evidence validated**;

cumulative integration through P02: **COMPLETED WITHOUT SEMANTIC RECONSTRUCTION**;

R1 disposition: **SUPERSEDED BY R2 FOR DOCUMENTARY/MACHINE-READABLE CORRECTIONS ONLY; SCIENTIFIC EVIDENCE UNCHANGED**.

7.29 Phase-specific machine-readable companion files

The following companion files are deterministic structured projections of this annex; they do not supersede the canonical Markdown:

machine_readable/P02_run_matrix.yaml — exact accepted stage/run matrix and A0/A4 terminal summaries;

machine_readable/P02_analysis_contract.yaml — metrics, statistical rules, A4 claim restrictions, Stage18S status and interpretation ceiling;

machine_readable/P02_external_artifact_access_contract.yaml — preferred workspace snapshot, archival fallback, immutable revisions and dual-secret interface.

If a machine-readable file and this Markdown disagree, the disagreement is a **freeze-critical Protocol defect** to be corrected before downstream use; neither side may be silently preferred.

7.30 Minimum Phase-Analysis input set

The accepted run exposes at least the following controlling analysis inputs through the Protocol handoff and runtime bundle:

```
runtime/analysis_inputs/a0_completion.json;
runtime/analysis_inputs/a4_completion.json;
runtime/analysis_inputs/a4_role_control_statistics.json;
runtime/analysis_inputs/a4_c4_c5_statistics.json;
runtime/analysis_inputs/a4_c4_c5_comparison_artifacts.jsonl;
runtime/analysis_inputs/a4_burden_source.csv;
runtime/analysis_inputs/stage18_resource_constrained_anchor_budget_deviation.json;
runtime/analysis_inputs/figure_table_source_manifest.json;
runtime/analysis_inputs/failure_negative_summary.json;
runtime/analysis_inputs/a0_closure_source_manifest.json;
runtime/analysis_inputs/a0_participant_metrics.csv;
runtime/analysis_inputs/a0_statistics.json;
runtime/analysis_inputs/a4_participant_metrics.csv;
runtime/analysis_inputs/a4_role_control_participant_comparisons.csv;
runtime/analysis_inputs/a4_c4_c5_participant_comparisons.csv;
runtime/analysis_inputs/low_label_curve_summary.json and low_label_metric_source.csv;
runtime/analysis_inputs/subject_profile_summary.json, subject_profile_metric_source.csv, and session_profile_metric_source.csv;
runtime/analysis_inputs/training_policy_challenger_participant_comparisons.jsonl, training_policy_challenger_seed_comparisons.jsonl, and training_policy_challenger_terminal_states.jsonl;
runtime/analysis_inputs/training_policy_sr_probability_calibration_candidates.jsonl;
runtime/analysis_inputs/stage18S_R1_combined_cell_effects.csv, stage18S_R1_participant_paired_effects.csv, stage18S_R1_three_repeat_anchor_stability.csv,
stage18S_R1_six_budget_mr00_sensitivity.csv, provenance summaries, and stage18S_balanced_sensitivity_R1_preexecution_freeze.json;
runtime/manifests/stage_artifacts/20_readiness.json and canonical Layer2ReadinessReport partition;
runtime/analysis_inputs/training_policy_challenger_completion.json;
runtime/analysis_inputs/training_policy_challenger_statistics.json;
runtime/analysis_inputs/training_policy_sr_probability_selection.json;
runtime/protocol_change_required/P02_PREEXECUTION_TRAINING_POLICY_AMENDMENT_R2.yaml;
runtime/handoffs/stage18S_R1_sensitivity_evidence.json for separately labeled supplemental sensitivity evidence.
```

Phase Analysis must consume primitive/source records and these governed summaries without Layer 10 recomputation or result-selection changes.

7.31 Phase-specific final no-drift rules

- Do not rewrite post-observation Stage 11/12 amendments as pre-registered.
- Do not merge Stage 18S into canonical Stage 18 or use it to erase the A4 deviation.
- Do not claim five-repeat A4 stability, dense deep-budget A4 curves, or full Build Book A4 equivalence.
- Do not omit failed/skipped/blocked/incompatible/non-applicable terminal states.
- Do not change P01 dataset/split/label/window/low-label membership identities in a downstream analysis of this run.
- Do not use test evidence for retrospective model/policy selection.
- Do not choose latest for private external artifacts; pin exact revisions and verify hashes/manifests.

Do not assume a single Hugging Face credential covers both pre-P02 and P02 assets.

Do not serialize, print, log, or publish any literal access token.

Do not elevate P02 decoder/readiness evidence into calibration, safety, deployment, policy, robustness or clinical claims.

Do not let Layer 10 rerun, retune, reclassify, or strengthen P02 evidence.

Do not merge this annex into the cumulative Protocol by paraphrasing away run IDs, deviations, limitations or supersession history.

END OF INTEGRATED PHASE 02 PROTOCOL RECORD (SOURCE ANNEX R2)

PART V — CUMULATIVE RUN MATRIX

The cumulative run matrix retains phase identity while eliminating separate current run-matrix authority files. P00 cells preserve their historical Mode-B engineering evidence classification. P01 rows represent the final accepted 00–26 lineage; failed attempts are retained in rerun history rather than erased.

P00 engineering run matrix — 19 preserved cells

Cell	Purpose	Command	Evidence status	Limitations	Rerun rule
P00-CELL-AUTHORITY-INTAKE	Verify authority/source identities and requirement disposition	python scripts/run_local_first_finalization_audit.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-SCHEMA-COVERAGE	Validate JSON Schema and record-family coverage	python scripts/run_static_checks.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-CONFIG-RESOLUTION	Resolve strict P00 configuration	python -m iharq.cli phase validate-inputs --phase P00 --profile configs/phases/p00.yaml	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-IDENTITY-JCS-HASH	Validate typed IDs, canonical serialization, SHA-256 and golden vectors	python -m pytest -q tests/test_canonical.py tests/test_lineage_lifecycle.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-VALID-FIXTURES	Accept all valid and integrated non-empirical fixtures	python -m pytest -q tests/test_valid_fixtures.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-MALFORMED-FIXTURES	Reject the complete malformed taxonomy	python -m pytest -q tests/test_negative_fixtures.py tests/test_audit1_negative_fixtures.py tests/test_audit2_negative_fixtures.py tests/test_audit3_negative_fixtures.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-VALIDATOR-TEST-COVERAGE	Run complete deterministic suite	python -m pytest -q -p no:cacheprovider	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-A0-A13-READINESS	Verify A0-A13 readiness and reject A14	python scripts/run_phase0_final_implementation_audit.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-L0-L3-INTEGRATION	Verify L0-L3 integration foundation	python scripts/run_official_layer_audit_1.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-POLICY-UPDATE	Verify update-enabled policy traceability	python scripts/run_official_layer_audit_2.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-FROZEN-EVALUATION	Verify frozen-evaluation immutability	python scripts/run_official_layer_audit_2.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-L8-STRESS-LINEAGE	Verify clean-to-stressed lineage and limitations	python scripts/run_official_layer_audit_3.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-L9-EMBODIMENT-PROXY	Verify simulation-only embodiment proxy contracts	python scripts/run_official_layer_audit_3.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-L10-READ-ONLY	Verify Layer 10 source-only behavior	python scripts/run_official_layer_audit_3.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-MANIFEST-RECONCILIATION	Regenerate and compare repository manifest	python scripts/reconcile_repository_manifest.py && python scripts/reconcile_repository_manifest.py --check	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-LOCAL-REPRODUCTION	Reproduce from clean isolated local copy	python scripts/run_local_reproduction.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change

Cell	Purpose	Command	Evidence status	Limitations	Rerun rule
P00-CELL-PACKAGE-INTEGRITY	Build and verify repository-ready archive	python -m iharq.cli package build --output protocol_package_test.zip && python -m iharq.cli package verify --archive protocol_package_test.zip	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-FUTURE-PHASE-CONTRACTS	Verify P00-P15 reusable contracts and L0-L10 foundations	python scripts/run_phase0_final_implementation_audit.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change
P00-CELL-NEXT-DOCUMENT-READINESS	Verify six downstream readiness packages	python scripts/run_local_first_finalization_audit.py	ENGINEERING_RETROSPECTIVE_MODE_B	NON_EMPIRICAL; NON_CLAIM_BEARING	Rerun after any affected source/config/schema/test/manifest change

P01 accepted execution stage matrix — 27 stages

Stage	Purpose	Status	Validity	Actual outputs	Repair/rerun history
P01-STAGE-00	Corrected bootstrap and persistent isolated worker	PASS	VALID_FINAL_ACCEPTED_LINEAGE	authority_manifest.json	—
P01-STAGE-01	Environment	PASS	VALID_FINAL_ACCEPTED_LINEAGE	environment_manifest.json	—
P01-STAGE-02	Project and input intake	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-03	Authority and configuration	PASS	VALID_FINAL_ACCEPTED_LINEAGE	config_snapshot/official_run_freeze_manifest.json	—
P01-STAGE-04	Phase 0 regression	PASS	VALID_FINAL_ACCEPTED_LINEAGE	reports/phase_01/tests/phase0_and_runtime_regression.json	—
P01-STAGE-05	Source resolution	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-06	Dataset registry	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-07	Pass 1: verified source acquisition and bounded loading	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	R49 local stale revision guard failed before worker submission; R50 same-session continuation submitted Stage 07 once; PASS
P01-STAGE-08	Metadata normalization	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-09	Label mapping	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-10	Preprocessing compilation	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-11	Split construction and frozen fit population	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-12	Low-calibration budgets	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-13	Pass 2A: bounded preprocessing fit	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-14	Adopt verified core and materialize matched A4 R2	PASS	VALID_FINAL_ACCEPTED_LINEAGE	external_artifact_pointers/derived_windows_dataset.json; reports/phase_01/storage/a4_derived_output_storage_actual_precommit.json	R45 canonical record-ID normalization episode and later R48/R49 A4 implementation/profile repairs; accepted Stage 14 materialized A4 R2 while core was adopted unchanged
P01-STAGE-15	Validate and commit the separate A4 R2 Dataset	PASS	VALID_FINAL_ACCEPTED_LINEAGE	external_artifact_pointers/derived_windows_dataset.json; external_artifact_pointers/a4_window_family_dataset.json	A4 persistence/remote manifest validation after R48/R49 hardening; PASS
P01-STAGE-16	Record validation	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-17	Leakage audit	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—

Stage	Purpose	Status	Validity	Actual outputs	Repair/rerun history
P01-STAGE-18	A0–A13 readiness	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	Original missing-module failure; R51/ R52 recovery-cell defects; R53 worker-env import repair; PASS
P01-STAGE-19	Cards	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-20	Manifests	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-21	Negative register	PASS	VALID_FINAL_ACCEPTED_LINEAGE	negative_and_failed_results/run_failures_and_blockers.json	—
P01-STAGE-22	P02 and later compatibility	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-23	Evidence sufficiency	PASS	VALID_FINAL_ACCEPTED_LINEAGE	reports/phase_01/preliminary_gate_evaluation.json	—
P01-STAGE-24	Repair metadata	PASS	VALID_FINAL_ACCEPTED_LINEAGE	reports/phase_01/repair_reentry.json	—
P01-STAGE-25	Final export preparation	PASS	VALID_FINAL_ACCEPTED_LINEAGE	—	—
P01-STAGE-26	Terminal decision and bundle export	PASS	VALID_FINAL_ACCEPTED_LINEAGE	/kaggle/working/iharq_p01_l1/IHARQ_P01_L1_Phase_Execution_Bundle_d03f0a7c869d.zip; /kaggle/working/iharq_p01_l1/IHARQ_P01_L1_Phase_Execution_Bundle_d03f0a7c869d.zip.sha256; /kaggle/working/iharq_p01_l1/IHARQ_P01_L1_GitHub_Ready_Repository_d03f0a7c869d_20260807143013-663eab13.zip; /kaggle/working/iharq_p01_l1/IHARQ_P01_L1_GitHub_Ready_Repository_d03f0a7c869d_20260807143013-663eab13.zip.sha256; reports/phase_01/repository_release_plan.json	Original worker packaging BLOCKED by secret scan; R54 repack/redaction repaired external release without duplicating Stage 26; final canonical stage record PASS

Cumulative stage/cell denominator: P00 = 19 historical engineering cells; P01 = 27 accepted execution stages. No P01 stage identity is duplicated in the final canonical lineage.

P02 accepted execution stage matrix — 26 canonical stage records

Accepted run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d. Runtime-successor identity remains explicit; final PASS status never erases amendments or scoped limitations.

Stage	Attempt	Status	Observability	Runtime successor
00	634837763ab9	SUCCESS	PASS	
01	d4ab0101ea08	SUCCESS	PASS	
02	c87fedef7bcd	SUCCESS	PASS	
03	5b6834384730	SUCCESS	PASS	
04	3ec63b778b7d	SUCCESS	PASS	
05	39df114163ce	SUCCESS	PASS	
06	879c816b6a5b	SUCCESS	PASS	
07	f96233a7477e	SUCCESS	PASS	
08	83b1345dfb42	SUCCESS	PASS	{'archive_status': 'PRIOR_FAILURES_ARCHIVED_AND_122_RE TRIED', 'id': 'P02-RUNTIME-SUCCESSOR-R1-CLASS-WEIGHT-APPLICABILITY-LIVE-CONTINUATION', 'prior_attempt_id': '43e766d443bc', 'scientific_configuration_changed': False, 'semantic_gate_artifact':

Stage	Attempt	Status	Observability	Runtime successor
				'stage_artifacts/08_semantic_gate_successor_R1.json'}
09	c31206f11c83	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R2-CANONICAL-LOW-CAL-BUDGET-ID-PARSER-LIVE-CONTINUATION', 'scientific_configuration_changed': False, 'semantic_audit': 'diagnostics/runtime_successor/G09_SEMANTIC_AUDIT_R1.json'}
10	8e4ca693efde	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R2-CANONICAL-LOW-CAL-BUDGET-ID-PARSER-LIVE-CONTINUATION', 'scientific_configuration_changed': False, 'semantic_audit': 'diagnostics/runtime_successor/G10_SEMANTIC_AUDIT_R1.json'}
11	77533b224dad	SUCCESS	PASS	
12	b9d3bbec230b	SUCCESS	PASS	{'FBCNet_author_horizon_provenance': {'IHARQ_preserved_differences': {'class_weight_policy': 'P02_R2_GOVERNED', 'continue_after_earlstop': False, 'selection_metric': 'validation_BACC', 'split_authority': 'P01_SUBJECT_GROUPED_ROLES', 'test_visibility': 'STRICTLY_AFTER_SELECTION'}, 'author_batch_size': 16, 'author_load_best_model': True, 'author_lr': 0.001, 'author_max_epochs': 1500, 'author_no_decrease_epochs': 200, 'source_commit': 'de1bbdd8a54cb1e466830e3d47070e0e56761a37', 'source_file': 'codes/classify/cv.py', 'source_repository': 'ravikiran-mane/FBCNet'}, 'P00_P01_mutated': False, 'P01_consumption_audit': 'diagnostics/runtime_successor/stage12_external_models_R6R5/P02_STAGE12_R6R5_P01_CONSUMPTION_AUDIT_R1.json', 'amendment_reason': 'PREDECLARED_R6R4_MODEL_SPECIFIC_HORIZON': FBCNET_1500_200_MATCHING_ORIGINAL_AUTHOR_HORIZON_AND_NODECREASE_WINDOW; DBCONFORMER_AND_CBRAMOD_RETAIN_150_40', 'artifact_id': 'P02-STAGE12-R6R5-FBCNET-AUTHOR-HORIZON-R1', 'author_centered_recipes': {'DNN-FBCNET': {'batch_size': 16, 'grad_clip': None, 'label_smoothing': 0.0, 'lr': 0.001, 'max_epochs': 1500, 'optimizer': 'Adam', 'patience': 200, 'recipe_id': 'R6-FBCNET-AUTHOR-CENTERED-R1', 'restore_best': True, 'scheduler': 'constant', 'selection': 'validation_BACC', 'weight_decay': 0.0}, 'DNN-SEQ': {'batch_size': 32, 'grad_clip': None, 'label_smoothing': 0.0, 'lr': 0.001, 'max_epochs': 150, 'optimizer': 'Adam', 'patience': 40, 'recipe_id': 'R6-DBCONFORMER-AUTHOR-CENTERED-R1', 'restore_best': True, 'scheduler': 'constant', 'selection': 'validation_BACC', 'weight_decay': 0.0}, 'SSL-CBRAMOD': {'batch_size': 64, 'body_lr': 0.0001, 'eta_min': 1e-06, 'grad_clip': 1.0, 'head_lr': 0.0005, 'input_scale': 10000.0,

Stage	Attempt	Status	Observability	Runtime successor
				'label_smoothing': 0.1, 'max_epochs': 150, 'optimizer': 'AdamW', 'patience': 40, 'recipe_id': 'R6-CBRAMOD-SOURCE-CENTERED-R1', 'restore_best': True, 'scheduler': 'cosine', 'selection': 'validation_BACC', 'source_sampling_hz': 160, 'target_sampling_hz': 200, 'weight_decay': 0.05}}, 'id': 'P02-RUNTIME-SUCCESSOR-R6R5-STAGE12-FBCNET-AUTHOR-HORIZON-1500-200', 'model_specific_horizon_policy': {'DNN-FBCNET': {'max_epochs': 1500, 'patience': 200, 'reason': 'PREDECLARED_R6R5_ORIGINAL_AUTHOR_HORIZON_AND_NODECREASE_WINDOW'}, 'DNN-SEQ': {'max_epochs': 150, 'patience': 40, 'reason': 'AUTHOR_CENTERED_R6R3_HORIZON_RETAINED'}, 'SSL-CBRAMOD': {'max_epochs': 150, 'patience': 40, 'reason': 'SOURCE_CENTERED_R6R3_HORIZON_RETAINED'}}}, 'pre_r6_archive': 'diagnostics/runtime_successor/stage12_external_models_R6R5/pre_r6r3_official_stage12_archive/20260812T005807Z', 'scientific_configuration_changed': True, 'scientific_training_policy_amendment': True, 'semantic_audit': 'diagnostics/runtime_successor/G12_R6R3_SEMANTIC_AUDIT.json', 'staged_closure_audit': 'diagnostics/runtime_successor/stage12_external_models_R6R5/G12_R6R5_STAGED_CLOSURE_AUDIT_R1.json', 'test_set_used_for_hyperparameter_selection': False}
13	01b2fde44053	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R7H7-STAGE11-EEGNET-AUTHOR-CENTERED-10000E-TRUE-P120-CONTINUATION', 'scientific_configuration_changed': False}
14	6f67192684bf	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R7H7-STAGE11-EEGNET-AUTHOR-CENTERED-10000E-TRUE-P120-CONTINUATION', 'scientific_configuration_changed': False}
15	0cedb7522386	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R7H7-STAGE11-EEGNET-AUTHOR-CENTERED-10000E-TRUE-P120-CONTINUATION', 'scientific_configuration_changed': False}
16	bb462cca110e	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R7H7-STAGE11-EEGNET-AUTHOR-CENTERED-10000E-TRUE-P120-CONTINUATION', 'scientific_configuration_changed': False, 'semantic_audit': 'diagnostics/runtime_successor/G16_SEMANTIC_AUDIT_R1.json'}
17	785dd4d86f13	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R7H7-STAGE11-EEGNET-AUTHOR-CENTERED-10000E-TRUE-P120-CONTINUATION', 'scientific_configuration_changed': False}
18	a91f9d90814c	SUCCESS	PASS	{'ablation_conditions_removed': False, 'anchor_budget_deep_claim_authorized': True, 'completed_A0_artifacts_mutated': False,

Stage	Attempt	Status	Observability	Runtime successor
				'deep_anchor_budget_ids': ['P01-L1-LOW-CAL-OFFICIAL-R2:1_PER_CLASS', 'P01-L1-LOW-CAL-OFFICIAL-R2:8_PER_CLASS', 'FULL_TRAIN'], 'deep_budget_grid_reduced': True, 'dense_deep_budget_curve_claim_authorized': False, 'execution_transport_revision': 'R10.3_DUAL_GPU_DIFFERENT_ABLATION_CELLS', 'external_first_checkpoint_route': True, 'full_buildbook_replication_equivalence': False, 'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'multi_seed_stability_claim_authorized': False, 'policy_sha256': 'a21655f9b52cfbf817b1b271f758739cf01b1e7a75e786c880ccdd1a3c58ef32', 'protocol_deviation_from_buildbook': True, 'protocol_deviation_id': 'P02-STAGE18-RESOURCE-CONSTRAINED-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'refit_repeat_indices_executed': [0], 'scientific_change_scope': 'A4_AUTHOR_SOURCE_REFIT_PLUS_SINGLE_REPEAT_PLUS_DEEP_ANCHOR_BUDGET_REDUCTION;R10_3_TRANSPORT_FIXES', 'scientific_configuration_changed': True, 'semantic_audit': 'diagnostics/runtime_successor/G18_R6R3_SEMANTIC_AUDIT.json', 'single_model_multi_gpu': False, 'test_outcome_influence': 'PROHIBITED'}
18U	6c35912e38f7	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'scientific_configuration_changed': False}
19	243091602665	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'scientific_configuration_changed': False}
20	b5a19c3412a2	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'scientific_configuration_changed': False}
21	3d6646e74d1e	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'scientific_configuration_changed': False}
22	6bad6ea9c5ff	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'scientific_configuration_changed': False}
23	0e3f7416ac5e	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'scientific_configuration_changed': False}
24	5027e6f7359e	SUCCESS	PASS	{'id': 'P02-RUNTIME-SUCCESSOR-R10-STAGE18-A4-ANCHOR-BUDGET-SINGLE-REPEAT-R1', 'scientific_configuration_changed': False}

Gate closure: G00–G24 plus G18U are PASS with empty blockers. Stage 18 remains sufficient only for the governed resource-constrained A4 scope; Stage 18S remains post-hoc descriptive sensitivity.

PART VI — CUMULATIVE ANALYSIS CONTRACT

No inferential statistics are invented for P00 or P01. P00 is deterministic engineering-foundation analysis; P01 is deterministic data-protocol/reproducibility closure. Downstream scientific/model analyses belong to later phases.

P00 allowed analyses

Analysis ID	Inventory	Method	Evidence ceiling	Scientific inference
P00-AN-ARTIFACT-COVERAGE	expected/discovered/validated/failed/excluded/invalid/accepted artifacts	exact counts and path reconciliation	ENGINEERING_FOUNDATION_CONFORMANCE	No inferential scientific claims
P00-AN-SCHEMA-CONFIG	schemas/configs/record families	catalog-to-file-to-validator coverage	ENGINEERING_FOUNDATION_CONFORMANCE	No inferential scientific claims
P00-AN-FIXTURE-RESULTS	valid/integrated/malformed fixtures	pass/reject counts with explicit denominator	ENGINEERING_FOUNDATION_CONFORMANCE	No inferential scientific claims
P00-AN-TEST-GATES	tests/validators/gates	deterministic result and gate evidence crosswalk	ENGINEERING_FOUNDATION_CONFORMANCE	No inferential scientific claims
P00-AN-INTEGRATION	L0-3, L3-7, L7-10, update, frozen, stress, embodiment, L10	identity/lineage/limitation preservation	ENGINEERING_FOUNDATION_CONFORMANCE	No inferential scientific claims
P00-AN-REPRODUCTION	runtime lock, environment snapshot, clean reproduction and package integrity	exact hashes and exit codes	ENGINEERING_FOUNDATION_CONFORMANCE	No inferential scientific claims

P00 prohibited analyses: scientific superiority; treatment effects; p-values; clinical endpoints; model/calibration/stress/simulator performance claims.

P01 allowed analyses

Analysis ID	Analysis	Unit	Denominator	Aggregation	Interpretation ceiling
P01-AC-001-SOURCE-INVENTORY	Source inventory, provenance, license and checksum reconciliation	source file/dataset	all activated source files/datasets	counts and exact hash reconciliation	data provenance/reproducibility only
P01-AC-002-SPLIT	Subject-group split allocation and disjointness	subject group and derived window	all admitted subjects and accepted core windows	role counts by dataset and global	split/leakage correctness only
P01-AC-003-DENOMINATOR	Accepted event/window denominator conservation	accepted source event	12,910 accepted parent events	expected/discovered/valid window counts by dataset/role	data-materialization closure only
P01-AC-004-QUALITY	Quality/validation closure	quality summary/window	489	flag/hard-invalid counts by dataset	quality annotation/validity only
P01-AC-005-LEAKAGE	Leakage and visibility checks	subject/event/window/budget membership	all assigned groups/windows/budget identities	deterministic boolean checks	absence of detected contract leakage under implemented checks
P01-AC-006-EXTERNAL	External artifact persistence and integrity	external artifact/shard/index	declared governed artifacts	version/hash/size/count reconciliation	reproducible retrieval/integrity only
P01-AC-007-ABLATION-READINESS	A0-A13 foundation readiness and A14 absence	ablation identity	exact A0-A13 set (14 identities)	readiness disposition per identity	foundation readiness only; no effectiveness
P01-AC-008-A4	A4 R2 data-substrate synchronization/feasibility accounting	parent event / registered A4 view	12,910 core parent events	matched counts/profile identity/feasibility boundary	future A4 substrate readiness; NOT A4 effectiveness
P01-AC-009-ENVIRONMENT	Environment and reproducibility amendment reconciliation	execution environment/profile	one accepted P01 execution environment	exact versions/resources/amendment classifications	execution reproducibility only
P01-AC-010-GATES-REPAIRS	Gate closure and failed/superseded attempt accounting	gate/stage/repair episode	16 gates; 27 accepted stages; all material repair episodes	status/blocker/repair classification	execution closure only

Global negative/missingness rule: preserve failures, invalid/unmatched/diagnostic evidence; missingness never becomes zero-success. **Global no-post-hoc rule:** no result-dependent relabeling, denominator changes, source replacement or retrospective preregistration; A4 R2 foundation is not effectiveness evidence.

P02 allowed analyses and frozen analysis-input contract

Primary metric: BACC; complementary: F1_MACRO; secondary: ACC; aggregation: PARTICIPANT_FIRST. A0 scope:

RAW_ACCEPT_ALL_DECODER_MEASUREMENT_UNDER_FROZEN_P01_INPUTS. A4 scope: RESOURCE_CONSTRAINED_ANCHOR_BUDGET_SINGLE_REPEAT_A4_ABLATION.

Stage 18S is post-hoc/descriptive only. Layer 0 remains required before public claim approval.

Canonical P02 Phase-Analysis inputs: 38 paths. The exact list is frozen in protocol_v1_0/machine_readable/analysis_contract.yaml and the preserved P02 source companion.

PART VII — CUMULATIVE ABLATION AND CONTROL REGISTER

The authoritative global ladder is exactly A0–A13. P00 created foundation identities but executed no empirical ablation. P01 produced Layer-1 foundation readiness for all A0–A13 and executed none scientifically. A14 is prohibited/absent.

ID	Official identity	P00 disposition	P01 foundation	Executed in P01	Downstream	Evidence limitation
A0	Raw Decoder / Accept-All Raw Decoder Reference	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A1	Calibrated Decoder / Calibration Visibility	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A2	Simple Registered Threshold / Confidence-Threshold Baseline	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A3	Uncertainty and Selective Prediction	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A4	Longer-Window, Multi-Window Voting/ Averaging and Ordinary Ensemble Controls	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	A4_R2_FUTURE_CONFIRMATORY_ONLY; R1_4S_PROFILE_INFEASIBLE
A5	IHARQ-lite / Rule-Based Evidence Verification	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A6	IHARQ + Evidence-Quality Estimator	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A7	IHARQ + RegimeRisk Temporal Trust	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A8	Learning-to-defer / Deferral Comparison	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A9	Supervised Adaptive-IHARQ / Adaptive Readiness Policy	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A10	Contextual Bandit / Simulation-Bounded Immediate-Reward Adaptation	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A11	Reinforcement-Learning Policy / Simulation-Bounded Sequential Policy Learning	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A12	StressForge Stress Tests / Controlled Stress Robustness	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE
A13	Layer 9 Simulation-Only Embodiment Demo	FOUNDATION_IDENTITY_ONLY; NOT_ACTIVE; NO_EMPIRICAL_CELL	FOUNDATION_READY	false	P02-P15	FOUNDATION_READINESS_IS_NOT_EFFECTIVENESS_EVIDENCE

A14 prohibition

A14 = PROHIBITED / ABSENT. There is no valid A14 selector, run, result or claim. Any occurrence of “A14” in this document is solely an explicit prohibition/absence statement. Local A12.x identities, where they exist in upstream architecture, remain A12.x and are not renamed A14.

A4 cumulative chronology

- 16. **PRE-RUN / EARLY PROPOSAL:** +0.0...+4.0 s A4 longer window (640 samples) was proposed.
- 17. **FEASIBILITY FAILURE:** valid parent event PhysioNetMI:104:0:8:event:24 had only 560 samples after cue; the 4.0 s profile required 80 nonexistent samples.
- 18. **GOVERNED REPAIR:** no padding, clipping, fabrication or event dropping was allowed.
- 19. **FINAL R2 FOUNDATION:** A4_LONG_MATCHED_3P5S_R2, +0.0...+3.5 s, 560 samples; A4_MULTI_3X2S_UNIFORM_0P75S_R2 with slices 0:320, 120:440, 240:560.
- 20. **MATCHED DENOMINATOR:** 12,910/12,910 core parent events.
- 21. **P01 EVIDENCE STATE:** FOUNDATION_READY; executed_in_p01 = false.
- 22. **FUTURE CONFIRMATORY CONTRACT:** downstream A4 use must inherit exact R2 identities; substitution requires governed amendment/invalidation analysis.

Current cumulative A0–A13 state after P02

Ablation	Official identity	P00	P01	P02	Current	Limitation
A0	Raw Decoder / Accept-All Raw Decoder Reference	P00 readiness only	P01 foundation ready; not executed	EXECUTED	EXECUTED	678 terminals including non-success states
A1	Calibrated Decoder / Calibration Visibility	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A2	Simple Registered Threshold / Confidence-Threshold Baseline	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A3	Uncertainty and Selective Prediction	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A4	Longer-Window, Multi-Window Voting/Averaging and Ordinary Ensemble Controls	P00 readiness only	P01 foundation ready; not executed	EXECUTED_WITH_RECORDED_DEVIATION	EXECUTED_WITH_RECORDED_DEVIATION	C0–C5 preserved; repeat0 + deep anchors; no full-grid equivalence
A5	IHARQ-lite / Rule-Based Evidence Verification	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A6	IHARQ + Evidence-Quality Estimator	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A7	IHARQ + RegimeRisk Temporal Trust	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A8	Learning-to-defer / Deferral Comparison	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A9	Supervised Adaptive-IHARQ / Adaptive Readiness Policy	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A10	Contextual Bandit / Simulation-Bounded Immediate-Reward Adaptation	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A11	Reinforcement-Learning Policy / Simulation-Bounded Sequential Policy Learning	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A12	StressForge Stress Tests / Controlled Stress Robustness	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A13	Layer 9 MyoSuite/OpenSim/static-replay Embodiment Demo	P00 readiness only	P01 foundation ready; not executed	NOT_APPLICABLE_IN_P02 / DOWNSTREAM	NOT_YET_ACTIVE_AFTER_P02	Official identity retained for downstream phase
A14	absent / prohibited	PROHIBITED	PROHIBITED	PROHIBITED	PROHIBITED	No A14 may be invented

A13 current official wording is harmonized to the finalized P02 identity. The shortened predecessor P01 structured label remains provenance, not current authority.

PART VIII — CUMULATIVE AMENDMENT, DEVIATION, SUPERSESSION, AND RERUN REGISTER

Material P01 repair/rerun history

Repair	Stage/surface	Classification	Defect	Science changed	Data changed	Rerun scope	Resolution
R45 / SAME-SESSION CORE-ADOPTION NORMALIZATION	14	CANONICALIZATION_FIX	Existing verified core adoption blocked because date-bearing upstream record IDs changed dependent semantic hashes despite otherwise equivalent scientific records.	False	False	fresh worker/bundle plus affected adoption path; preserved persistent checkpoints	Dependency-order full-record equivalence with exact upstream-ID mapping plus exact remote manifest hash; core Dataset not mutated/reuploaded.
R46	runtime overlay	IMPLEMENTATION_BUG_FIX	Missing datetime/timezone runtime imports.	False	False	affected runtime path only	Required imports added.
R47	A4 profile canonicalization	CANONICALIZATION_FIX	Float literals in hash-bearing A4 profile could cause governed representation drift.	False	False	profile serialization/validation only	Governed decimal strings used for hash-bearing seconds fields.
R48	14-15 A4 materialization/persistence	IMPLEMENTATION_BUG_FIX + PERSISTENCE_FIX	A4 child interface/set handling and storage identity/read verification gaps; resumability and exact remote manifest closure hardened.	False	False	A4 child/interface and persistence path	Child interface corrected; reader/storage identity closure, synthetic E2E, resumable subject checkpoints and remote manifest verification added.
R49 / P01-L1-A4-WINDOW-FAMILY-FREEZE-R2	14-15 A4 profile	SCIENTIFIC_CONTRACT_CHANGE (A4 ALTERNATIVE ONLY)	Original proposed A4 +0.0..+4.0 s window is impossible for one valid released parent event (PhysioNetMI:104:0:8:event:24) without 80 nonexistent samples.	True	True	A4 materialization path with exact 12,910 matched parent events	R2 freezes +0.0..+3.5 s 560-sample tensor and three registered 2 s slices; no drop/pad/clip; confirmatory use prohibited until Protocol sync and downstream execution.
R50	07	INTEGRATION_FIX	Stale local DISPLAY_REVISION guard expected R42 while live runtime was R49; Stage 07 had not yet been submitted.	False	False	submit Stage 07 once using live worker; no 00-06 replay	Revision guard corrected; Stage 07 PASS.
R51-R53 / P01-L1-R53-STAGE18-WORKER-ENV-IMPORT-PROBE-R1	18	INTEGRATION_FIX	A4 readiness wrapper imported write_json from nonexistent module; two recovery-cell validation mistakes then occurred (malformed shim newline and notebook-kernel import probe outside worker PYTHONPATH).	False	False	Stage 18 only after worker-environment import probe	Compatibility shim re-exported authoritative manifests.write_json and was probed under exact worker_env; Stage 18 PASS and readiness artifact verified.
P01-L1-R54-FINAL-EXPORT-SECRET-REDACTION-R1	26 external release closure	SECURITY/PACKAGING_FIX	Live Kaggle credential was serialized into environment_amendment and the secret scanner correctly blocked the contaminated export.	False	False	redaction/repack/final integrity only; original worker Stage 26 not duplicated	Secret-like environment values redacted; final bundle/repository rebuilt; exact-token scan, manifests and checksums PASS; contaminated failed release is non-authoritative.

Protocol/document supersession and structural consolidation

Change	Type	Predecessor	Successor	Scientific effect	Current disposition
Master R2 → R3	GOVERNANCE_MIGRATION	IHARQ-PROTOCOL-V1-MASTER-R2	IHARQ-PROTOCOL-V1-MASTER-R3	NONE	V6.1 current workflow; R2 historical
P00 R1 → R2	INDEPENDENT_AUDIT_SUCCESOR	IHARQ-PROTOCOL-V1-P00-ANNEX-R1	IHARQ-PROTOCOL-V1-P00-ANNEX-R2	NONE	R2 P00 content preserved
P01 Annex R1	PHASE_ADDITION	none	IHARQ-PROTOCOL-V1-P01-ANNEX-R1	Records actual accepted P01; no post-hoc effectiveness claim	Internalized into cumulative Protocol

Change	Type	Predecessor	Successor	Scientific effect	Current disposition
Single-file consolidation	DOCUMENT_STRUCTURE_CONSOLIDATION	Master R3 + P00 R2 + P01 R1 + separate structured surfaces	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P01-R1	NONE	One current authority; predecessors historical/source only

Amendment classification principles

- Only the A4 R1 → R2 **alternative profile** is a SCIENTIFIC_CONTRACT_CHANGE, limited to the future A4 control; the official P01 core did not change.
- Python 3.12 compatibility is EXECUTION_COMPATIBILITY_CHANGE.
- adaptive-disk policy is RESOURCE_POLICY_CHANGE.
- R45/R46/R47/R48/R50/R51–R53 are canonicalization/implementation/integration/persistence fixes as recorded.
- R54 is SECURITY/PACKAGING_FIX.
- No listed non-A4 repair changed dataset membership, labels, split, preprocessing, official core window, metric, core denominator or claim ceiling.

Material P02 amendment/rerun history

Event	Classification	Scientific/config change	Current disposition
P02-PREEXEC-TRAINING-POLICY-AMENDMENT-R2	pre-test training-policy resolution	YES, before relevant final test evaluation	frozen supporting policy
Stage 11 author-centered EEGNet successors	post-observation scientific amendment + rerun	YES	R7H7 canonical; predecessor evidence preserved
Stage 12 author/source-centered successors	post-observation scientific policy/config amendment + rerun	YES	R6R5 canonical; blocked/skipped branches preserved
Stage 18 resource-constrained execution	Protocol deviation / compute-resource reduction	execution surface reduced	accepted restricted A4 scope; not full replication
Stage 18S	post-hoc sensitivity	supplemental only	descriptive; canonical Stage18 unchanged
Stage 24 low-disk HF release/finalization replay	packaging/release repair	NO scientific change	archival fallback + accepted replay

Cumulative integration event: P00/P01 cumulative predecessor + standalone P02 R2 → IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1. Classification: document-structure and machine-contract integration; scientific change: NONE.

PART IX — CUMULATIVE ENVIRONMENT AND REPRODUCIBILITY REGISTER

Phase/state	Environment	Status/consequence
P00 historical execution	Python 3.13.5 exact verified local runtime; portable cross-version lock incomplete	Mode-B engineering foundation; nonblocking portability limitation
P01 pre-run intent	Kaggle image intent; Python >=3.11,<3.12; exact package pins; original 60 GiB minimum disk	Historical intended environment
P01 accepted execution	Python 3.12.13; package pins matched; P01-L1-KAGGLE-ENV-FREEZE-R5	Accepted execution compatibility amendment; scientific values unchanged
P01 resource amendment	P01-L1-KAGGLE-ADAPTIVE-DISK-R1; observed free disk 18.94 GiB	Resource-policy amendment; no scientific effect
P01 release security	R54 redaction/repack after secret scanner blocked contaminated Stage-26 export	Final package token scan PASS; contaminated release non-authoritative

Deterministic/runtime identities must be carried with reproductions. Environment differences never silently authorize scientific-contract changes.

P02 accepted runtime and reconstruction

Phase/state	Environment / identity rule	Status/consequence
P02 accepted execution	Exact environment manifest + runtime-successor chain	Accepted; reproduce with recorded Python/package/GPU/runtime identities
P02 notebook reconstruction	Integrated implementation notebook + post-restart continuation + stage ledgers	Distinct run-config/freeze/source/config/stage-plan/notebook hash classes preserved
P02 release/security	Private HF persistence; symbolic credentials only	Literal-token publication prohibited

PART X — CUMULATIVE DATASET, SPLIT, MODEL, AND METHOD STATE

Dataset and split overview through P02

P02 introduces no replacement dataset/split authority; it consumes the exact accepted P01 substrate. Detailed hashes, licenses, subject/window denominators and preprocessing remain in the P01/P02 phase records.

Dataset	First active	Split	P01 role	P02 role	Current / limitation
PhysioNetMI	P01	P01-L1-SPLIT-OFFICIAL-R2	source/provenance + binary MI core windows	inherited for A0/A4 decoders	ACTIVE; run-context/rate/channel limitations preserved
BNCI2014_001	P01	P01-L1-SPLIT-OFFICIAL-R2	binary left/right benchmark source	inherited for A0/A4 decoders	ACTIVE; exclusions/EOG/license limitations preserved
Lee2019_MI	P01	P01-L1-SPLIT-OFFICIAL-R2	binary left/right target companion	inherited for A0/A4 decoders	ACTIVE; labeled offline/train scope + resampling/session lineage preserved

Canonical split: P01-L1-SPLIT-OFFICIAL-R2, seed 20260804, subject-atomic 60/20/10/10; accepted core windows 7,589 train / 2,655 calibration / 1,283 validation / 1,383 test. Low-label budgets remain 1/2/4/8/16/32 events per class with one inherited frozen membership subset per budget.

Model/method overview through P02

Branch	Family	Terminal state	Evidence role	A4
SAN-MAJ	sanity	3 SUCCESS	reference	No
SAN-STRAT	sanity	60 SUCCESS	reference	No
SAN-PERM	sanity diagnostic	300 SUCCESS	diagnostic negative control	No
SAN-PRIOR	sanity diagnostic	3 SUCCESS	diagnostic prior control	No
DIAG-LOGVAR	diagnostic	3 SUCCESS	diagnostic baseline	No
CLS-CSP-LDA	classical	18 SUCCESS / 3 FAILED	A0 + A4	Yes
CLS-FBCSP-LR	classical	3 SUCCESS	A0 + A4	Yes
RIE-TS-LR	Riemannian	21 SUCCESS	A0 + A4	Yes
RIE-EA-TS	Riemannian	3 SUCCESS	A0 + A4 when admitted	Conditional
RIE-MDM	Riemannian diagnostic	3 SUCCESS	diagnostic only	No

Branch	Family	Terminal state	Evidence role	A4
DNN-EEGNET	compact neural	105 SUCCESS	A0 + A4; mixed Stage11 lineage	Yes
DNN-FBCNET	conditional neural	15 SUCCESS	A0 + A4 if admitted	Conditional
DNN-SEQ	conditional sequence	15 SUCCESS	A0 + A4 if admitted	Conditional
DNN-EGTC	fallback	15 CONDITIONAL_SKIP	fallback not activated	No
SSL-CBRAMOD	compact SSL	105 SUCCESS	A0; qualified A4	Qualified
SSL-REVE	SSL diagnostic	3 DEPENDENCY_BLOCKED	diagnostic blocked	No

PART XI — CUMULATIVE EXTERNAL ARTIFACT AND REPRODUCIBILITY-ACCESS REGISTER

P00 has no Protocol-critical oversized numerical Dataset pointer. P01 externalizes the large numerical arrays while preserving compact governed manifests/pointers in the project state. A handle alone is insufficient; exact revision and manifest hash are mandatory.

Artifact	Provider	Handle	Provider rev	Logical rev	Format	Access	Manifest SHA-256	Local copy state
P01-L1-DERIVED-WINDOWS-d03f0a7c869dd95a-20260806222242-68a91473	Kaggle	csthv999z/iharq-p01-l1-derived-windows-d03f0a7c869d-20260806222242-68a91473	2	1	LOSSLESS_HDF5_SUBJECT_SHARDS	PRIVATE	dc21f418e9a1add62adb346f627d5d729735a5f1ecaa1d771f6fa5cfede627e1	SHARDS_DELETED_AFTER_VERIFIED_KAGGLE_BLOB_UPLOAD
P01-L1-A4-DERIVED-WINDOWS-4cd080393345e8aa-20260807143013-663eab13	Kaggle	csthv999z/iharq-p01-l1-a4-d03f0a7c-9a7c6108	1	2	LOSSLESS_HDF5_MATCHED_3P5S_SUBJECT_SHARDS_WITH_REGISTERED_VIRTUAL_MULTIWINDOW_VIEWS	PRIVATE	29124d59907610b1545e0a2b5d1811a4d4a2bf1df64cad49aa880f4721c47305	A4_SHARDS_DELETED_AFTER_VERIFIED_KAGGLE_BLOB_UPLOAD

The official core Dataset remains immutable/reused; the A4 Dataset is a separate alternative-window substrate. Neither may be silently regenerated or substituted by a downstream consumer.

P02 private Hugging Face sources

Role	Repository/dataset	Immutable revision	Access	Integrity / use
Preferred direct continuation	Csthv/p02-phase02-final-whole-working-20260814t052120z-b890ff92	bc14961e14f2e48690e55df3577014275f9cbf30	IHARQ_HF_TOKEN_P02	verify working-tree manifest + per-object index; preferred workspace substrate
Archival/fallback release	Csthv/iharq-p02-phase02-r2-P02-L2-OFFICIAL-RUN-R4DY-20260814t051352z	257a407b5c7ea37b6c620863ee261c010c8f197c	IHARQ_HF_TOKEN_P02	verify exact uploaded object size/SHA; valid archival transport reconstruction

Credential classes: IHARQ_HF_TOKEN_PRE_P02 for pre-P02 private Hugging Face assets and IHARQ_HF_TOKEN_P02 for P02 HF assets. A single-token assumption is prohibited; literal values are prohibited; mutable latest is prohibited. P01 Protocol-critical numerical arrays remain Kaggle-governed and are not rewritten as HF assets.

PART XII — CUMULATIVE LIMITATION, EVIDENCE-CEILING, HANDOFF, AND CURRENT-STATE REGISTER

Limitation ID	Scope	Origin	Description	Evidence consequence	Claim consequence	Status
P00-ENGINEERING-FOUNDATION-ONLY	P00	P00	P00 evidence is Mode-B engineering/retrospective foundation evidence and contains no empirical claim-bearing cell.	ENGINEERING_FOUNDATION_CONFORMANCE only	No scientific effectiveness claim	ACTIVE_HISTORICAL_LIMIT
P00-PORTABLE-LOCK-INCOMPLETE	P00	P00	Portable registry-resolved cross-version uv.lock incomplete; exact verified Python 3.13.5 runtime lock available.	Cross-version portability not proven	No effect on P00 engineering validation already executed	NONBLOCKING
PUBLIC_EEG_ONLY	P01	P01	P01 uses selected public EEG datasets only.	Bounds P01 evidence interpretation/reproduction	Must be inherited by downstream claim/evidence products	ACTIVE
NON_CLINICAL	P01	P01	P01 evidence is non-clinical research data/protocol evidence.	Bounds P01 evidence interpretation/reproduction	Must be inherited by downstream claim/evidence products	ACTIVE
NO_DEPLOYMENT_CLAIM	P01	P01	P01 does not establish deployment safety/effectiveness.	Bounds P01 evidence interpretation/reproduction	Must be inherited by downstream claim/evidence products	ACTIVE
PRIVATE_KAGGLE_EXTERNAL_ARTIFACT_ACCESS	P01	P01	Core and A4 numerical datasets require private Kaggle access and exact revision/hash verification.	Bounds P01 evidence interpretation/reproduction	Must be inherited by downstream claim/evidence products	ACTIVE
A4_R2_RETROSPECTIVE_FEASIBILITY_ORIGIN_NOT_CONFIRMATORY_IN_P01	P01	P01	A4 R2 emerged from feasibility repair and is future-confirmatory only; no P01 effectiveness result.	Bounds P01 evidence interpretation/reproduction	Must be inherited by downstream claim/evidence products	ACTIVE
ACTUAL_RUNTIME_PYTHON_3_12_13_WITH_COMPATIBILITY_AMENDMENT	P01	P01	Accepted P01 execution used Python 3.12.13 under documented compatibility amendment.	Bounds P01 evidence interpretation/reproduction	Must be inherited by downstream claim/evidence products	ACTIVE
ADAPTIVE_DISK_RESOURCE_AMENDMENT	P01	P01	Accepted P01 execution used adaptive disk policy rather than original 60 GiB preflight.	Bounds P01 evidence interpretation/reproduction	Must be inherited by downstream claim/evidence products	ACTIVE

Cumulative evidence ceilings

- **P00:** engineering-foundation conformance only; no empirical scientific-effectiveness claim.
- **P01:** provenance, intake correctness, labels/exclusions, split/leakage integrity, preprocessing/window materialization, quality, lineage, denominator closure, external persistence and downstream readiness.
- **Not established through P01:** decoder superiority, calibration benefit, A4 effectiveness, clinical benefit, deployment safety, policy-learning benefit, stress robustness, temporal-trust benefit, embodiment/control effectiveness.
- Layer 0, not this Protocol, later approves/qualifies/blocks claim wording.

P02 active limitations after cumulative integration

- public benchmark binary-MI research evidence only; non-clinical and no deployment claim
- all P02 conclusions inherit exact P01 labels/split/preprocessing/core/A4 windows/source availability
- one frozen low-label membership subset per budget; no repeated subset-resampling uncertainty
- non-success branches/terminals remain in denominators and evidence inventories
- Stage 11/12 accepted evidence includes transparent post-observation author/source-informed amendments
- Stage 11 primary EEGNet is mixed grandfathered P500/R6 lineage; R7 TRUE_P120 is diagnostic challenger only
- A4 only refit repeat 0; deep anchor budgets 1/class, 8/class, FULL_TRAIN; no five-repeat/dense-curve/full-replication claim
- Stage 18S is post-hoc descriptive sensitivity only; canonical Stage18/G18 unchanged
- probability/ROC-AUC semantics conditional on genuine compatible score/probability outputs
- final notebook is a post-restart continuation; Stages 00–18 reconstructed from canonical ledgers/artifacts

- P02 HF repositories private; remote access requires authorization and exact immutable verification
- mixed pre-P02/P02 private HF access uses separate symbolic credential classes
- accepted results tied to recorded environment/runtime-successor identities
- calibration, uncertainty, abstention, policy, temporal/stress robustness, simulation and embodiment remain downstream/unestablished

Cumulative evidence ceilings through P02

- P00: engineering-foundation conformance only; no empirical scientific-effectiveness claim.
- P01: data/provenance/split/preprocessing/window/quality/lineage/denominator/external-persistence/downstream-readiness evidence; no decoder-effectiveness claim.
- P02: raw accept-all decoder/baseline measurement, low-label/profile evidence, and governed A4 ordinary-control evidence only within the resource-constrained scope.
- A4: no five-repeat stability, dense deep-budget response-curve, or full-original-Build-Book replication-equivalence claim.
- Stage 18S: descriptive/post-hoc sensitivity only. Layer 0 remains claim-disposition authority.

Historical downstream analysis and phase-handoff contracts at P01 closure

Phase 1 Evidence, Results, and Interpretation Report

May consume this Protocol, the accepted P01 execution bundle, the embedded analysis contract and exact run identities. It owns comprehensive results/findings/interpretation/candidate claims. It may not alter this Protocol retrospectively.

Layer 0

Consumes evidence ceilings, limitations and stable evidence identities. It owns claim approval/qualification/blocking and may not change measurements.

Evidence Map

Consumes stable run/record/artifact IDs and Layer-0 dispositions to map claims to evidence/manuscript locations. It does not redefine Protocol contracts.

Layer 10

Consumes read-only governed records/manifests/results. It renders cards/tables/figures/reproducibility views and must never hide negative/diagnostic evidence or recompute hidden experimental logic.

P01 → P02 technical consumption contract

P02 may consume the exact verified core Dataset pointer and, for A4 only under an explicit downstream Protocol/analysis cell, the exact A4 R2 Dataset/profile. It must inherit the canonical label maps, subject split, preprocessing, window profiles, config identity, parent-event matching keys and limitations. It is explicitly prohibited from silent relabeling, rewindowing, split mutation, test leakage, denominator substitution or A4 profile substitution. Any such change requires a governed amendment and descendant invalidation/rerun assessment.

Forward rule for future phases

P02 and every later phase extend this same cumulative Protocol v1.0 document. Phase-specific content remains clearly sectioned internally, but no future phase creates a competing separately authoritative Protocol file. Historical revisions may be archived by content hash. Only one current Protocol revision is authoritative at a time.

Downstream contracts through P02

Phase-02 Evidence, Results, and Interpretation Report may consume the cumulative Protocol, accepted P02 execution bundle and exact analysis-input contract; it may not rewrite Protocol history or expand A4/18S claim scope.

Layer 0 controls candidate-claim disposition; the Evidence Map binds reviewed claims to exact evidence; Layer 10 consumes read-only governed evidence and may not rerun, retune, rematch or strengthen it.

P02 → P03 continuation contract

P03 may consume accepted P02 raw scores/probabilities and model-family evidence for calibration/uncertainty/selective-prediction work. It must preserve P01/P02 lineage, canonical/superseded boundaries, class order, score semantics, failures, A4 limitations and external-artifact integrity. Any new transformation is a new governed downstream artifact.

Current canonical state at end of P02

Surface	Current state
completed phases	P00, P01, P02
current cumulative Protocol	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1
P00	frozen engineering foundation
P01	accepted L1 data/split/preprocessing/window substrate

Surface	Current state
P02 run	P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d
P02 Stage 11	accepted mixed grandfathered/source-informed EEGNet lineage
P02 Stage 12	accepted author/source-centered successor; later revalidation no retraining implied
A0	COMPLETE; 678 terminals including explicit non-success states
A4	COMPLETE FOR GOVERNED RESOURCE-CONSTRAINED SINGLE-REPEAT/ANCHOR-BUDGET SCOPE
Stage 18S	PASS; POST-HOC DESCRIPTIVE SENSITIVITY ONLY
evidence sufficiency	G23 PASS; G24 PASS; blockers []
preferred P02 source	Csthv/p02-phase02-final-whole-working-20260814t052120z-b890ff92@bc14961e14f2e48690e55df3577014275f9cbf30
P02 archival fallback	Csthv/iharq-p02-phase02-r2-P02-L2-OFFICIAL-RUN-R4DY-20260814t051352z@257a407b5c7ea37b6c620863ee261c010c8f197c
continuation	P03 extends this same cumulative Protocol authority

APPENDIX A — STRUCTURED CUMULATIVE PROTOCOL METADATA

Field	Value
protocol_id	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1
protocol_version	v1.0
document_revision	R1
coverage	P00–P02
status	FROZEN_WITH_EXPLICIT_NONBLOCKING_LIMITATIONS
source cumulative P00/P01	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P01-R1
source P02	IHARQ-PROTOCOL-V1-P02-ANNEX-R2
next extension	P03

APPENDIX B — STRUCTURED CUMULATIVE RUN MATRIX

Canonical companion: protocol_v1_0/machine_readable/run_matrix.yaml. It preserves all P00/P01 predecessor rows and nests the exact P02 R2 run matrix under phase P02. P02: 26 canonical stages, 26 gate records, A0 terminal cells 678, A4 full-Build-Book replication equivalence false.

APPENDIX C — STRUCTURED CUMULATIVE ANALYSIS CONTRACT

Canonical companion: protocol_v1_0/machine_readable/analysis_contract.yaml. It preserves P00/P01 analysis authority and nests the exact P02 R2 analysis contract with 38 canonical inputs, participant-first aggregation, BACC primary metric, explicit A4 claim locks and Stage18S post-hoc status.

APPENDIX D — STRUCTURED CUMULATIVE ABLATION REGISTER

Canonical companion: protocol_v1_0/machine_readable/ablation_register.yaml. Current A0–A13 identities are explicit; P01 legacy label differences remain traceable; A14 is PROHIBITED_ABSENT.

APPENDIX E — STRUCTURED EXTERNAL ARTIFACT REGISTER

Canonical companion: protocol_v1_0/machine_readable/external_artifact_access_contract.yaml. P01 Kaggle pointers remain provider-specific; P02 preferred workspace and archival HF repositories are pinned to exact revisions with dual symbolic credential interfaces and mutable-latest prohibition.

APPENDIX F — SOURCE / AUTHORITY UTILIZATION AND MERGE-SOURCE MATRIX

Source	Merge role	Treatment	Status
Cumulative Protocol through P01	primary backbone	P00/P01 phase records and predecessor cumulative state preserved	USED_PRIMARY
P02 Annex R2	primary P02 source	integrated without semantic reconstruction; source annex retained	USED_PRIMARY

Source	Merge role	Treatment	Status
P02 R2 machine companions	primary structured source	nested into cumulative YAMLS; source copies retained	USED_PRIMARY
Governance/authorities	supporting terminology/ownership	used only where merge ambiguity required; closed science not reopened	USED_SUPPORTING

APPENDIX G — CROSS-PHASE MERGE CONSISTENCY MATRIX

Cross-check	Final state	Status
P00 preservation	complete P00 record retained	PASS
P01 preservation	complete P01 record retained	PASS
P02 preservation	complete validated P02 record retained	PASS
Stage11/12	amendment/rerun history not flattened	PASS
A4	resource scope/claim lock preserved	PASS
Stage18S	post-hoc status preserved	PASS
A14	prohibited	PASS
historical vs current	P01 closure vs post-P02 current state distinguished	PASS
machine-readable parity	phase-indexed cumulative contracts	PASS
future P03	extensible without redesign	PASS

APPENDIX H — HUMAN / STRUCTURED NO-DRIFT VALIDATION

The final master, phase projections and cumulative machine-readable companions must agree on phase/run IDs, status vocabulary, ablation states, limitations, amendment history, external pointers, credential classes and analysis restrictions. Deterministic validation reports are packaged under validation/.

APPENDIX I — PROTOCOL FREEZE, PRESERVATION, AND INTEGRITY MANIFEST

Field	Frozen value
current_protocol_id	IHARQ-PROTOCOL-V1-CUMULATIVE-THROUGH-P02-R1
coverage	P00–P02
P00 preserved	YES
P01 preserved	YES
P02 preserved	YES
Stage11/12 history preserved	YES
A4 limitation preserved	YES
Stage18S post-hoc preserved	YES
failures/superseded preserved	YES
A14 prohibited	YES
next extension	P03

FINAL CURRENT-STATE SUMMARY THROUGH P02

P00, P01 and P02 are represented inside one current cumulative Protocol authority. P00 remains the engineering foundation; P01 remains the governed data/split/preprocessing/window substrate; P02 is the accepted Layer-2 decoder/baseline measurement phase with transparent Stage11/12 amendment history, complete A0 accounting, A4 evidence sufficient only for its governed resource-constrained scope, Stage18S post-hoc descriptive sensitivity, explicit non-success evidence, and immutable external-artifact access rules. The cumulative analysis authority and machine-readable companions preserve phase boundaries and are designed for direct P03 extension.

IHARQ_PROTOCOL_V1_THROUGH_P02_MERGE_STATUS = COMPLETE